Numeracy

Short Course

Student Workbook

- **Topic 1: Personal Identity and Community**
  Chapters 1–8 | Weeks 1–9
- **Topic 2: Workplace and Employment**
  Chapters 9–15 | Weeks 10–18

Aligned to the QCAA Senior Short Course: Numeracy Syllabus 2025 v1.2

Student Information

Name: _____

Class: _____

Teacher: _____

Numeracy Short Course — Student Workbook

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How to Use This Workbook

This workbook is your main resource for Numeracy. It contains everything you need for both topics, both assessment tasks, and your final examination. Keep it safe and bring it to every lesson.

Worked Examples	Read each worked example carefully before attempting the exercises. The step-by-step working shows you exactly how to set out your answers.
Practice Exercise	These questions follow the same type as the worked examples. Attempt them after each set of worked examples.
Application Exercise	These questions apply the skills in real workplace or community contexts. They are slightly more challenging.
Chapter Review	Completed at the end of each chapter. The final two questions are multi-step problems.
Skills Checklist	Rate your confidence at the end of each chapter and identify areas to revise.
Open-Book Exam	This workbook is permitted in Assessment Task 2. Tab key pages now so you can find formulas quickly during the exam.

Always show your working. A correct answer without working may not receive full marks.

Box Types in This Workbook

Chapter Information	Appears at the start of each chapter. Shows the week, topic and skills area.
Key Formulas	Contains all the formulas you need for that chapter. Tab these pages!
Worked Example	A fully worked problem with step-by-step method.
Step 1 / Step 2 . . .	Individual steps within a worked example.
Practice Exercise	Questions for you to complete. Write your answers in the spaces provided.
Chapter Review	End-of-chapter review questions covering all skills.

Course Overview

Week	Lessons	Chapter / Content	Topic	Skills
1	1–4	Ch 1: Numbers in Everyday Life	T1	N&A
2	5–8	Ch 2: Fractions, Decimals and %	T1	N&A
3	9–12	Ch 3: Rates, Ratios and Comparisons	T1	N&A
4	13–16	Ch 4: Length, Perimeter and Area	T1	M&S
5	17–20	Ch 5: Shapes, Area and Maps	T1	M&S
6	21–24	Ch 6: Time and Angles	T1	M&S
7	25–28	Ch 7: Data and Graphs	T1	S&P
8	29–32	Ch 8: Chance and Probability	T1	S&P
9	33–36	**Assessment Task 1: Project**	T1	—
10	37–40	Ch 9: Numbers at Work	T2	N&A
11	41–44	Ch 10: Workplace Numbers	T2	N&A
12	45–48	Ch 11: 3D Shapes and Volume	T2	M&S
13	49–52	Ch 12: Mass, Capacity and Temperature	T2	M&S
14	53–56	Ch 13: Workplace Measurement	T2	M&S
15	57–60	Ch 14: Data at Work	T2	S&P
16	61–64	Ch 15: Probability and Trends at Work	T2	S&P
17	65–68	Exam revision (teacher-led)	—	—
18	69–72	**Assessment Task 2: Examination**	T2	—

Skills Areas:

N&A	Number and Algebra
M&S	Measurement and Space
S&P	Statistics and Probability

Assessment:

Task 1 (Week 9)	Individual Project — Topic 1: Personal Identity and Community
Task 2 (Week 18)	Open-Book Examination — 5 min perusal, 45 min working time

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TOPIC ONE

1 Personal Identity and Community

Chapters 1–8 | Weeks 1–9

In this topic you will develop numeracy skills in the context of everyday life, personal finances, community spaces and civic participation. You will work with numbers, measurement, shapes, data and probability in situations relevant to yourself and your community.

CHAPTERS IN THIS TOPIC

Ch 1: Numbers in Everyday Life
Ch 2: Fractions, Decimals and %
Ch 3: Rates, Ratios and Comparisons
Ch 4: Length, Perimeter and Area
Ch 5: Shapes, Area and Maps
Ch 6: Time and Angles
Ch 7: Data and Graphs
Ch 8: Chance and Probability

Assessment Task 1: Project

1. Numbers in Everyday Life

Chapter Information

Chapter:	1 — Numbers in Everyday Life
Weeks:	Week 1
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Lesson reference:	Lessons 1–4

Numbers are part of every decision we make. When you check your bank balance, buy a bus ticket, watch the footy scores, or plan a weekend trip with friends, you are using numbers. Australia's population of more than 26 million people, the distance of 1,763 km between Sydney and Brisbane, the price of a movie ticket — these are all whole numbers that appear in everyday life. In this chapter, you will build confidence with reading, writing, ordering and calculating with whole numbers up to the millions.

Learning Objectives

By the end of this chapter you will be able to:

- Read and write whole numbers up to millions in words and numerals
- Identify the place value and value of any digit in a whole number
- Order and compare whole numbers from smallest to largest (and largest to smallest)
- Add, subtract, multiply and divide whole numbers
- Use a calculator to perform and check calculations
- Estimate answers by rounding before calculating

Key Terms

- **Whole number:** A number with no decimal or fraction part (0, 1, 2, 3, . . .)
- **Digit:** Any of the ten symbols used to write numbers: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- **Place value:** The value of a digit based on *where* it sits in a number
- **Sum:** The result of adding two or more numbers
- **Difference:** The result of subtracting one number from another
- **Product:** The result of multiplying two or more numbers
- **Quotient:** The result of dividing one number by another

- **Estimate:** An approximate answer, found by rounding numbers before calculating
- **BODMAS:** The order in which operations must be performed — Brackets, Orders, Division, Multiplication, Addition, Subtraction
- **Operation:** A mathematical process — addition ($+$), subtraction ($-$), multiplication ($\times$), division ($\div$)

Key Formulas and Rules

- **BODMAS order of operations:**
  1. **B**rackets — calculate anything inside brackets first
  2. **O**rders — powers and square roots
  3. **D**ivision and **M**ultiplication — left to right
  4. **A**ddition and **S**ubtraction — left to right
- **Rounding rule:** Look at the digit *to the right* of the rounding position. If it is 5 or more, round up. If it is 4 or less, round down.
- **Estimation tip:** Always estimate *before* you calculate. If your calculator answer is far from your estimate, recheck your working.
- **Place value chart:**

Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
1,000,000	100,000	10,000	1,000	100	10	1

Worked Examples

Worked Example 1 — Reading and Writing Whole Numbers

Write the number 4,367,092 in words.

Solution:

Step 1: Split the number into groups of three, starting from the right.

$$4 \quad | \quad 367 \quad | \quad 092$$

Step 2: Read each group and add the period name.

- 4 → *four million*
- 367 → *three hundred and sixty-seven thousand*
- 092 → *ninety-two*

Step 3: Combine.

Answer: four million, three hundred and sixty-seven thousand, and ninety-two

Reverse example: Write **"two million, fifty thousand, and six"** as a numeral.
Answer: **2,050,006** (zeros hold the place for missing values)

Worked Example 2 — Place Value

For the number 3,854,206, state the place value and value of the digit 8.

Solution:

Step 1: Write each digit in the place value chart.

Millions	Hundred Thous.	Ten Thous.	Thousands	Hundreds	Tens	Ones
3	**8**	**5**	**4**	**2**	**0**	**6**

Step 2: Locate the digit 8.
The digit 8 is in the **hundred thousands** column.

Step 3: Calculate its value.

$8 \times 100{,}000 = \mathbf{800{,}000}$

Answer: The place value of 8 is **hundred thousands**. Its value is **800,000**.

Worked Example 3 — Ordering Whole Numbers

Order these numbers from smallest to largest: **7,840** **7,408** **7,084** **7,480**

Solution:

Step 1: Compare the thousands digits. All four numbers start with 7 — the thousands digits are equal.

Step 2: Compare the hundreds digits.

- 7,**8**40 → hundreds digit = 8
- 7,**4**08 → hundreds digit = 4
- 7,**0**84 → hundreds digit = 0
- 7,**4**80 → hundreds digit = 4

The smallest hundreds digit is 0, so 7,084 is the smallest number.

Step 3: For 7,408 and 7,480 (both have hundreds digit 4), compare the tens digits: $0 < 8$, so $7{,}408 < 7{,}480$.

Answer (smallest to largest): 7,084 < 7,408 < 7,480 < 7,840

Worked Example 4 — Addition and Subtraction (Shopping Budget)

Ben goes to Coles. He buys milk for \$3.80, bread for \$4.50, cheese for \$6.90 and apples for \$4.20. He pays with a \$20 note. How much change does he receive?

Solution:

Step 1: Add all items to find the total cost.

$$\$3.80 + \$4.50 + \$6.90 + \$4.20 = \$19.40$$

Step 2: Subtract the total from the amount paid.

$$\$20.00 - \$19.40 = \$0.60$$

Answer: Ben receives **60 cents** change.

Note: Even though cents (decimals) appear here, the same process applies to whole numbers. Always list items clearly and add step by step.

Worked Example 5 — Multiplication and Division (Tickets)

A music festival sells adult tickets for \$45 each. A total of 236 adult tickets are sold on opening day. The total revenue is to be shared equally among 4 performers. Calculate: (a) the total ticket revenue, and (b) the amount each performer receives.

Solution:

Part (a) — Total revenue:

$$236 \times \$45 = \$10{,}620$$

Check: $200 \times 45 = 9{,}000$ and $36 \times 45 = 1{,}620$, so $9{,}000 + 1{,}620 = \$10{,}620$ ✓

Part (b) — Each performer's share:

$$\$10{,}620 \div 4 = \$2{,}655$$

Answer: Each performer receives **\$2,655**.

Worked Example 6 — Estimate Then Calculate

A school tuck shop sells 387 lunch packs each week. Each lunch pack costs \$8. Estimate the weekly revenue, then use a calculator to find the exact amount.

Solution:

Step 1: Estimate — round each number to a convenient value.

$$387 \approx 400 \qquad \$8 \approx \$8 \quad \text{(already simple)}$$

$$\text{Estimate} = 400 \times 8 = \$3{,}200$$

Step 2: Calculate (using calculator).

$$387 \times 8 = \$3{,}096$$

Step 3: Check.
The exact answer ($3,096) is close to the estimate ($3,200). The difference is $104 — reasonable given the rounding. The calculation is *likely correct*.

Answer: Estimated weekly revenue = **$3,200**. Exact revenue = **$3,096**.

Tip: If your estimate and calculator answer are very far apart (e.g., thousands apart when you expected hundreds), re-enter the calculation — you may have pressed a wrong key.

Practice Exercise

Practice Exercise — Numbers in Everyday Life

Instructions: Complete all questions. Show your working. You may use a calculator where indicated.

Part A — Reading and Writing Whole Numbers

1. Write the number **4,507** in words.

2. Write the number **1,030,060** in words.

3. Write **"sixty-two thousand, three hundred and fifteen"** as a numeral.

4. Write **"nine million, four hundred thousand, and seven"** as a numeral.

Part B — Place Value

5. What is the **place value** (position name) of the digit **8** in the number 285,604?

6. What is the **value** of the digit **3** in the number 1,307,250? (e.g., write 300,000 if the 3 is in the hundred thousands place)

7. In the number **5,924,831**, write down the digit that is in the:

(a) ten thousands place _____

(b) hundreds place _____

(c) millions place _____

Part C — Ordering Numbers

8. Order these numbers from **smallest to largest**:

45,678 45,768 45,087 45,870

_____

9. Order these numbers from **largest to smallest**:

1,200,000 1,020,000 1,002,000 1,200,200

_____

Part D — Calculations

10. Calculate: $3{,}456 + 2{,}897$

_____

_____

11. Calculate: $12{,}500 - 4{,}836$

_____

_____

12. Calculate: 365×24

_____

_____

_____

13. Calculate: $2{,}808 \div 12$

_____

_____

_____

_____

Part E — Estimate Then Calculate

14. Estimate $489 + 312$ by rounding both numbers to the nearest hundred. Then find the exact answer.

My estimate: _____

Exact answer: _____

15. Estimate 67×48 by rounding both numbers to the nearest ten. Then use a calculator to find the exact answer.

My estimate: _____

Calculator answer: _____

16. Use BODMAS to evaluate: $(12 + 8) \times 5 - 14 \div 2$

Application Exercise

Application Exercise — Everyday Number Problems

Instructions: Read each problem carefully. Identify the operation(s) needed, show all working, and write a sentence answer where indicated.

1. Mia goes shopping and buys a jacket for \$89, a T-shirt for \$34 and a pair of jeans for \$67. How much does she spend in total?

Answer: Mia spends _____ in total.

2. Jake starts the month with \$245 in his bank account. He earns \$320 doing odd jobs and spends \$198 on groceries. How much does he have left?

Answer: Jake has _____ left.

3. The 2024 population of Brisbane is approximately 2,582,000. The population of Hobart is 238,000. How many more people live in Brisbane than in Hobart?

Answer: _____ more people live in Brisbane than Hobart.

4. The road distance from Sydney to Melbourne is 877 km. Sam drives from Sydney to Melbourne and then back to Sydney. What total distance does she drive?

Answer: Sam drives a total of _____.

5. A cinema sells tickets at \$18 each. If 264 people attend a Friday night session, what is the total ticket revenue?

Answer: Total revenue is _____.

6. A school gym class has 4 teams with 7 students in each team. The teacher then adds 3 more students to the class.

(a) How many students were in the class originally?

(b) How many students are in the class after the 3 are added?

7. Jordan downloads 4 apps. Three apps each use 340 MB of data, and one app uses 1,200 MB. How much total data is used? Give your answer in MB.

Answer: _____ MB of data is used.

8. An athletics club has 252 members divided equally into 9 training groups.

(a) How many members are in each group?

(b) If 3 more groups are added (with the same number of members each), how many members will the club have in total?

9. Zara earns \$875 per week and saves \$125 of it each week. After 8 weeks, how much has she saved?

Answer: Zara has saved _____ after 8 weeks.

10. A council park covers an area of 3,200 square metres. It is divided into 8 equal sections for different activities. What is the area of each section?

Answer: Each section has an area of _____ m^2.

11. The AFL grand final attracted an attendance of 100,024 people.

(a) Round this to the nearest thousand.

(b) Round this to the nearest ten thousand.

12. A mobile phone plan includes 15,000 MB of data per month. Liam uses 11,350 MB in month 1 and 2,840 MB in month 2.

(a) How much data has Liam used in total across both months?

(b) Has he stayed within his monthly limit for each month? Explain using numbers.

Chapter Review

Chapter 1 Review — Mixed Questions

Instructions: This review covers all skills from Chapter 1. Show all working. A calculator may be used for Questions 9–12.

1. Write the number **1,204,750** in words.

2. What is the **value** of the digit **6** in the number 2,637,100?

3. Order these numbers from **largest to smallest**:

303,040 330,400 300,340 303,400

4. Calculate: $5{,}874 + 3{,}649$

5. Calculate: $20{,}000 - 8{,}375$

6. Calculate: 48×125

7. Calculate: $4{,}995 \div 15$

8. Estimate $923 + 478$ by rounding both numbers to the nearest hundred. Then find the exact answer.

My estimate: **Exact answer:**

9. A family drives from Adelaide to Darwin, a distance of 3,027 km. They make the same return trip. What is the total distance they travel?

Answer: _____

10. *(Multi-step)* A cricket club orders 15 sets of cricket gear. Each set costs \$285. The club currently has \$3,500 in its account.

(a) What is the total cost of all 15 sets?

(b) How much more money does the club need?

Answer: The club needs an extra _____.

11. *(Multi-step)* Amira earns \$640 per week at a supermarket. Each week she pays \$185 rent, \$120 for food and \$75 for transport.

(a) How much does Amira have left over each week after paying her expenses?

(b) How much will she have left over across a 12-week school term?

Answer: Amira has _____ left over each week, and _____ over 12 weeks.

12. A school fundraiser sold 345 chocolate bars at \$3 each and 128 raffle tickets at \$5 each. What was the total amount raised?

Answer: Total raised = _____.

Chapter 1 Complete!

Skills checklist — tick each one when you feel confident:

- ☐ Read and write whole numbers in words
- ☐ Identify place value of any digit
- ☐ Order and compare whole numbers
- ☐ Add and subtract whole numbers
- ☐ Multiply and divide whole numbers
- ☐ Estimate by rounding
- ☐ Use a calculator and check with an estimate
- ☐ Apply number skills to real-world problems

2. Fractions, Decimals and Percentages

Chapter Information

Chapter:	2 — Fractions, Decimals and Percentages
Weeks:	Week 2
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Lesson reference:	Lessons 5–8

Fractions, decimals and percentages appear everywhere in daily life. A sale sign says "25% off". A nutrition label shows that a biscuit contains $\frac{3}{10}$ of your daily sugar. Your phone warns you that battery is at 0.15 (15%). These are all different ways of expressing the same idea — a part of a whole. In this chapter, you will move fluently between fractions, decimals and percentages, and use them to solve real problems like finding sale prices and calculating GST.

Learning Objectives

By the end of this chapter you will be able to:

- Recognise and use common fractions including proper fractions
- Find equivalent fractions and simplify fractions to their simplest form
- Read, write and compare decimals to two decimal places
- Convert between fractions, decimals and percentages
- Calculate a percentage of an amount
- Find discounted prices and calculate GST (10%)

Key Terms

- **Numerator:** The top number of a fraction — tells you how many parts you have
- **Denominator:** The bottom number of a fraction — tells you how many equal parts the whole is divided into
- **Proper fraction:** A fraction where the numerator is smaller than the denominator (e.g., $\frac{3}{4}$)
- **Equivalent fractions:** Fractions that have the same value but different numerators and denominators (e.g., $\frac{1}{2} = \frac{4}{8}$)
- **Simplest form:** A fraction where the numerator and denominator share no common factor other than 1

- **Decimal:** A number that uses a point to separate whole number parts from parts less than one (e.g., 0.75)
- **Decimal place:** The position of a digit after the decimal point (tenths, hundredths, thousandths)
- **Percentage:** A number expressed as a fraction of 100, written with a % symbol
- **Discount:** An amount taken off the original price of an item
- **Sale price:** The price you pay after a discount has been applied
- **GST (Goods and Services Tax):** A 10% tax added to the price of most goods and services in Australia

My Conversion Reference Table — Complete the missing values

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{4}$		
$\frac{3}{4}$		
$\frac{1}{10}$		
$\frac{1}{5}$		
$\frac{2}{5}$		
$\frac{1}{3}$		
	0.75	
		80%
		35%

Key: Given value | Blank — write your answer here

Tip: Keep this table handy — you can bring it into your open-book examination.

Key Formulas and Rules

- **Fraction to decimal:** Divide the numerator by the denominator.

$$\frac{3}{4} = 3 \div 4 = 0.75$$

- **Decimal to percentage:** Multiply by 100.

$$0.75 \times 100 = 75\%$$

- **Percentage to decimal:** Divide by 100.

$$75\% \div 100 = 0.75$$

- **Percentage of an amount:**

$$\text{Percentage of amount} = \text{Amount} \times \frac{\text{Percentage}}{100}$$

- **Discount and sale price:**

$$\text{Discount} = \text{Original price} \times \frac{\text{Discount }\%}{100} \qquad \text{Sale price} = \text{Original price} - \text{Discount}$$

Quick method: $\text{Sale price} = \text{Original price} \times \left(1 - \frac{\text{Discount }\%}{100}\right)$

- **GST (10%):**

$$\text{GST amount} = \text{Pre-GST price} \times 0.10 \qquad \text{Total price} = \text{Pre-GST price} \times 1.10$$

- **Common equivalents to memorise:**

$50\% = 0.5 = \frac{1}{2}$ $\quad 25\% = 0.25 = \frac{1}{4}$ $\quad 75\% = 0.75 = \frac{3}{4}$

$10\% = 0.1 = \frac{1}{10}$ $\quad 20\% = 0.2 = \frac{1}{5}$ $\quad 1\% = 0.01 = \frac{1}{100}$

Worked Examples

Worked Example 1 — Simplifying a Fraction

Simplify $\frac{12}{16}$ to its simplest form.

Solution:

Step 1: Find the Highest Common Factor (HCF) of 12 and 16.

Factors of 12: 1, 2, 3, **4**, 6, 12
Factors of 16: 1, 2, **4**, 8, 16

HCF = **4**

Step 2: Divide both numerator and denominator by the HCF.

$$\frac{12}{16} = \frac{12 \div 4}{16 \div 4} = \frac{3}{4}$$

Answer: $\frac{12}{16} = \frac{3}{4}$ (simplest form, since 3 and 4 share no common factor other than 1)

Check: $\frac{3}{4}$ cannot be simplified further because the only common factor of 3 and 4 is 1.

Worked Example 2 — Equivalent Fractions

(a) Find three fractions equivalent to $\frac{2}{3}$. **(b) Find the missing value: $\frac{2}{3} = \frac{?}{12}$**

Solution:

Part (a): Multiply both numerator and denominator by the same number.

$$\frac{2}{3} = \frac{2 \times 2}{3 \times 2} = \frac{4}{6} \qquad \frac{2}{3} = \frac{2 \times 3}{3 \times 3} = \frac{6}{9} \qquad \frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$$

Part (b): The denominator has been multiplied by 4 (since $3 \times 4 = 12$).
Multiply the numerator by the same number: $2 \times 4 = \mathbf{8}$.

$$\frac{2}{3} = \frac{8}{12}$$

Worked Example 3 — Converting Fraction to Decimal and Percentage

Convert $\frac{3}{4}$ to a decimal and a percentage.

Solution:

Step 1: Fraction to decimal.
Divide the numerator by the denominator.

$$\frac{3}{4} = 3 \div 4 = \mathbf{0.75}$$

Step 2: Decimal to percentage.
Multiply by 100.

$$0.75 \times 100 = \mathbf{75}\%$$

Answer: $\frac{3}{4} = 0.75 = 75\%$

Worked Example 4 — Converting Percentage to Decimal and Fraction

Convert 35% to a decimal and to a fraction in simplest form.

Solution:

Step 1: Percentage to decimal.
Divide by 100.

$$35\% \div 100 = \mathbf{0.35}$$

Step 2: Percentage to fraction.
Write the percentage over 100, then simplify.

$$35\% = \frac{35}{100}$$

HCF of 35 and 100 is 5.

$$\frac{35}{100} = \frac{35 \div 5}{100 \div 5} = \frac{7}{20}$$

Answer: $35\% = 0.35 = \frac{7}{20}$

Worked Example 5 — Calculating a Percentage of an Amount

Calculate 20% of $350.

Solution:

Method 1 — Multiply by the decimal:

$$20\% \text{ of } \$350 = 0.20 \times 350 = \$\mathbf{70}$$

Method 2 — Build from 10%:
10% of $350 = $35
20% = 2 × 10% = 2 × $35 = **$70**

Answer: 20% of $350 = **$70**

Both methods give the same answer. Method 2 is a useful mental maths strategy for common percentages (10%, 20%, 5%, 25%).

Worked Example 6 — Finding a Discounted Price and GST

(a) A pair of shoes originally costs $120. They are on sale at **30% off**. What is the sale price?
(b) A repair kit costs $85 before GST. How much GST is added, and what is the total price including GST?

Solution:

Part (a) — Discount:

Step 1: Calculate the discount amount.

$$\text{Discount} = 30\% \times \$120 = 0.30 \times 120 = \$36$$

Step 2: Subtract from original price.

$$\text{Sale price} = \$120 - \$36 = \$\mathbf{84}$$

Quick method: Sale price = 70% *of $120* = 0.70 × 120 = $84 ✓

Part (b) — GST:

Step 1: Calculate GST (10%).

$$\text{GST} = 10\% \times \$85 = 0.10 \times 85 = \$8.50$$

Step 2: Add GST to original price.

$$\text{Total} = \$85 + \$8.50 = \$\mathbf{93.50}$$

Quick method: Total $= 110\%$ *of \$85* $= 1.10 \times 85 = \$93.50$ ✓

Practice Exercise

Practice Exercise — Fractions, Decimals and Percentages

Instructions: Show all working. You may use a calculator where indicated.

Part A — Simplifying Fractions

1. Simplify $\frac{6}{9}$ to its simplest form. Show the HCF.

2. Simplify $\frac{15}{25}$ to its simplest form.

3. Simplify $\frac{18}{24}$ to its simplest form.

Part B — Equivalent Fractions

4. Find the missing numerator: $\frac{3}{5} = \frac{?}{20}$

5. Find the missing denominator: $\frac{2}{7} = \frac{6}{?}$

6. Are $\frac{4}{10}$ and $\frac{6}{15}$ equivalent? Show your working.

Part C — Ordering Fractions and Decimals

7. Order these values from **smallest to largest** by converting all to decimals first:

$$\frac{3}{4} \qquad \frac{1}{2} \qquad \frac{2}{3} \qquad \frac{1}{4}$$

Part D — Conversions

8. Complete the conversion table below.

Fraction	Decimal	Percentage
$\frac{1}{2}$	_____	_____
$\frac{3}{5}$	_____	_____
_____	0.25	_____
_____	_____	40%
_____	0.08	_____

Part E — Calculating Percentages

9. Calculate 15% of $200.

10. Calculate 25% of \$80.

11. Calculate 40% of 350 items.

12. Convert 0.6 to a fraction (simplest form) and a percentage.

13. Convert 85% to a decimal and a fraction in simplest form.

14. A class has 28 students. $\frac{1}{4}$ of the class is absent today. How many students are absent?

15. Which is greater: 65% or $\dfrac{2}{3}$? Show your working.

Application Exercise

Application Exercise — Real-World Fractions, Decimals and Percentages

Instructions: Read each problem carefully. Show all working and write a sentence answer.

1. Woolworths has 25% off a jacket originally priced at \$48. What is the sale price?

Answer: The sale price is _____.

2. A water bottle holds 750 mL. Tom has drunk $\frac{3}{5}$ of it. How many millilitres has he drunk? How much is left?

Answer: Tom has drunk _____ and has _____ left.

3. Layla scored 34 out of 40 on a maths test. What percentage did she score?

Answer: Layla scored _____%.

4. A phone plan includes 10 GB of data. Amara has used 6.5 GB.

(a) What fraction of her data has she used? Give your answer in simplest form. (Hint: 1 GB = 1, so 6.5 GB out of 10 GB = $\frac{6.5}{10} = \frac{13}{20}$)

(b) What percentage of her data has she used?

(c) What percentage remains?

5. A laptop costs $1,299. A store is selling it for 20% off. What is the sale price?

Answer: The sale price is _____.

6. A nutrition label shows that one serve of cereal contains 12 g of fat. The daily recommended intake of fat is 65 g. What percentage of the daily recommended fat is in one serve? Round to one decimal place.

Answer: One serve provides _____% of the recommended daily fat.

7. A plumber charges $340 for a job (before GST). GST of 10% is added.

(a) How much is the GST?

(b) What is the total price the customer pays?

8. Milo scored 45 out of 50 on a science test and 38 out of 40 on a maths test. Which score is the higher percentage? Show all working.

Answer: The higher score is _____.

9. At a school fete, $\frac{3}{8}$ of the students bought a sausage sizzle. There are 240 students at the school.

(a) How many students bought a sausage sizzle?

(b) What percentage of students did NOT buy a sausage sizzle?

10. A cinema ticket costs \$22. This price already **includes** GST of 10%.

(a) What was the price of the ticket before GST? (Hint: \$22 = 110% of the pre-GST price)

(b) How much of the \$22 is the GST component?

11. A clothing store has **everything 30% off**. Chloe wants to buy a dress for \$85 and shoes for \$120.

(a) What is the discount on the dress?

(b) What is the discount on the shoes?

(c) What is the total cost of both items after the discount?

Answer: Total cost = _____.

12. At a school sports day, $\frac{2}{5}$ of students chose swimming and 35% chose athletics. The rest chose cricket.

(a) What percentage chose swimming?

(b) What percentage chose cricket?

(c) Write the cricket percentage as a fraction in simplest form.

Chapter Review

Chapter 2 Review — Mixed Questions

Instructions: This review covers all skills from Chapter 2. Show all working. A calculator may be used for Questions 9–12.

1. Simplify $\frac{24}{36}$ to its simplest form.

2. Write $\frac{3}{8}$ as a decimal and as a percentage.

3. Write 45% as a decimal and as a fraction in simplest form.

4. Find the missing value: $\frac{5}{8} = \frac{?}{40}$

5. Order these values from **smallest to largest**. Show your conversions.

$$0.4 \qquad \frac{3}{8} \qquad 42\% \qquad \frac{1}{3}$$

6. Calculate 35% of $260.

7. Luca scored 27 out of 30 on a quiz. What percentage did he score?

8. Calculate the GST (10%) on a $340 item. What is the total price including GST?

9. A mobile phone plan is $65 per month. The provider offers a 15% loyalty discount. What is the new monthly cost?

Answer: New monthly cost = _____.

10. *(Multi-step)* A jacket is marked at $180. It is on sale for **25% off**. You must also pay 10% GST on the sale price.

(a) What is the sale price before GST?

(b) How much GST is added to the sale price?

(c) What is the final price you pay?

Answer: Final price = _____.

11. *(Multi-step)* A bottle of juice contains 1.25 L (1,250 mL). Tom drinks $\frac{2}{5}$ of the bottle in the morning and 30% of the *original* bottle in the afternoon.

(a) How many mL does Tom drink in the morning?

(b) How many mL does Tom drink in the afternoon?

(c) How many mL of juice is left?

Answer: There is _____ mL of juice left.

12. A store advertises a television for \$799. A different store sells the same model for \$849 but offers a 10% discount. Which store gives the better price? Show all working to justify your answer.

Answer: _____ gives the better price.

Chapter 2 Complete!

Skills checklist — tick each one when you feel confident:

- ☐ Simplify fractions to simplest form
- ☐ Find equivalent fractions
- ☐ Order fractions and decimals
- ☐ Convert fraction ↔ decimal ↔ percentage
- ☐ Calculate a percentage of an amount
- ☐ Find a discounted price
- ☐ Calculate GST (10%)
- ☐ Solve multi-step percentage problems

3. Rates, Ratios and Comparisons

Chapter Information

Chapter:	3 — Rates, Ratios and Comparisons
Weeks:	Week 3
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Lesson reference:	Lessons 9–12

Which supermarket deal is better value — 400 g of chips for \$3.60 or 700 g for \$5.95? How long will a 450 km road trip take at 100 km/h? If a punch recipe for 4 people needs 3 cups of juice and 5 cups of lemonade, how much do you need for a party of 24? All of these questions involve comparing quantities. In this chapter, you will use rates to compare values, ratios to describe and scale mixtures or groups, and bring together everything from Chapters 1 and 2 to solve multi-step problems.

Learning Objectives

By the end of this chapter you will be able to:

- Understand what a rate is and calculate rates from given quantities
- Use unit rates to compare the value of different-sized products (best buy)
- Write ratios from descriptions and simplify ratios using the HCF
- Scale ratios up and down to solve practical problems
- Share amounts in a given ratio
- Choose the right tool — rate, ratio or percentage — to solve a problem

Key Terms

- **Rate:** A comparison of two quantities with *different* units, written as one quantity per unit of another (e.g., \$4.50/kg, 80 km/h, 12 L/100 km)
- **Unit rate:** A rate where the second quantity is 1 (e.g., \$1.80 *per* kilogram)
- **Unit price:** The cost per single unit of measurement — used to find the best buy
- **Per:** Means "for each" or "for every one" — the key word in any rate
- **Ratio:** A comparison of two (or more) quantities of the *same* type using the colon symbol or the word "to" (e.g., 3:2)
- **Equivalent ratio:** Two ratios that represent the same comparison (e.g., 2:4 and 1:2)

- **Simplest form (ratio):** A ratio where both parts share no common factor other than 1
- **Scale:** Multiply or divide both parts of a ratio by the same number to make it bigger or smaller
- **Proportion:** A statement that two ratios are equal
- **Best buy:** The product option that gives the lowest cost per unit of measurement

Key Formulas and Rules

- **Rate:**

$$\text{Rate} = \frac{\text{Quantity}}{\text{Unit}} \qquad \text{e.g., Speed} = \frac{\text{Distance}}{\text{Time}}$$

- **Speed, Distance, Time triangle:**

Distance (D)	
Speed (S)	Time (T)

$$D = S \times T \qquad S = \frac{D}{T} \qquad T = \frac{D}{S}$$

- **Unit price:**

$$\text{Unit price} = \frac{\text{Total cost}}{\text{Quantity}}$$

- **Simplifying a ratio:** Divide both parts by their Highest Common Factor (HCF). e.g., $12 : 8 \xrightarrow{\div 4} 3 : 2$
- **Scaling a ratio:** Multiply *or* divide both parts by the same number. e.g., $3 : 5 \xrightarrow{\times 4} 12 : 20$
- **Sharing in a ratio:**
  1. Find the total number of parts.
  2. Divide the total amount by the total parts (= value of 1 part).
  3. Multiply by each share.

Choosing the Right Tool — Rates, Ratios or Percentages?

Situation	Use this tool
Comparing different-sized packages to find best value	**Unit rate / unit price**
Comparing speeds, flows or consumption	**Rate** (km/h, L/100 km, etc.)
Describing a mixture (paint, concrete, recipe)	**Ratio**
Splitting a total among people or groups	**Ratio** (share)
Finding a portion of a whole	**Fraction or percentage**
Calculating a discount, tax, or increase	**Percentage**

Worked Examples

Worked Example 1 — Calculating a Rate (Speed)

A car travels 240 km in 3 hours. Calculate its average speed in km/h.

Solution:

Step 1: Identify the formula.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Step 2: Substitute and calculate.

$$S = \frac{240}{3} = \mathbf{80}\ \text{km/h}$$

Answer: The car's average speed is **80 km/h**.

Extension: How long would it take to travel 340 km at 80 km/h?
$T = D \div S = 340 \div 80 = 4.25$ hours $= 4$ hours and 15 minutes

Worked Example 2 — Unit Price and Best Buy

A supermarket sells yoghurt in two sizes: 600 g for \$4.80 and 1 kg for \$7.60. Which is the better buy?

Solution:

Step 1: Calculate the unit price (cost per 100 g) for each option.

$$600\,\text{g option:} \quad \frac{\$4.80}{600} \times 100 = \$0.80 \text{ per } 100\,\text{g}$$

$$1\,\text{kg option:} \quad \frac{\$7.60}{1000} \times 100 = \$0.76 \text{ per } 100\,\text{g}$$

Step 2: Compare. $\$0.76 < \$0.80$
Answer: The **1 kg tub** is the better buy at \$0.76 per 100 g (4 cents cheaper per 100 g).

Tip: Always compare the same unit. Here we used "per 100 g" — you could also use "per gram" or "per kg", as long as both options are expressed the same way.

Worked Example 3 — Writing and Simplifying a Ratio

A youth group has 18 girls and 12 boys. Write the ratio of girls to boys and simplify it.

Solution:

Step 1: Write the ratio in the order stated.

$$\text{Girls} : \text{Boys} = 18 : 12$$

Step 2: Find the HCF of 18 and 12.
Factors of 18: 1, 2, 3, **6**, 9, 18
Factors of 12: 1, 2, 3, **6**, 12 $\Rightarrow$ HCF = 6

Step 3: Divide both parts by 6.

$$18 : 12 = \frac{18}{6} : \frac{12}{6} = \mathbf{3 : 2}$$

Answer: The ratio of girls to boys is **3:2**.

Check: $3 + 2 = 5$ *parts total. Girls* $= \frac{3}{5}$ *of the group* $= \frac{3}{5} \times 30 = 18$ ✓

Worked Example 4 — Scaling a Ratio (Recipe)

A smoothie recipe for 4 people uses 3 cups of frozen mango, 2 cups of yoghurt and 1 cup of orange juice. How much of each ingredient is needed for 20 people?

Solution:

Step 1: Find the scale factor.

$$\text{Scale factor} = \frac{20}{4} = 5$$

The recipe needs to be multiplied by 5.

Step 2: Multiply each ingredient by 5.

Ingredient	For 4 people	For 20 people
Frozen mango	3 cups	$3 \times 5 = \mathbf{15}$ cups
Yoghurt	2 cups	$2 \times 5 = \mathbf{10}$ cups
Orange juice	1 cup	$1 \times 5 = \mathbf{5}$ cups

Answer: For 20 people: 15 cups mango, 10 cups yoghurt, 5 cups orange juice.

Worked Example 5 — Sharing in a Ratio

Amira and Jordan win a prize of \$450 and agree to share it in the ratio 2:3. How much does each person receive?

Solution:

Step 1: Find the total number of parts.

$$2 + 3 = 5 \text{ parts}$$

Step 2: Find the value of 1 part.

$$\$450 \div 5 = \$90 \text{ per part}$$

Step 3: Multiply by each person's share.

$$\text{Amira (2 parts):} \quad 2 \times \$90 = \$\mathbf{180}$$

$$\text{Jordan (3 parts):} \quad 3 \times \$90 = \$\mathbf{270}$$

Answer: Amira receives **\$180** and Jordan receives **\$270**.
Check: $\$180 + \$270 = \$450$ ✓

Practice Exercise

Practice Exercise — Rates, Ratios and Comparisons

Instructions: Show all working. You may use a calculator where indicated.

Part A — Rates

1. A car travels 300 km in 4 hours. What is its average speed in km/h?

2. Emma reads 45 pages in 3 hours. What is her reading rate in pages per hour?

3. A tap fills a 180 L tank in 4 hours. How many litres does it fill per hour?

4. A cyclist rides at 24 km/h. How far does she travel in 2.5 hours?

Part B — Unit Price and Best Buy

5. A 500 g tub of yoghurt costs \$3.50. What is the price per 100 g?

6. Which is better value? 2 L of orange juice for \$4.40 or 1.5 L for \$3.45? Show your unit price calculations.

Answer: The better buy is _____.

Part C — Writing and Simplifying Ratios

7. A bag contains 5 red marbles and 3 blue marbles. Write the ratio of red to blue.

8. In a class of 25 students, 10 are absent today. Write and simplify the ratio of absent to present students.

9. Simplify each ratio.

(a) 15 : 25 _____

(b) 24 : 36 _____

(c) 9 : 12 : 15 _____

Part D — Scaling Ratios

10. A ratio is 3 : 4. Scale it up so that the first part equals 12.

11. A recipe uses flour and sugar in the ratio 3 : 1. If you use 12 cups of flour, how many cups of sugar do you need?

12. Cordial is mixed with water in the ratio 1 : 4. To make 2.5 L of drink, how much cordial and how much water is needed?

Part E — Conversion Review

13. Write 0.35 as a fraction in simplest form and as a percentage.

14. Write $\frac{3}{8}$ as a decimal and as a percentage.

Application Exercise

Application Exercise — Real-World Rates, Ratios and Comparisons

Instructions: Read each problem carefully. Identify the correct tool (rate, ratio, or percentage), show all working, and write a sentence answer.

1. A supermarket sells chips in two sizes: 400 g for \$3.60 or 700 g for \$5.95. Which bag is better value per 100 g? Show your calculations.

Answer: The better buy is the _____ bag.

2. A cyclist completes a 45 km track in 1.5 hours. What is her average speed in km/h?

Answer: Her average speed is _____ km/h.

3. A family is driving from Brisbane to Cairns, a distance of 1,700 km, at an average speed of 100 km/h.

(a) How many hours will the trip take?

(b) If they stop for a 30-minute lunch break and two 15-minute rest stops, what is the total travel time from start to finish?

Answer: Total time = _____.

4. A pancake recipe for 4 people requires 2 cups of flour, 1 cup of milk and 3 eggs. Scale the recipe to serve 12 people.

Answer (for 12 people): flour _____, milk _____, eggs _____.

5. White and blue paint are mixed in the ratio 5 : 2. A painter uses 10 L of white paint.

(a) How many litres of blue paint are needed?

(b) What is the total volume of paint mixed?

6. Jess and Sam share a prize of \$600 in the ratio 3 : 2. How much does each person receive?

Answer: Jess receives _____ and Sam receives _____.

7. Concrete is made from cement, sand and gravel in the ratio 1 : 3 : 5. A builder needs 18 kg of sand.

(a) How many kilograms of cement are needed?

(b) How many kilograms of gravel are needed?

(c) What is the total mass of concrete produced?

_____

8. Two mobile data plans are available:

- Plan A: 15 GB for \$45
- Plan B: 20 GB for \$65

Which plan gives more data per dollar? Show your working.

_____

_____

_____

_____

Answer: _____ gives more data per dollar.

9. A prize of \$1,200 is shared among three friends in the ratio 2 : 3 : 4. How much does each person receive?

_____

_____

_____

_____

_____

Answer: Person 1: _____, Person 2: _____, Person 3: _____.

10. Orange juice is sold in two sizes at a school canteen: 250 mL for \$1.80 or 600 mL for \$4.20.

(a) Calculate the unit price (per 100 mL) for each size.

_____

_____

_____

(b) Which is the better buy?

_____

(c) If Liam buys 3 of the smaller bottles instead of 1 of the larger, how much more does he spend?

11. A fruit punch is made using apple juice and ginger ale in the ratio $3:5$. Altogether, 4 L of punch is to be made.

(a) How many litres of apple juice are needed?

(b) How many litres of ginger ale are needed?

(c) What fraction of the punch is apple juice? Write it in simplest form.

12. Queensland has approximately 5 million people and New South Wales has approximately 8 million people.

(a) Write the ratio of Queensland's population to NSW's population in simplest form.

(b) What fraction of the *combined* population lives in Queensland? Write as a decimal rounded to 2 decimal places.

(c) What percentage of the combined population lives in NSW? Round to 1 decimal place.

Chapter Review

Chapter 3 Review — Topic 1 Number & Algebra Summary (Chapters 1–3)

Instructions: This review covers rates, ratios and comparisons, and brings together all Number & Algebra skills from Topic 1. Show all working. A calculator may be used for Questions 9–14.

1. A car travels 360 km in 4.5 hours. Calculate its average speed.

2. A 750 mL bottle of water costs \$2.85. What is the price per 100 mL?

3. Which is better value: 1 kg of coffee for \$6.20 or 1.5 kg for \$8.70? Show your unit price working.

Answer: _____ is better value.

4. A class has 18 adults and 12 children. Write the ratio of adults to children in simplest form.

5. Find the missing value: $4 : 7 = ? \ : 35$

6. Write 72% as a decimal and as a fraction in simplest form.

7. Write $\frac{5}{6}$ as a decimal (use your calculator; round to 3 decimal places) and as a percentage.

8. In a ratio $2:5$, the total amount is 70. Find the value of each part.

9. A car uses 8 L of fuel per 100 km. How much fuel is needed for a 450 km trip?

Answer: _____ litres of fuel needed.

10. A recipe for 6 people needs: 3 cups of rice, 2 cups of beans and 1 cup of spice mix. Scale the recipe to serve 15 people.

Answer (for 15): rice _____, beans _____, spice _____.

11. *(Multi-step)* Tom and Lucy share a cleaning job. Tom works 3 hours and Lucy works 5 hours. They are paid a total of \$192 split in the same ratio as the hours they worked. How much does each person earn?

Answer: Tom earns _____, Lucy earns _____.

12. *(Multi-step)* Sam earns \$720 per week. He spends $\frac{1}{3}$ on rent and 25% on food, and saves the rest.

(a) How much does Sam spend on rent each week?

(b) How much does Sam spend on food each week?

(c) How much does Sam save each week?

(d) What fraction of his income does Sam save? Simplify your answer.

13. Two internet plans are compared:

- Plan X: 50 GB for \$60
- Plan Y: 80 GB for \$95

Which plan gives more data per dollar? Show full working.

Answer: _____ gives more data per dollar.

14. *(Multi-step)* A punch recipe uses lemonade and juice in the ratio $3 : 5$. Altogether, 6.4 L of punch is needed for a party.

(a) How many litres of lemonade are needed?

(b) How many litres of juice are needed?

(c) Juice costs \$2.50 per litre and lemonade costs \$1.80 per litre. What is the total cost of ingredients?

Answer: Total cost = _____.

Chapter 3 Complete!

Skills checklist — tick each one when you feel confident:

- ☐ Calculate a rate from given quantities
- ☐ Use Speed = Distance ÷ Time
- ☐ Find unit prices to compare value
- ☐ Identify the best buy
- ☐ Write a ratio from a description
- ☐ Simplify a ratio using HCF
- ☐ Scale a ratio up or down
- ☐ Share an amount in a given ratio
- ☐ Choose rate, ratio, or percentage correctly
- ☐ Solve multi-step problems combining all skills

4. Length, Perimeter and Area

Chapter Information

Chapter:	4 — Length, Perimeter and Area
Weeks:	Week 4
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Lesson reference:	Lessons 13–16

Measurement appears in almost every home and community task. How much fencing does a garden need? How many square metres of carpet will cover a bedroom floor? How much paint is required to cover a wall? These questions all require calculating length, perimeter or area. In this chapter you will work with metric units, calculate the perimeter and area of rectangles, squares and triangles, and estimate the area of irregular shapes using a grid.

Learning Objectives

By the end of this chapter you will be able to:

- Identify and use metric units for length: millimetres (mm), centimetres (cm), metres (m) and kilometres (km)
- Convert between metric length units in both directions
- Calculate the perimeter of rectangles, squares and triangles
- Calculate the area of rectangles and squares
- Calculate the area of triangles
- Estimate the area of irregular shapes by counting squares on a grid

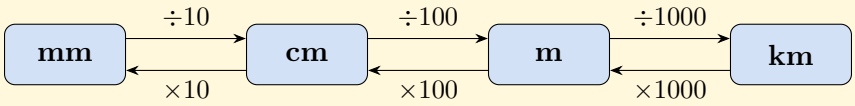
Key Terms

- **Millimetre (mm):** The smallest common metric unit of length; 10 mm = 1 cm
- **Centimetre (cm):** Used for small everyday measurements; 100 cm = 1 m
- **Metre (m):** The base unit of length; used for room sizes, fabric, sporting distances
- **Kilometre (km):** Used for large distances; 1 km = 1,000 m
- **Perimeter:** The total distance around the outside of a shape
- **Area:** The amount of flat surface inside a shape, measured in square units (cm^2, m^2)
- **Length (L):** The longer side of a rectangle

- **Width (W):** The shorter side of a rectangle (also called breadth)
- **Base (B):** The bottom side of a triangle
- **Height (H):** The perpendicular distance from the base to the opposite vertex of a triangle (must be at right angles to the base)
- **Square units:** Units used to measure area — cm^2 (square centimetres), m^2 (square metres)

Key Formulas and Conversions

Length unit conversions:



Perimeter formulas:

$$P_{\text{rectangle}} = 2L + 2W \qquad P_{\text{square}} = 4S \qquad P_{\text{triangle}} = a + b + c$$

Area formulas:

$$A_{\text{rectangle}} = L \times W \qquad A_{\text{square}} = S^2 \qquad A_{\text{triangle}} = \frac{B \times H}{2}$$

*Remember: Area is always measured in **square units** (cm^2, m^2, etc.). Perimeter is measured in **linear units** (cm, m, etc.).*

Worked Examples

Worked Example 1 — Converting Length Units

(a) Convert 3.5 km to metres. (b) Convert 420 cm to metres. (c) Convert 85 mm to cm.

Solution:

(a) Moving from km → m: *multiply* by 1,000.

$$3.5 \text{ km} \times 1{,}000 = \mathbf{3{,}500} \text{ m}$$

(b) Moving from cm → m: *divide* by 100.

$$420 \text{ cm} \div 100 = \mathbf{4.2} \text{ m}$$

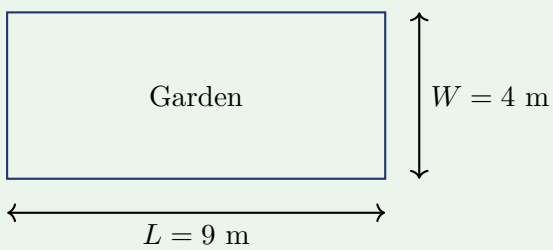
(c) Moving from mm → cm: *divide* by 10.

$$85 \text{ mm} \div 10 = \mathbf{8.5} \text{ cm}$$

*Memory tip: Moving to a **larger** unit → divide. Moving to a **smaller** unit → multiply.*

Worked Example 2 — Perimeter of a Rectangle

A rectangular garden measures 9 m long and 4 m wide. How much fencing is needed to go all the way around it?



Solution:

Step 1: Write the formula.

$$P = 2L + 2W$$

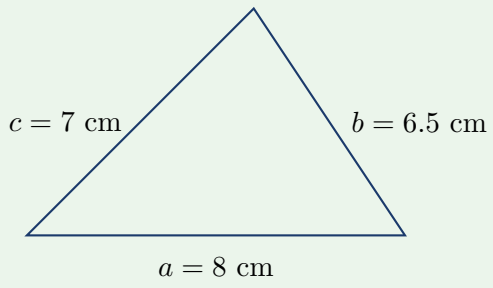
Step 2: Substitute $L = 9$ and $W = 4$.

$$P = 2(9) + 2(4) = 18 + 8 = \mathbf{26} \text{ m}$$

Answer: 26 metres of fencing is needed.

Worked Example 3 — Perimeter of a Triangle

Find the perimeter of the triangle below.



Solution:

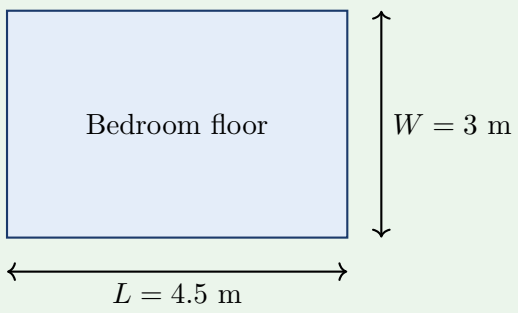
Step 1: Use the formula $P = a + b + c$.

$$P = 8 + 6.5 + 7 = \mathbf{21.5} \text{ cm}$$

Answer: The perimeter of the triangle is **21.5 cm**.

Worked Example 4 — Area of a Rectangle

A bedroom measures 4.5 m by 3 m. What area of carpet is needed to cover the floor?



Solution:

Step 1: Use the formula $A = L \times W$.

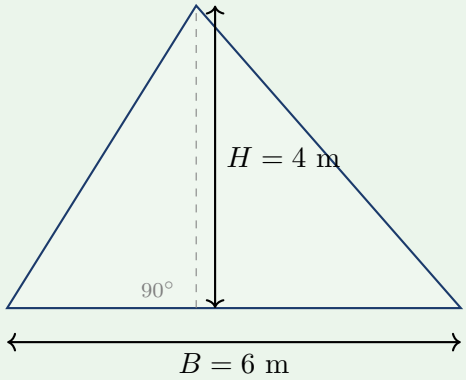
$$A = 4.5 \times 3 = \mathbf{13.5} \text{ m}^{\mathbf{2}}$$

Answer: 13.5 m^2 of carpet is needed.

Note: If the carpet costs \$42 per m^2, the total cost $= 13.5 \times \$42 = \$567$.

Worked Example 5 — Area of a Triangle

A triangular garden bed has a base of 6 m and a perpendicular height of 4 m. Find its area.



Solution:

Step 1: Use the formula $A = \dfrac{B \times H}{2}$.

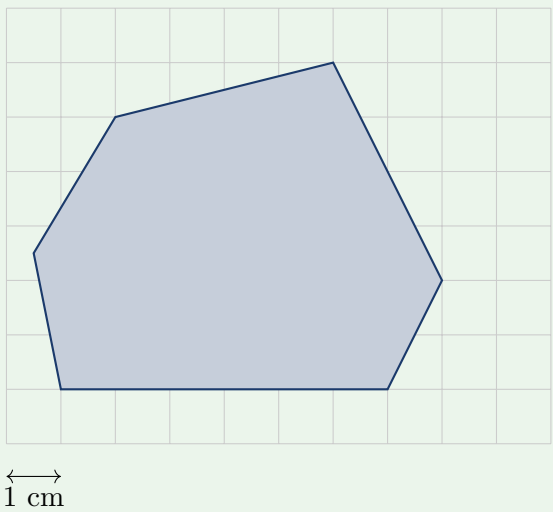
$$A = \frac{6 \times 4}{2} = \frac{24}{2} = \mathbf{12\ m^2}$$

Answer: The area of the triangular garden bed is **12 m**^{2}.

*Important: The height (H) must be the **perpendicular** height — the dashed line at right angles to the base, not the slanted side.*

Worked Example 6 — Area by Counting Squares

Estimate the area of the shaded shape below. Each small square represents 1 cm^2.



Solution:

Step 1: Count the squares that are *completely inside* the shape.
Whole squares $\approx$ **26**

Step 2: Count the squares that are *partially inside* the shape and estimate each as $\frac{1}{2}$ a square.
Partial squares $\approx 10 \rightarrow 10 \times \frac{1}{2} = \mathbf{5}$

Step 3: Add them together.

$$\text{Estimated area} \approx 26 + 5 = \mathbf{31}\ \text{cm}^{\mathbf{2}}$$

Answer: The area of the shape is approximately **31 cm**^{2}.

Note: Answers within ± 3 cm^2 are acceptable for counting square estimates.

Practice Exercise

Practice Exercise — Length, Perimeter and Area

Instructions: Show all working. Include units in every answer.

Part A — Length Conversions

1. Convert each measurement to the unit shown.

(a) 3.5 km = _____ m

(b) 450 cm = _____ m

(c) 82 mm = _____ cm

(d) 2.4 m = _____ cm

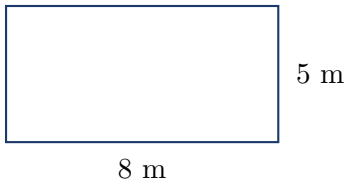
(e) 0.6 km = _____ m

(f) 1,250 m = _____ km

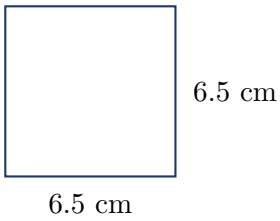
Part B — Perimeter

2. Calculate the perimeter of each shape.

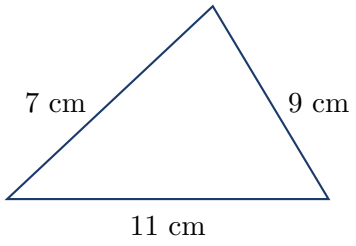
(a) A rectangle with $L = 8$ m and $W = 5$ m.



(b) A square with side length 6.5 cm.



(c) A triangle with sides 7 cm, 9 cm and 11 cm.

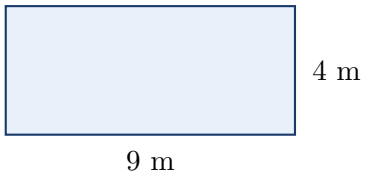


(d) A rectangle with $L = 12.5$ m and $W = 4$ m.

Part C — Area

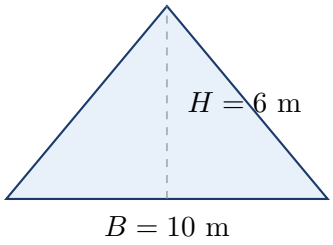
3. Calculate the area of each shape. Remember to include units^2.

(a) A rectangle: $L = 9$ m, $W = 4$ m.



(b) A square with side 7 cm.

(c) A triangle: $B = 10$ m, $H = 6$ m.

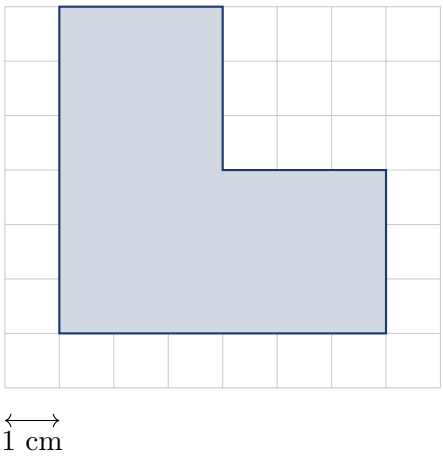


(d) A triangle: $B = 8.4$ cm, $H = 5$ cm.

Part D — Counting Squares

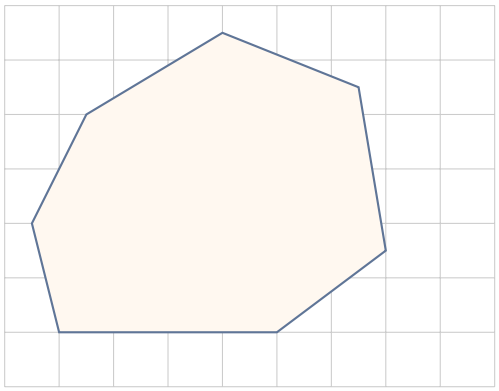
4. Find the area of each shaded shape by counting squares. Each square = 1 cm^2.

(a) Count the squares inside the L-shaped region.



Whole squares inside: _____ Area = _____ cm^2

(b) Estimate the area of the irregular shape. Count whole squares and half-squares separately.



$\overleftrightarrow{1\text{ cm}}$

Whole squares: _____ Partial squares: _____ $\times \frac{1}{2} =$ _____

Estimated area $\approx$ _____ cm^2

Application Exercise

Application Exercise — Real-World Length, Perimeter and Area

Instructions: Draw a diagram where helpful, show all working, and write a sentence answer. Include units in every answer.

1. A rectangular garden bed is 8 m long and 5 m wide. A gardener wants to build a timber border around the entire edge.

(a) What length of timber is needed?

(b) Timber costs \$12 per metre. What is the total cost?

Answer: Total cost = _____.

2. A bedroom measures 4.5 m by 3.8 m. Carpet costs \$45 per m^2.

(a) What is the area of the bedroom floor?

(b) What is the total cost of the carpet?

Answer: Cost = _____.

3. A triangular garden bed has a base of 7 m and a perpendicular height of 5 m.

(a) Find the area of the garden bed.

(b) Mulch costs \$8 per m^2. How much will it cost to cover the garden bed?

Answer: Cost = _____.

4. A school oval is rectangular, measuring 110 m long and 68 m wide.

(a) What is the perimeter of the oval?

(b) What is the area of the oval?

5. A rectangular wall is 4 m wide and 2.5 m high. It has one window measuring 1.2 m by 0.9 m. A painter needs to paint the wall but not the window.

(a) What is the total area of the wall?

(b) What is the area of the window?

(c) What area needs to be painted?

Answer: Area to be painted = _____.

6. A bathroom floor is 3 m by 3 m. Square tiles measure 20 cm × 20 cm each and cost \$3.50 each.

(a) How many tiles fit along one side of the room? (Convert 3 m to cm first.)

(b) How many tiles are needed in total?

(c) What is the total cost of the tiles?

Answer: _____ tiles needed; total cost = _____.

7. A fence is needed around a rectangular yard 15 m by 10 m. One 15 m side backs onto a wall and does not need fencing. A gate 1.2 m wide is included in one of the 10 m sides.

(a) How many metres of fencing (not including the gate) are needed?

(b) Fencing costs \$18 per metre and the gate costs \$320. What is the total cost?

Answer: Total cost = _____.

8. A rectangular kitchen floor measures 3.6 m by 2.4 m. Square tiles are 30 cm × 30 cm and cost \$4.80 each.

(a) What is the area of the kitchen floor in m^2?

(b) What is the area of one tile in m^2?

(c) How many tiles are needed?

(d) What is the total cost?

Answer: _____ tiles; total cost = _____.

9. A triangular sail on a small boat has a base of 3.6 m and a perpendicular height of 5.2 m. What is the area of the sail?

Answer: Area = _____.

10. A rectangular park is 20 m × 15 m. Inside the park, a triangular flower bed (base 6 m, height 5 m) is paved. What area of the park remains as grass?

Answer: Grass area = _____.

11. A square vegetable patch has a perimeter of 28 m.

(a) What is the side length of the patch?

(b) What is the area of the patch?

Answer: Area = _____.

12. A swimming pool is rectangular, 25 m long and 10 m wide. A tiled path 1 m wide runs all the way around the outside of the pool.

(a) What are the outer dimensions of the path (length and width including the path)?

(b) What is the area of the outer rectangle (pool + path)?

(c) What is the area of the tiled path only?

Answer: Path area = _____.

Chapter Review

Chapter 4 Review — Length, Perimeter and Area

Instructions: Show all working and include units in every answer.

1. Complete the conversion table.

mm	cm	m
	45	
75		
		2.8

m	km
2,800	
	0.35

= given value = write your answer here

2. Calculate the perimeter of a rectangle with $L = 7$ m and $W = 4.5$ m.

3. Calculate the perimeter of a square with side 9.5 cm.

4. A triangle has sides of 5 cm, 7 cm and 9 cm. Find its perimeter.

5. Calculate the area of a rectangle with $L = 11$ m and $W = 6$ m.

6. Calculate the area of a square with side 8.5 cm.

7. A triangle has a base of 14 m and a perpendicular height of 9 m. Find its area.

8. A triangular road sign has a base of 60 cm and a height of 52 cm. Find its area in cm^2.

9. *(Multi-step)* A rectangular living room measures 5.2 m by 4 m.

(a) What is the floor area?

(b) Floorboards cost \$38 per m^2 for materials and \$12 per m^2 for installation. What is the total cost?

Answer: Total cost = _____.

10. *(Multi-step)* A rectangular backyard is 18 m long and 12 m wide. The owner wants to:

- Lay lawn over the whole yard — lawn costs \$7 per m^2
- Fence all four sides — fencing costs \$22 per metre

(a) What is the area of the yard?

(b) What is the cost of the lawn?

(c) What is the perimeter of the yard?

(d) What is the cost of the fencing?

(e) What is the total cost of both the lawn and fencing?

Answer: Total = _____.

11. A square picture frame (outer edge) has a side of 40 cm. The picture inside has a side of 32 cm. What is the area of the frame border (the part that is frame, not picture)?

Answer: Frame area = _____.

12. *(Select the correct formula)* A gardener is planning two new garden beds:

- Bed A: a rectangle, 8 m × 3.5 m
- Bed B: a triangle, base 10 m, height 6 m

Which bed has the larger area, and by how much?

Answer: _____ has the larger area by _____ m^2.

Chapter 4 Complete!

Skills checklist — tick each one when you feel confident:

- ☐ Convert between mm, cm, m and km
- ☐ Calculate perimeter of a rectangle
- ☐ Calculate perimeter of a square
- ☐ Calculate perimeter of a triangle
- ☐ Calculate area of a rectangle
- ☐ Calculate area of a square
- ☐ Calculate area of a triangle
- ☐ Estimate area by counting squares
- ☐ Solve cost problems using area or perimeter
- ☐ Solve multi-step measurement problems

5. Shapes, Area and Maps

Chapter Information

Chapter:	5 — Shapes, Area and Maps
Weeks:	Week 5
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Lesson reference:	Lessons 17–20

Every neighbourhood has shapes in it — rectangular parks, square buildings, circular fountains. Every map has a scale that lets you calculate real distances from a tiny drawing. In this chapter you will identify and describe common 2D shapes, convert between area units, and use a map of the fictional suburb of *Springvale* to practise reading coordinates, following directions and calculating distances.

Learning Objectives

By the end of this chapter you will be able to:

- Identify and describe properties of triangles, rectangles, squares and circles
- Convert between area units: mm^2, cm^2, m^2 and km^2
- Read and interpret a map using a scale, key and compass directions
- Use grid coordinates to describe and find locations
- Calculate real distances from map measurements using a scale

Key Terms

- **Vertex (vertices):** A corner point where two sides of a shape meet
- **Right angle:** An angle of exactly 90° — shown by a small square in diagrams
- **Line of symmetry:** A line that divides a shape into two identical mirror-image halves
- **Area unit conversion:** Changing an area measurement from one square unit to another (e.g., cm^2 to m^2)
- **Map scale:** A ratio that compares a distance on the map to the real distance (e.g., 1:10,000 means 1 cm on map = 10,000 cm = 100 m in real life)
- **Map key (legend):** A list explaining what each symbol on the map represents
- **Compass:** An instrument (or diagram) showing the four main directions: North (N), South (S), East (E) and West (W)

- **Grid reference (coordinates):** A letter-number pair used to identify a square on a grid map (e.g., C4 = column C, row 4)
- **Scale factor:** The number you multiply by to convert a map distance to a real distance

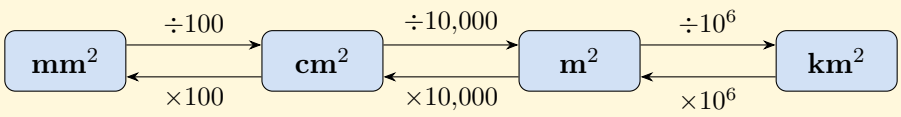
Shape Properties Reference Table — Complete the missing cells

Shape	Sides	Equal sides	Right angles	Lines of symmetry	Other properties
Square	4	_____	_____	_____	All sides equal
Rectangle	_____	2 pairs	4	_____	Opposite sides parallel
Equilateral triangle	_____	_____	0	_____	All angles = 60°
Right-angled triangle	3	_____	_____	0	One angle = 90°
Isosceles triangle	3	_____	0	_____	2 equal angles
Circle	0	—	—	_____	No straight sides

Tip: This table is a useful reference in your open-book examination.

Key Formulas and Rules

Area unit conversions:



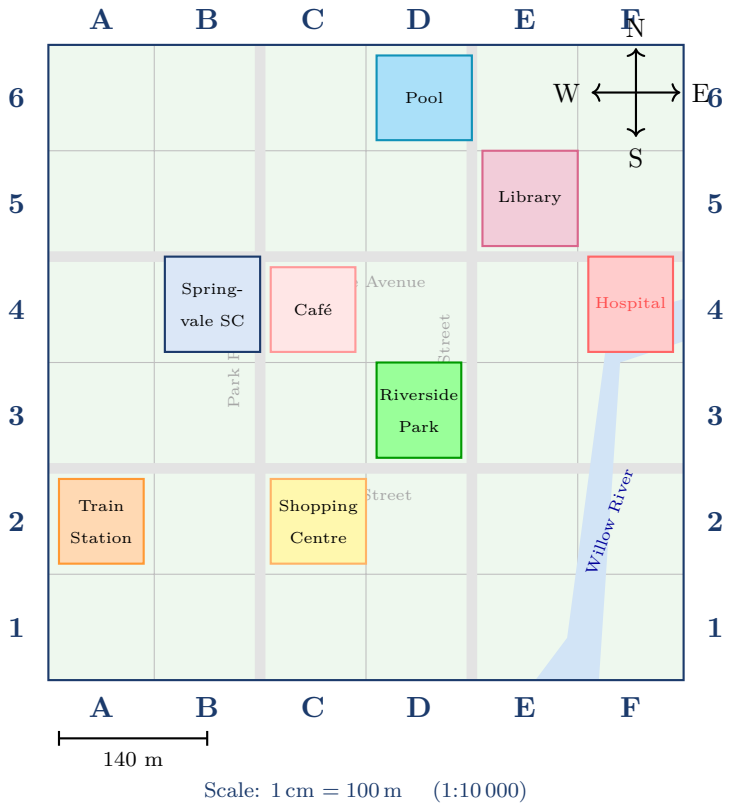
Why are the numbers so large? Because a square metre is a 100 cm × 100 cm square, so $1\,\text{m}^2 = 100 \times 100 = 10{,}000\,\text{cm}^2$.

Map scale formula:

$$\text{Actual distance} = \text{Map measurement (cm)} \times \text{Scale factor}$$

Always convert the final answer to the unit asked for (m or km).

Springvale Neighbourhood Map — used throughout this chapter



- Train Station
- Shopping Centre
- Riverside Park
- Springvale SC (School)
- Café
- Library
- Hospital
- Swimming Pool

Worked Examples

Worked Example 1 — Shape Properties

Name each shape described and state two of its properties.

(a) *"I have 4 equal sides and 4 right angles."*

(b) *"I have 3 equal sides and 3 equal angles of 60°."*

(c) *"I have no straight sides and infinite lines of symmetry."*

Solution:

(a) 4 equal sides + 4 right angles ⇒ **Square**
Properties: all sides equal; diagonals bisect each other at right angles.

(b) 3 equal sides + all angles 60° ⇒ **Equilateral triangle**
Properties: 3 lines of symmetry; all interior angles equal.

(c) No straight sides + infinite symmetry ⇒ **Circle**
Properties: every point on the edge (circumference) is the same distance from the centre; no vertices.

Worked Example 2 — Converting Area Units

(a) Convert 2.5 m^2 to cm^2. (b) Convert 45,000 cm^2 to m^2. (c) Convert 3 km^2 to m^2.

Solution:

(a) $m^2 \to cm^2$: multiply by 10,000.

$$2.5 \times 10{,}000 = \mathbf{25{,}000}\ \mathrm{cm}^{\mathbf{2}}$$

(b) $cm^2 \to m^2$: divide by 10,000.

$$45{,}000 \div 10{,}000 = \mathbf{4.5}\ \mathrm{m}^{\mathbf{2}}$$

(c) $km^2 \to m^2$: multiply by 1,000,000.

$$3 \times 1{,}000{,}000 = \mathbf{3{,}000{,}000}\ \mathrm{m}^{\mathbf{2}}$$

Worked Example 3 — Reading Grid Coordinates

Use the Springvale map to answer: (a) What is located at grid reference D3? (b) What is the grid reference of the Library? (c) In which direction would you travel to go from the School to the Café?

Solution:

(a) Grid reference D3: column D, row 3 ⇒ **Riverside Park**

(b) The Library is in column E, row 5 ⇒ **E5**

(c) School is at B4; Café is at C4 (same row, one column to the right).
Moving from B to C is moving **East**.

Worked Example 4 — Calculating Real Distance from a Map Scale

On the Springvale map, the scale is 1:10,000 (1 cm = 100 m). The measured distance on the map between the Train Station and the Shopping Centre is 2 cm. What is the actual distance?

Solution:

Step 1: Use the scale formula.

$$\text{Actual distance} = \text{Map measurement} \times \text{Scale factor}$$

Step 2: Substitute.

$$\text{Actual distance} = 2\,\text{cm} \times 10{,}000 = 20{,}000\,\text{cm}$$

Step 3: Convert to metres (divide by 100).

$$20{,}000\,\text{cm} \div 100 = \mathbf{200}\,\text{m}$$

Answer: The actual distance is **200 metres**.

Worked Example 5 — Describing a Route Using Compass Directions

Describe a route from the Train Station (A2) to the Library (E5) on the Springvale map, using compass directions.

Solution:

Step 1: Identify start and end points.
Start: Train Station, grid A2 End: Library, grid E5

Step 2: Describe the movement column by column and row by row.

- From A to E: move 4 columns to the right ⇒ travel **East**
- From row 2 to row 5: move 3 rows upward ⇒ travel **North**

Sample route: "From the Train Station, travel **East** along Oak Street, then turn **North** on High Street, continuing to the Library at E5."

Note: There are many valid routes. The key is to use compass direction words and grid references.

Practice Exercise

Practice Exercise — Shapes, Area and Maps

Instructions: Show all working. Use the Springvale map for map questions.

Part A — Shape Properties

1. Name the shape described in each clue.

(a) I have 4 sides, opposite sides are equal and parallel, and I have 4 right angles, but not all sides are equal. _____

(b) I have 3 sides. Two sides are equal in length and two angles are equal. _____

(c) I have 4 equal sides but no right angles. (Hint: not a square) _____

(d) I have no sides and no angles, but a perfectly round boundary. _____

2. How many lines of symmetry does each shape have?

(a) Square _____

(b) Rectangle (not a square) _____

(c) Equilateral triangle _____

(d) Circle _____

Part B — Area Unit Conversions

3. Convert each area to the unit shown.

(a) $3\,\text{m}^2 =$ _____ cm^2

(b) $80{,}000\,\text{cm}^2 =$ _____ m^2

(c) $5\,\text{km}^2 =$ _____ m^2

(d) $250\,\text{mm}^2 =$ _____ cm^2

(e) $0.4\,\text{m}^2 =$ _____ cm^2

(f) $2{,}500{,}000\,\text{m}^2 =$ _____ km^2

Part C — Grid References (use the Springvale map)

4. Name the location at each grid reference.

(a) A2 _____

(b) C2 _____

(c) B4 _____

(d) D6 _____

5. Write the grid reference for each location.

(a) Hospital _____

(b) Riverside Park _____

(c) Café _____

(d) Swimming Pool _____

Part D — Compass Directions (use the Springvale map)

6. State the compass direction of travel for each journey.

(a) From Café (C4) to Library (E5): _____

(b) From Hospital (F4) to School (B4): _____

(c) From Train Station (A2) to Swimming Pool (D6): _____

(d) From Library (E5) to Shopping Centre (C2): _____

Part E — Map Scale Calculations

Use the Springvale map scale: 1 cm = 100 m (scale 1:10,000).

7. Convert each map distance to a real-world distance in metres.

(a) Map distance: 1.5 cm _____

(b) Map distance: 3.2 cm _____

(c) Map distance: 0.8 cm _____

8. A real distance of 450 m is to be shown on the Springvale map. How many centimetres on the map does this equal?

_____

_____

_____

Application Exercise

Application Exercise — Shapes, Area and Maps in Context

Instructions: Use the Springvale map where indicated. Show all working and write a sentence answer.

Questions 1–7 use the Springvale map (scale: 1 cm = 100 m).

1. Mia walks from Springvale School (B4) to the Café (C4) and then to Riverside Park (D3). Describe her route using compass directions for each leg of the journey.

_____

_____

_____

2. The distance from the Train Station (A2) to the Shopping Centre (C2) measures 2 cm on the map.

(a) What is the actual distance in metres?

_____

_____

(b) If Liam walks at 80 m per minute, how long does this walk take?

_____

_____

3. A student measures the distance from the Hospital (F4) to the Library (E5) on the map and gets 1.4 cm. What is the actual distance in metres?

_____

_____

Answer: _____

4. Jordan needs to get from the Swimming Pool (D6) to the Train Station (A2) to catch a train.

(a) Plan a route from D6 to A2 using compass directions.

_____

(b) The route measures approximately 4.5 cm on the map. What is the actual distance?

5. Which two locations on the Springvale map are furthest apart? Estimate the map distance between them and calculate the real-world distance.

6. Riverside Park (D3) is roughly rectangular in shape and occupies one full grid square on the map. Each grid square on the map is 1.4 cm × 1.4 cm.

(a) What is the actual side length of the park in metres?

(b) What is the actual area of the park in m^2?

7. A new café is to be built halfway between the Train Station (A2) and the School (B4). Describe where this new location would appear on the map using a grid reference (it may fall between two grid squares — describe it as best you can).

Questions 8–12 are general shapes and area questions.

8. A floor plan shows a bedroom measuring 4.5 cm long and 3 cm wide. The floor plan has a scale of 1:100 (1 cm = 1 m).

(a) What are the actual dimensions of the bedroom?

(b) What is the actual floor area of the bedroom?

9. A football field has an area of 7,140 m^2. Convert this area to km^2. Round to 4 decimal places.

Answer: _____ km^2

10. A square courtyard has an area of 625 m^2.

(a) What is the side length of the courtyard?

(b) Convert the area of the courtyard to cm^2.

11. A rectangular room is shown on a floor plan with scale 1:50 (1 cm = 0.5 m). On the plan the room measures 7 cm by 5 cm.

(a) What are the actual dimensions of the room?

(b) What is the actual area of the room?

12. A triangular garden feature has a base of 4 m and a height of 3 m. A gardener converts the area to cm^2 for a tile order. What is the area in cm^2?

Answer: _____ cm^2

Chapter Review

Chapter 5 Review — Shapes, Area and Maps

Instructions: Show all working. Use the Springvale map where needed.

1. Complete the sentence: A rectangle has _____ sides, _____ right angles and _____ lines of symmetry.

2. Name a 2D shape that has exactly 1 line of symmetry.

_____

3. Convert: $6.2\,\text{m}^2 =$ _____ cm^2

4. Convert: $350{,}000\,\text{cm}^2 =$ _____ m^2

5. Convert: $1.5\,\text{km}^2 =$ _____ m^2

6. Using the Springvale map, what is located at grid reference F4?

_____

7. Using the Springvale map, in which direction would you travel to go from the Shopping Centre (C2) to the Hospital (F4)?

_____

_____

8. A map distance of 3.5 cm represents a real distance on a 1:10,000 map. What is the actual distance in metres?

_____

_____

9. On a 1:50,000 map, two towns are 4 cm apart. What is the actual distance between them in kilometres?

_____

_____

_____

Answer: _____ km

10. A house block is 18 m wide and 30 m long.

(a) What is the area of the block in m^2?

(b) Convert this area to cm^2.

11. *(Multi-step)* A room is shown on a floor plan with scale 1:100 (1 cm = 1 m). The room is 6 cm × 4 cm on the plan. Flooring costs \$52 per m^2.

(a) What are the real dimensions of the room?

(b) What is the real floor area?

(c) What is the total cost of flooring?

Answer: Total cost = _____.

12. *(Multi-step)* Using the Springvale map, plan a round trip from the School (B4) that visits the Library (E5) and the Swimming Pool (D6) before returning to the School. State the compass direction for each leg of the journey and estimate the total map distance. Calculate the real-world total distance.

Answer: Estimated total distance ≈ _____.

Chapter 5 Complete!

Skills checklist — tick each one when you feel confident:

- ☐ Identify 2D shapes by their properties
- ☐ State sides, right angles and lines of symmetry
- ☐ Convert between mm^2, cm^2, m^2 and km^2
- ☐ Read grid references on a map
- ☐ Use compass directions (N, S, E, W)
- ☐ Calculate real distances from map measurements
- ☐ Interpret a floor plan using a scale
- ☐ Solve multi-step map and area problems

6. Time and Angles

Chapter Information

Chapter:	6 — Time and Angles
Weeks:	Week 6
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Lesson reference:	Lessons 21–24

Time management is a fundamental life skill. Catching a bus, booking a doctor's appointment, planning a school assessment schedule — all of these require reading times, calculating durations and interpreting timetables. Angles are equally practical: a carpenter measures angles to cut timber, a navigator uses angles to plan a route, and a clock face divides a full circle into measurable angles. In this chapter you will work fluently with both 12-hour and 24-hour time, read timetables and calendars, and identify and classify angles.

Learning Objectives

By the end of this chapter you will be able to:

- Read and write time in both 12-hour (am/pm) and 24-hour format
- Convert between 12-hour and 24-hour time
- Calculate elapsed time (duration) between a start and end time
- Read and interpret a bus timetable to plan journeys
- Calculate the number of days between two dates using a calendar
- Identify, name and classify angles: acute, right, obtuse, straight, reflex and full rotation

Key Terms

- **12-hour time:** Time expressed using am (before noon) and pm (after noon), with hours from 1 to 12
- **24-hour time:** Time expressed using four digits with hours from 00 to 23 (no am/pm needed)
- **am:** Ante meridiem — the period from midnight (12:00 am) to just before noon (11:59 am)
- **pm:** Post meridiem — the period from noon (12:00 pm) to just before midnight (11:59 pm)

- **Elapsed time:** The total time that passes between a start time and an end time (also called duration)
- **Timetable:** A printed schedule showing departure and arrival times for transport services
- **Departure:** The time a service leaves a stop
- **Arrival:** The time a service reaches a stop
- **Angle:** The amount of turn between two lines that meet at a point, measured in degrees (°)
- **Acute angle:** An angle greater than 0° and less than 90°
- **Right angle:** An angle of exactly 90° (marked with a small square)
- **Obtuse angle:** An angle greater than 90° and less than 180°
- **Straight angle:** An angle of exactly 180° (a straight line)
- **Reflex angle:** An angle greater than 180° and less than 360°
- **Full rotation:** A complete turn of 360°

Key Formulas and Rules

12-hour → 24-hour:

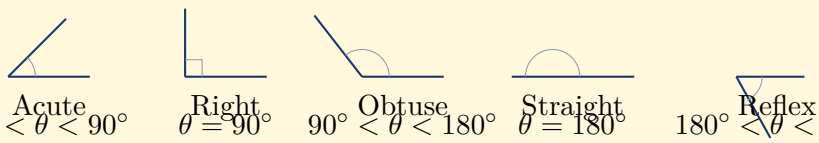
- am times: keep the same (write with 4 digits)
  e.g., 9:15 am = **09:15**
- Noon: 12:00 pm = **12:00**
- pm times: add 12 to the hours
  e.g., 3:45 pm = $3 + 12 =$ **15:45**
- Midnight: 12:00 am = **00:00**

24-hour → 12-hour:

- 00:xx to 11:59 → am (replace 00 with 12 for midnight)
- 12:xx → 12:xx pm (noon)
- 13:00 to 23:59 → subtract 12 and add pm
  e.g., 19:30 = $19 - 12 =$ **7:30 pm**

Elapsed time = End time − Start time (use the "jump method": jump to the next hour, then count hours, then count remaining minutes)

Angle classifications:



Acute $0° < \theta < 90°$ | Right $\theta = 90°$ | Obtuse $90° < \theta < 180°$ | Straight $\theta = 180°$ | Reflex $180° < \theta < 360°$

Clock angles: There are 360° in a full clock face. The minute hand moves **6° per minute**; the hour hand moves **30° per hour**.

Springvale Bus Timetable — used throughout this chapter

Route 72 — Springvale to City Central *All times in 24-hour format*

Service	Springvale	Park Road	Shopping Centre	Riverside	City Central
706	06:15	06:22	06:30	06:38	06:50
720	07:00	07:08	07:16	07:25	07:38
735	08:30	08:38	08:47	08:56	09:10
750	10:15	10:22	10:30	10:39	10:52
760	13:40	13:47	13:55	14:04	14:17
775	16:20	16:28	16:37	16:46	17:00
790	18:05	18:12	18:20	18:29	18:42

Route 72R — City Central to Springvale *All times in 24-hour format*

Service	City Central	Riverside	Shopping Centre	Park Road	Springvale
801	07:20	07:33	07:41	07:50	07:57
815	09:00	09:13	09:21	09:30	09:37
830	12:30	12:43	12:51	13:00	13:07
845	15:00	15:13	15:21	15:30	15:37
860	17:30	17:43	17:51	18:00	18:07
875	19:15	19:28	19:36	19:45	19:52

Services operate Monday to Friday only. Times shown are scheduled; allow extra time during peak hour.

June 2026 Calendar — used for date calculation questions

Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Worked Examples

Worked Example 1 — Converting Between 12-Hour and 24-Hour Time

Convert each time as indicated.

(a) 3:45 pm → 24-hour

(b) 11:20 am → 24-hour

(c) 19:30 → 12-hour

(d) 00:45 → 12-hour

Solution:

(a) 3:45 pm: pm → add 12 to hours. $3 + 12 = 15$ ⇒ **15:45**

(b) 11:20 am: am → keep the same, write with 4 digits. ⇒ **11:20**

(c) 19:30: hours > 12 → subtract 12 and add pm. $19 - 12 = 7$ ⇒ **7:30 pm**

(d) 00:45: 00 means midnight → **12:45 am**
(Note: 00:00 = 12:00 am midnight; 12:00 = 12:00 pm noon)

Worked Example 2 — Calculating Elapsed Time

A bus departs at 08:20 and arrives at 11:05. How long is the journey?

Solution — using the "jump method":

Step 1: Jump from 08:20 to the next whole hour (09:00).

$$09{:}00 - 08{:}20 = 40 \text{ minutes}$$

Step 2: Jump from 09:00 to 11:00 (count whole hours).

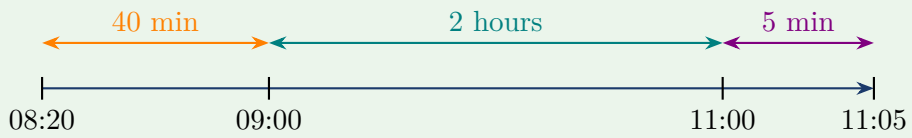
$$11{:}00 - 09{:}00 = 2 \text{ hours}$$

Step 3: Jump from 11:00 to the arrival time 11:05.

$$11{:}05 - 11{:}00 = 5 \text{ minutes}$$

Step 4: Add the parts together.

$$40 \text{ min} + 2 \text{ hr} + 5 \text{ min} = \mathbf{2} \text{ hours and } \mathbf{45} \text{ minutes}$$



Worked Example 3 — Reading the Bus Timetable

Use the Route 72 timetable to answer: (a) How long does Service 735 take from Springvale to City Central? (b) Amira needs to reach City Central before 10:00. Which service should she catch from Springvale? (c) How long does she wait at Riverside if she misses Service 735 and takes the next service?

Solution:

(a) Service 735: departs Springvale 08:30, arrives City Central 09:10.
Elapsed time: 09:10 − 08:30 = **40** minutes

(b) She needs to arrive before 10:00.
Service 735 arrives at 09:10 ✓ Service 750 arrives at 10:52 ×
She should catch **Service 735**, departing Springvale at **08:30**.

(c) If she misses 735, the next service is 750.
Service 735 passes Riverside at 08:56; Service 750 passes at 10:39.
Wait time: 10:39 − 08:56 = 1 hr 43 min
*(Jump: 08:56 → 09:00 = 4 min; 09:00 → 10:00 = 1 hr; 10:00 → 10:39 = 39 min; total = **1 hr 43 min**)*

Worked Example 4 — Days Between Dates

Using the June 2026 calendar, how many days are there from June 8 to June 25?

Solution:

Method 1 — Subtraction:

$$25 - 8 = \mathbf{17} \text{ days}$$

(We count from June 8 to June 25 but do not include June 8 itself — we count the days that pass.)

Method 2 — Counting on the calendar:
June 8 (Mon) → June 15 (Mon) = 7 days
June 15 → June 22 (Mon) = 7 days
June 22 → June 25 (Thu) = 3 days
Total: $7 + 7 + 3 =$ **17** days ✓

Across months: From June 20 to July 10.
June 20 to June 30 = 10 days remaining in June
July 1 to July 10 = 10 days into July
Total: $10 + 10 =$ **20** days

Worked Example 5 — Identifying and Classifying Angles

Classify each angle and state whether it is acute, right, obtuse, straight, reflex or a full rotation.

(a) 55° (b) 90° (c) 145° (d) 180° (e) 310° (f) 360°

Solution:

(a) 55° $0° < 55° < 90°$ ⇒ **Acute**
(b) 90° Exactly 90° ⇒ **Right angle**
(c) 145° $90° < 145° < 180°$ ⇒ **Obtuse**
(d) 180° Exactly 180° (straight line) ⇒ **Straight angle**
(e) 310° $180° < 310° < 360°$ ⇒ **Reflex**
(f) 360° Complete turn ⇒ **Full rotation**

Worked Example 6 — Clock Angles

What angle does the minute hand make between the 12 and the 6 on a clock? What about between the 12 and the 3?

Solution:

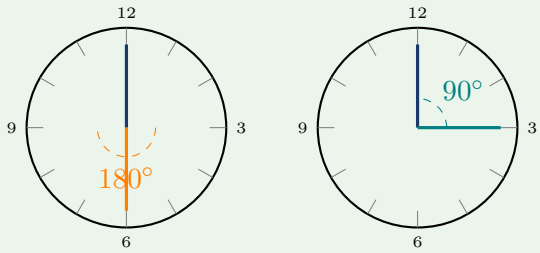
Key fact: A clock face = 360°, divided into 12 equal sections.
Each section (from one number to the next) = $360° \div 12 =$ **30°**

12 to 6: The minute hand travels 6 sections.

$$6 \times 30^\circ = \mathbf{180^\circ} \quad \text{(a straight angle)}$$

12 to 3: The minute hand travels 3 sections.

$$3 \times 30^\circ = \mathbf{90^\circ} \quad \text{(a right angle)}$$



12 to 6 12 to 3

Practice Exercise

Practice Exercise — Time and Angles

Instructions: Show all working. Use the timetable and calendar provided in this chapter.

Part A — Time Conversions

1. Convert each time to 24-hour format.

(a) 7:30 am _____

(b) 1:15 pm _____

(c) 11:50 pm _____

(d) Midnight (12:00 am) _____

(e) 12:30 pm (noon) _____

2. Convert each time to 12-hour format (include am or pm).

(a) 06:45 _____

(b) 14:30 _____

(c) 21:05 _____

(d) 00:20 _____

(e) 12:00 _____

Part B — Elapsed Time

3. Calculate the elapsed time between each pair of times. Use the jump method and show your steps.

(a) Start: 09:15 End: 11:45

_____

_____

_____

(b) Start: 07:50 End: 10:20

(c) Start: 13:35 End: 16:10

(d) Start: 22:40 End: 02:15 (next day)

Part C — Timetable Reading

Use the Route 72 timetable.

4. What time does Service 760 depart Springvale and arrive at City Central?

5. How long does Service 790 take from Springvale to City Central?

6. What is the first service from City Central to Springvale that departs after 3:00 pm?

7. How long does Service 830 take from City Central to Springvale?

Part D — Date Calculations

Use the June 2026 calendar.

8. How many days from June 3 to June 19?

_____

9. A school project is due on June 25. Today is June 10. How many days does the student have to complete it?

_____

_____

10. A camp starts on June 12 and runs for 5 days. What date does it end?

_____

_____

11. What day of the week is June 17, 2026?

_____

Part E — Angle Classification

12. Classify each angle as acute, right, obtuse, straight, reflex or full rotation.

(a) 75° _____

(b) 180° _____

(c) 210° _____

(d) 90° _____

(e) 115° _____

(f) 360° _____

13. Draw an example of each angle type in the space below.

Acute angle	Right angle	Obtuse angle
Straight angle	Reflex angle	Full rotation

Application Exercise

Application Exercise — Time and Angles in Everyday Life

Instructions: Use the timetable and calendar where indicated. Show all working and write a sentence answer.

1. Mia's school day runs from 8:45 am to 3:10 pm. How long is the school day?

Answer: The school day is _____ long.

2. Jake works from 09:30 to 17:15 on Monday.

(a) How many hours and minutes does he work?

(b) He earns $24.50 per hour. Calculate his total pay for the day (round to the nearest 15 minutes for the calculation).

Answer: Jake earns _____.

3. Using the Route 72 timetable, Priya needs to travel from Springvale to the Shopping Centre to meet a friend at **10:45**.

(a) Which service should she catch from Springvale to arrive before 10:45?

(b) How many minutes early will she arrive?

4. Sam travels from Springvale to City Central on Service 775, does some shopping, then returns on Service 860.

(a) What time does he arrive at City Central?

(b) What time does he depart City Central on the return?

(c) How long does Sam have in the city for shopping?

Answer: Sam has _____ for shopping.

5. A soccer match at Springvale oval starts at **14:30** and lasts **1 hour 50 minutes** (including half-time). What time does it finish? Write the answer in both 24-hour and 12-hour format.

Answer: 24-hour: _____ 12-hour: _____

6. Using the June 2026 calendar, Zara starts a 3-week fitness challenge on June 2.

(a) On what date does the challenge end?

(b) What day of the week does it end?

7. School holidays begin on June 27 and students return to school on July 14. How many days are the school holidays? (June has 30 days.)

Answer: School holidays are _____ days long.

8. A library book was borrowed on June 5 and must be returned within 21 days. What is the return date?

Answer: Return by _____.

9. A clock shows 6:00.

(a) What angle does the minute hand make between the 12 and the 6? Name the angle type.

(b) At 3:00, the minute hand points to 12 and the hour hand points to 3. What angle is formed between them going clockwise? Name the angle type.

10. A carpenter cuts a piece of timber at 45° to make a picture frame corner.

(a) Classify this angle.

(b) When two 45° cuts are joined, what angle is formed at the corner of the frame?

11. A bus route turns right at an intersection. A right turn is a rotation of 90°.

(a) If the bus turns right twice in a row, what total angle has it turned?

(b) After two right turns, is the bus now facing in the opposite direction to where it started? Explain.

12. A multi-leg bus journey: Jayden travels from Springvale to Riverside on Service 720, waits, then catches Service 830R (City Central to Springvale direction) from Riverside back to Park Road.

(a) What time does Jayden arrive at Riverside on Service 720?

(b) Service 830R passes Riverside at 12:43. How long does Jayden wait at Riverside?

Answer: Jayden waits _____.

Chapter Review

Chapter 6 Review — Time and Angles

Instructions: Show all working. Use the timetable and calendar where needed.

1. Convert to 24-hour time: 4:20 pm _____

2. Convert to 24-hour time: 12:15 am (midnight) _____

3. Convert to 12-hour time: 17:45 _____

4. Convert to 12-hour time: 03:30 _____

5. Calculate the elapsed time from 10:40 to 13:15.

_____

_____

_____

6. A movie starts at 7:15 pm and runs for 2 hours and 35 minutes. What time does it end? Give your answer in both 12-hour and 24-hour format.

_____

_____

Answer: 12-hour: _____ 24-hour: _____

7. Using the timetable, which Route 72 service from Springvale arrives at City Central closest to 7:30 am?

_____

_____

8. Using the timetable, how long does Service 845 take from City Central to Springvale?

_____

_____

9. Using the June 2026 calendar, how many Mondays are there in June?

_____

10. A school assignment was given on June 4 and is due June 22. How many days does the student have?

_____

11. Classify each angle and write one real-world example of where you might see it.

(a) 35°: Classification _____ Real-world example: _____

(b) 90°: Classification _____ Real-world example: _____

(c) 200°: Classification _____ Real-world example: _____

12. *(Multi-step)* Lena catches the 07:00 service from Springvale, does errands at City Central, then returns on Service 830.

(a) What time does she arrive at City Central?

(b) What time does Service 830 depart City Central?

(c) How long does Lena have at City Central for her errands?

Answer: Lena has _____ at City Central.

13. *(Multi-step)* A school excursion leaves school at 08:45 and returns at 15:20. Students must complete a 45-minute activity booklet on the bus (both ways combined).

(a) How long is the total excursion?

(b) After completing the activity booklet, how much excursion time remains?

Answer: _____ of excursion time remains.

Chapter 6 Complete!

Skills checklist — tick each one when you feel confident:

- ☐ Convert 12-hour time to 24-hour time
- ☐ Convert 24-hour time to 12-hour time
- ☐ Calculate elapsed time using the jump method
- ☐ Read a timetable for departure and arrival
- ☐ Plan a journey using a timetable
- ☐ Calculate days between dates using a calendar
- ☐ Classify angles (acute, right, obtuse, straight, reflex)
- ☐ Calculate clock face angles
- ☐ Solve multi-step time problems

Chapter 7 — Data and Graphs

Stage: Stage 4 — Student Workbook
Week: Week 7
Lessons: Lessons 25–28
Strand: Topic 1: Number, Algebra, Measurement, Space, Statistics & Probability
Sub-topic: Statistics & Probability

Introduction

Every day, data is collected and displayed to help us make decisions — from sports statistics to weather forecasts, from sales reports to health surveys. Being able to read, draw, and interpret graphs is one of the most useful numeracy skills you can have.

In this chapter you will collect data using tally charts and frequency tables, draw bar graphs and line graphs, and describe patterns and trends from displays. All of the skills in this chapter appear regularly in real workplaces and in the QCAA Numeracy examination.

Learning Objectives — by the end of this chapter you will be able to:

- Collect and organise data using tally marks and frequency tables
- Read and interpret bar graphs and column graphs
- Draw a bar graph from a frequency table
- Read and interpret line graphs, including identifying trends
- Draw a line graph from a data table
- Calculate the mean (average) from a small data set
- Describe patterns, trends and conclusions from graphs using mathematical language

Key Terms

Term	Definition
Data	Information collected for a purpose, often as numbers or categories
Tally chart	A quick way to count data using groups of five (four lines + one diagonal)
Frequency table	A table that shows how often each value or category occurs
Frequency	The number of times a value appears in a data set
Bar graph	A graph using rectangular bars to compare categories
Column graph	A bar graph with vertical (upright) bars
Line graph	A graph using connected points to show how data changes over time
Trend	The general direction data moves over time (increasing, decreasing, steady)
Mean (average)	The sum of all values divided by the number of values
Scale	The numbered intervals on the axis of a graph
Title	A label that describes what a graph is showing
Axis	The horizontal (x) or vertical (y) reference line on a graph

Key Formulas and Rules

Data and Graphs — Key Rules

Tally marks

~~||||~~ = 5 ~~||||~~ || = 7

Count in groups of 5, then add remainders.

Mean (average)

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Bar / Column graph	Line graph
• Used to compare categories	• Used to show change over time
• Bars do *not* touch (categories)	• Points are connected with lines
• Both axes must have a title	• Both axes must have a title
• Equal bar widths and gaps	• Time goes on the horizontal axis

Parts of a good graph (SALT): **S**cale **A**xes labelled **L**egend (if needed) **T**itle

Worked Examples

Worked Example 1 — Tally Chart and Frequency Table

Situation: A Year 10 class was asked to name their favourite after-school activity. The raw responses were:

Sport, Gaming, Social Media, Reading, Sport, Music, Social Media, Gaming, Sport, Social Media, Reading, Sport, Music, Social Media, Gaming, Sport, Sport, Social Media, Gaming, Music

Task: Complete the tally chart and frequency table.

Step 1: Go through the list one item at a time, adding a tally mark for each response.
Step 2: Count the tallies, writing the total in the frequency column.
Step 3: Check: all frequencies must add to the total number of responses (20).

Activity	Tally	Frequency
Sport	𝍸 |	6
Gaming	||||	4
Social Media	𝍸 |	6
Reading	||	2
Music	|||	3
Total		**21**

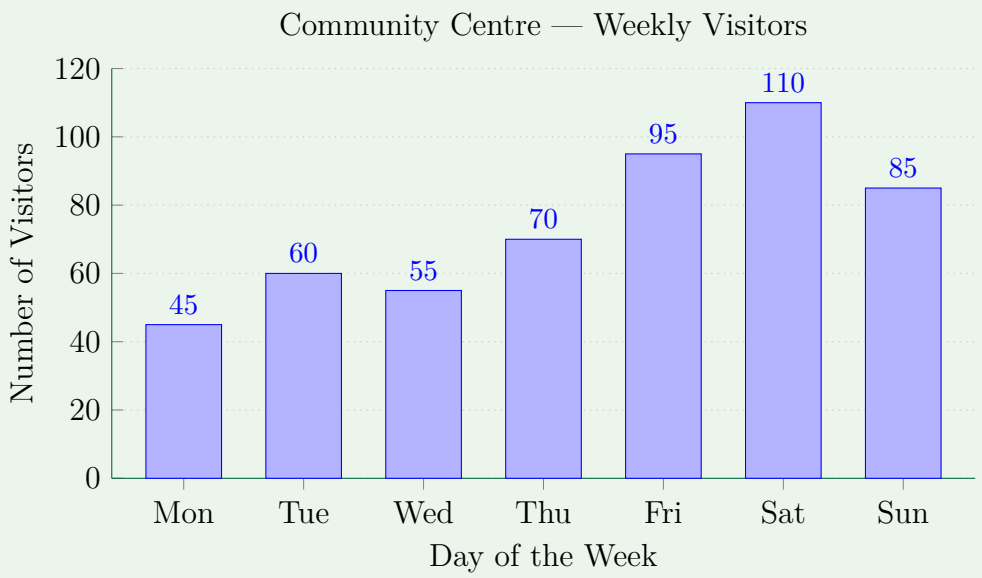
Wait — the total is 21, but there are only 20 responses. Recount. Music should be 3, but re-reading the list: Music appears at positions 6, 13, 20 = 3 times. Gaming: 2, 8, 15, 19 = 4 times. Social Media: 3, 7, 10, 14, 18 = 5 times, not 6. Correcting:

Activity	Tally	Frequency
Sport	𝍸 |	6
Gaming	||||	4
Social Media	𝍸	5
Reading	||	2
Music	|||	3
Total		**20**

Key point: Always check your total equals the number of data values.

Worked Example 2 — Reading a Bar Graph

The following bar graph shows the number of visitors to a local community centre each day for one week.



Questions and answers:

(a) **Which day had the most visitors?** Saturday (110 visitors)

(b) **How many more visitors on Saturday than Monday?** $110 - 45 = 65$ more visitors

(c) **What is the total number of visitors for the week?**

$$45 + 60 + 55 + 70 + 95 + 110 + 85 = 520 \text{ visitors}$$

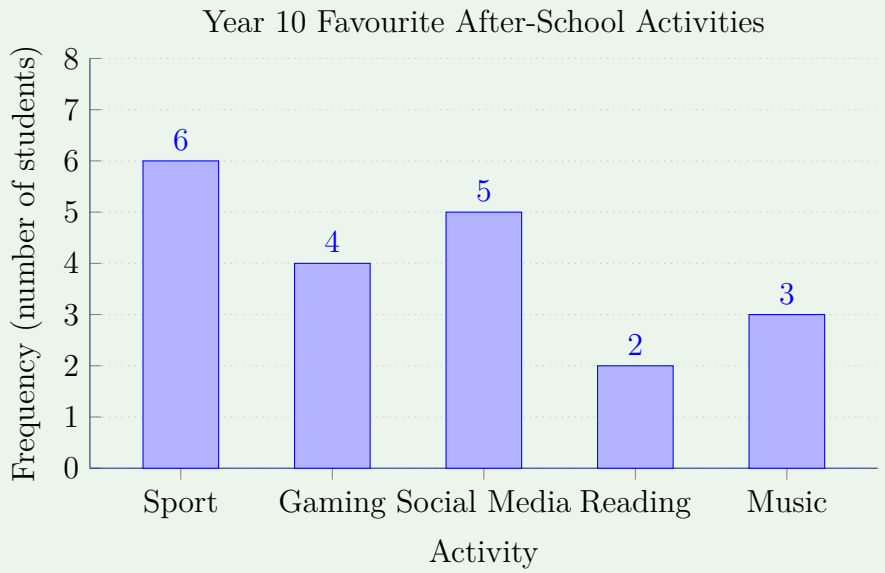
(d) **What is the mean (average) number of visitors per day?**

$$\text{Mean} = \frac{520}{7} \approx 74.3 \text{ visitors}$$

Worked Example 3 — Drawing a Bar Graph

Task: Draw a bar graph for the after-school activities frequency table from Worked Example 1.

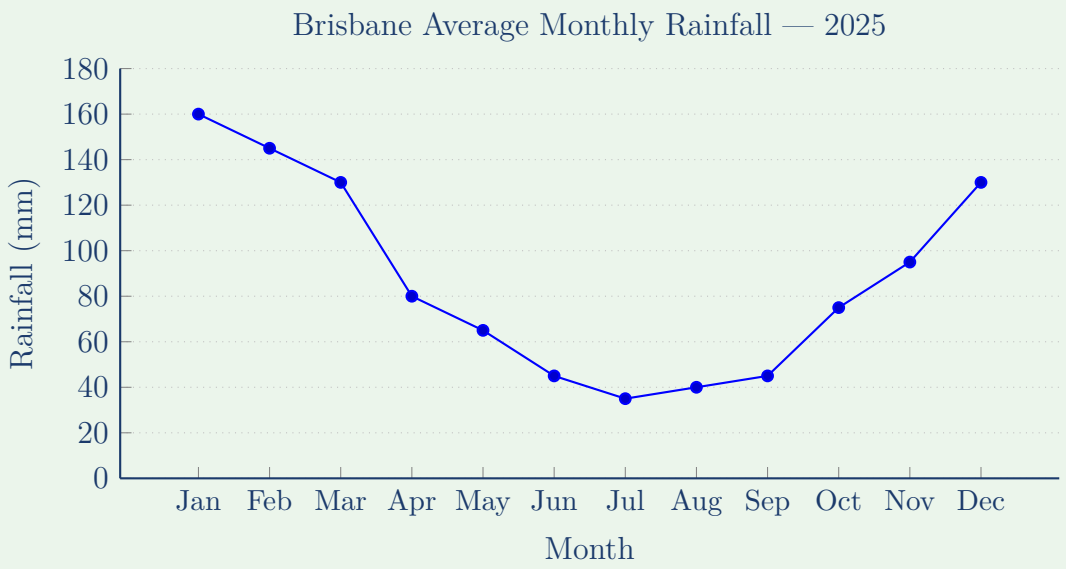
Step 1: Draw and label the horizontal axis (categories) and vertical axis (frequency).
Step 2: Choose a scale. The highest frequency is 6, so count by 1s up to 8.
Step 3: Draw bars of equal width. Leave equal gaps between bars.
Step 4: Write a title above the graph.



Remember SALT: **S**cale ✓ **A**xes labelled ✓ **L**egend (not needed — only one group) ✓ **T**itle ✓

Worked Example 4 — Reading a Line Graph

The line graph below shows Brisbane's average monthly rainfall (mm) for 2025.



Questions and answers:

(a) **Which month had the highest rainfall?** January (160 mm)

(b) **Which month had the lowest rainfall?** July (35 mm)

(c) **Describe the trend from January to July.** Rainfall *decreases steadily* from January to July.

(d) **Describe the trend from July to December.** Rainfall *increases* from July to December.

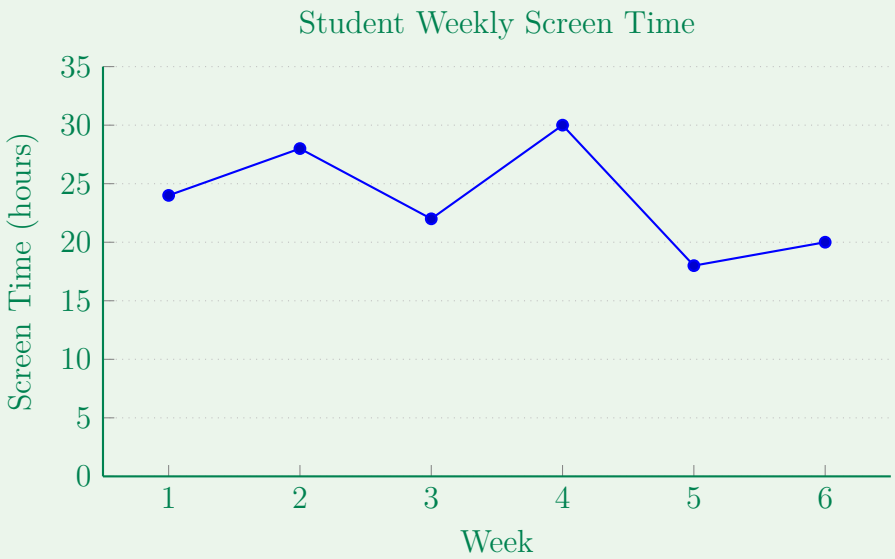
(e) **What is the difference between the wettest and driest months?** $160 - 35 =$ 125 mm

Worked Example 5 — Drawing a Line Graph

Task: The table shows a student's weekly screen time (hours) over 6 weeks. Draw a line graph.

Week	1	2	3	4	5	6
Screen Time (h)	24	28	22	30	18	20

Step 1: Draw the axes. Time (weeks) on the horizontal axis; screen time on the vertical axis.
Step 2: Choose a scale. Data ranges from 18–30, so scale from 0–35, counting by 5s.
Step 3: Plot each point. Week 1 → 24, Week 2 → 28, etc.
Step 4: Join the points with straight lines.
Step 5: Add a title and axis labels.



Describe the trend: The screen time *fluctuates* (goes up and down) with no clear overall trend. The highest point was Week 4 (30 hours) and the lowest was Week 5 (18 hours).

Worked Example 6 — Calculating the Mean

Task: The daily temperatures (°C) in Brisbane for one week were: 28, 31, 27, 33, 29, 35, 30. Calculate the mean temperature.

Step 1: Add all the values.

$$28 + 31 + 27 + 33 + 29 + 35 + 30 = 213$$

Step 2: Divide by the number of values (7 days).

$$\text{Mean} = \frac{213}{7} = 30.43\ldots \approx 30.4\ \text{°C (1 decimal place)}$$

Answer: The mean daily temperature for the week was approximately **30.4°C**.

Quick check: The mean should be somewhere in the middle of the data set (between 27 and 35). ✓

Practice Exercise

Practice Exercise — Data and Graphs

Part A — Tally Charts and Frequency Tables

1. A group of 20 students was asked how many text messages they sent yesterday. The responses were:

3, 7, 2, 5, 8, 3, 7, 10, 2, 5, 7, 3, 8, 5, 7, 2, 10, 3, 5, 7

Complete the frequency table:

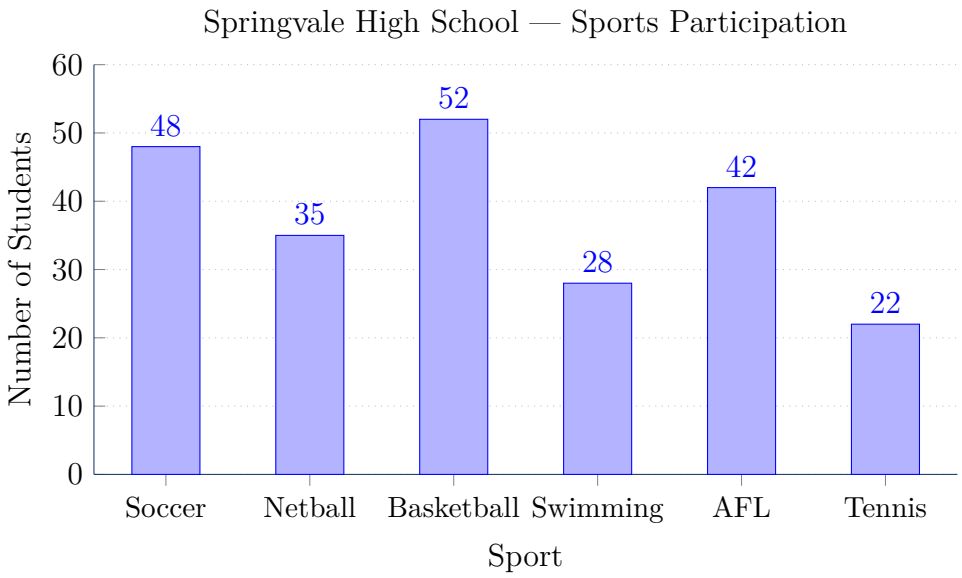
Messages	Tally	Frequency
2		
3		
5		
7		
8		
10		
Total		

2. Using the frequency table above, what was the most common number of text messages sent?

3. What fraction of students sent 7 or more text messages? Write your answer in simplest form.

Part B — Reading Bar Graphs

Questions 4–7 refer to the bar graph below showing sports participation at Springvale High School.



4. Which sport has the most participants?

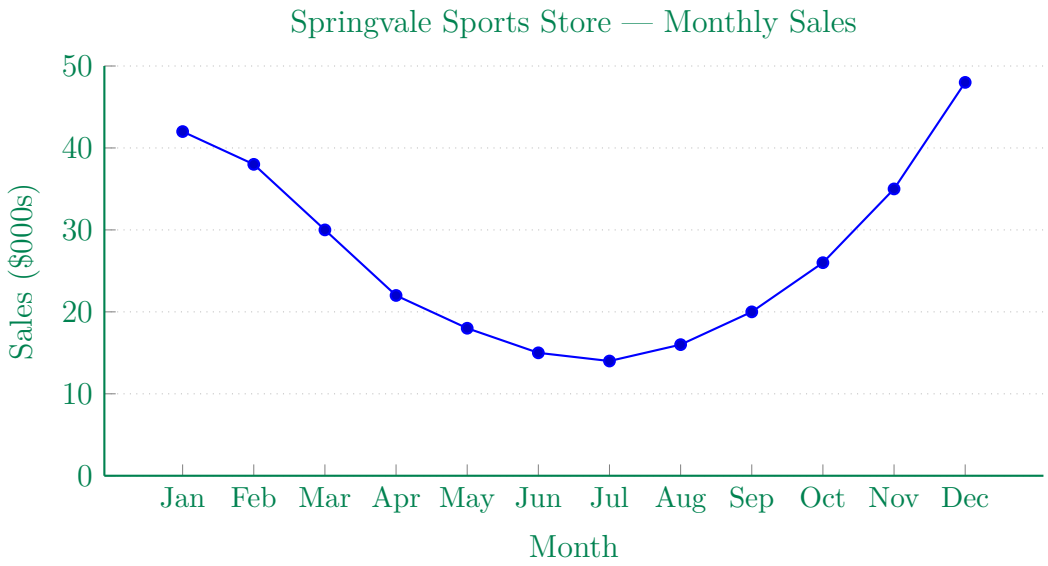
5. How many more students play Basketball than Tennis?

6. What is the total number of students represented in the graph?

7. What percentage of the total students play Soccer? (Round to the nearest whole percent.)

Part C — Reading Line Graphs

Questions 8–11 refer to the line graph below showing monthly sales at a sports store.



8. Which month had the lowest sales?

9. What was the value of sales in November? (Remember: the axis is in $000s)

10. Describe the general trend in sales from January to July in one sentence.

11. Suggest one reason why sales might be higher in December than in July.

Part D — Drawing Graphs

12. The table shows the number of goals scored by a netball team in each game of the season.

Game	1	2	3	4	5	6	7	8
Goals	32	28	35	30	38	42	36	44

Draw a line graph on the grid below. Remember: title, labelled axes, suitable scale.

13. Using your line graph, describe the overall trend in the team's goal scoring across the season.

Part E — Calculating the Mean

14. Five friends compared how many km they walked last week: 8, 12, 7, 15, 8. Calculate the mean number of km walked.

15. A student scored the following marks out of 20 on six quizzes: 14, 18, 12, 16, 15, 17.

(a) Calculate the mean quiz score.

(b) The student scored 19 on the seventh quiz. Will this raise or lower the mean? Explain.

Application Exercise

Application Exercise — Real-World Data and Graphs

1. *(Tally and frequency)* A café in Springvale recorded drink orders during one morning. Complete the tally chart for the following 24 orders:

Coffee, Tea, Coffee, Juice, Coffee, Smoothie, Tea, Coffee, Juice, Coffee, Coffee, Tea, Smoothie, Coffee, Juice, Tea, Coffee, Coffee, Smoothie, Juice, Tea, Coffee, Juice, Coffee

Drink	**Tally**	**Frequency**
Coffee		
Tea		
Juice		
Smoothie		
Total		

2. Using your frequency table from Question 1:

(a) What fraction of drinks ordered were coffee? Simplify your answer.

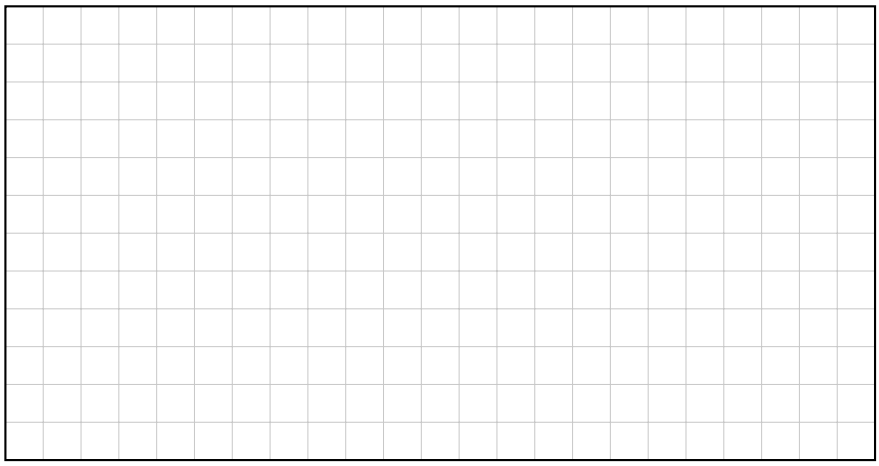
_____

(b) What percentage of drinks ordered were juice?

_____

_____

3. *(Bar graph — draw)* Draw a bar graph for the café drink data from Question 1 on the grid below.



4. *(Line graph — read and interpret)* The table shows Queensland's monthly tourism visitor numbers (thousands) for 2025:

Month	Jan	Feb	Mar	Apr	May	Jun
Visitors (000s)	420	390	350	310	280	260

Month	Jul	Aug	Sep	Oct	Nov	Dec
Visitors (000s)	265	290	330	370	400	445

(a) Plot the data as a line graph on the grid below.

(b) Which month had the lowest number of visitors?

(c) Describe the trend in visitor numbers from June to December.

(d) A tourism manager says: *"We should run marketing campaigns in May and June to boost winter visitor numbers."* Does the data support this suggestion? Explain.

5. *(Mean)* A Queensland farmer recorded the number of crates of mangoes picked each day for a week:

Day	Mon	Tue	Wed	Thu	Fri	Sat
Crates	18	24	21	27	22	30

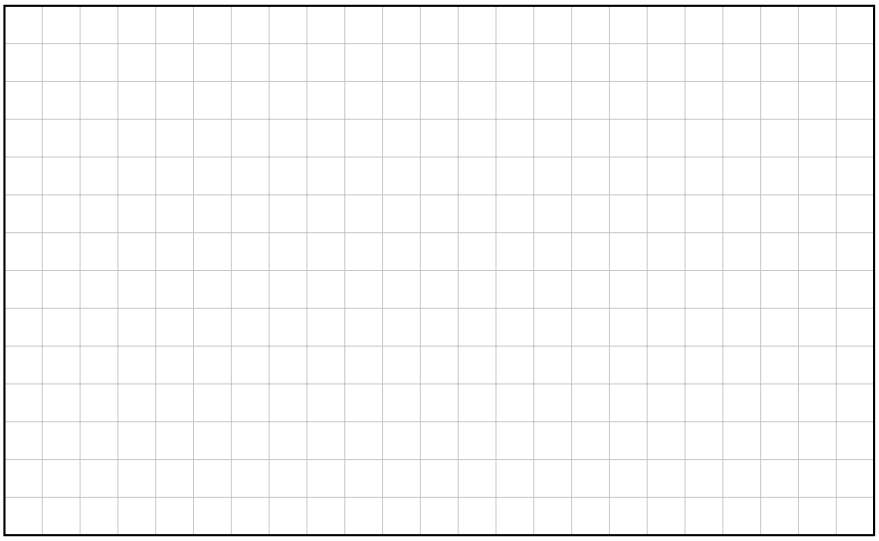
(a) Calculate the mean number of crates picked per day.

(b) The farmer aims to pick an average of at least 24 crates per day. Does this week's data meet the target?

6. *(Comparing graphs)* Two local soccer clubs recorded their membership numbers over 5 years:

Year	2021	2022	2023	2024	2025
Riverside FC	85	90	102	115	128
Northside FC	130	118	110	105	98

(a) On the grid below, plot *both* clubs as two separate lines on one graph. Label each line.

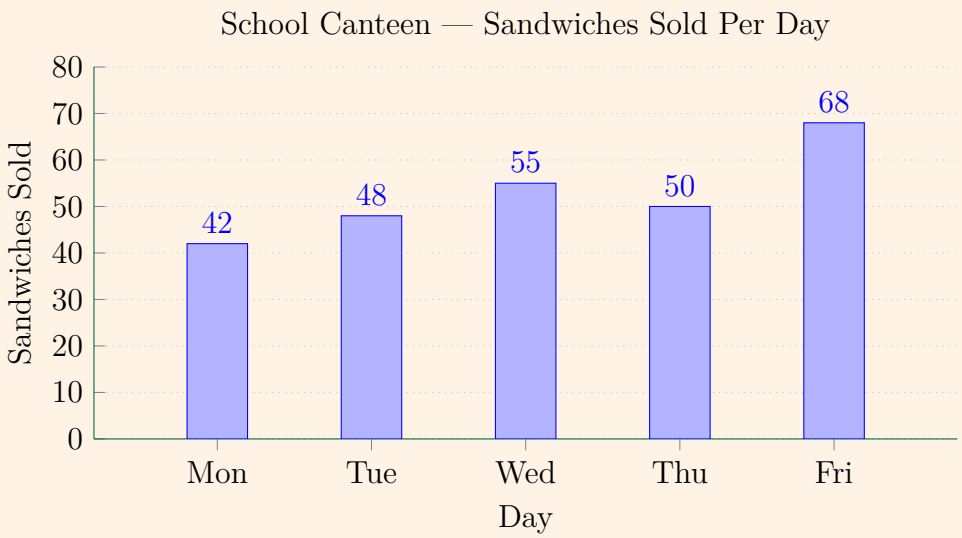


(b) Calculate the mean membership for each club over the 5 years.

(c) Describe the difference in trends between the two clubs.

(d) Based on the trend, predict which club will have more members in 2026.

7. *(Interpreting a graph — extended)* A school canteen manager uses a graph to help decide how many sandwiches to prepare. The bar graph shows the number sold each day last term:



(a) Calculate the mean number of sandwiches sold per day.

(b) The canteen prepares 10% more than the mean each day. How many sandwiches should be prepared each day? (Round to the nearest whole number.)

(c) On a Thursday next term, a 50-student excursion means fewer students at school. Would you expect the Thursday sales to be above or below the mean? Explain.

_____

_____

Chapter Review

Chapter 7 Review — Data and Graphs

1. Write the number of tally marks represented by each of the following:

(a) 𝍸 ||| (a group of five + three) _____

(b) 𝍸 𝍸 | (two groups of five + one) _____

2. A class survey found the following number of pets per household:

0, 1, 2, 0, 3, 1, 0, 2, 1, 1, 0, 2, 3, 1, 0, 4, 1, 2, 0, 1

Complete the frequency table:

Pets	Tally	Frequency
0		
1		
2		
3		
4		
Total		

3. Using the frequency table from Question 2:

(a) What is the most common number of pets?

_____

(b) What fraction of households have 2 or more pets? Simplify your answer.

_____

_____

4. Refer to the bar graph in Practice Exercise Part B (Sports Participation). If 15 new students join Tennis, what would the new total number of students in the graph be?

_____

_____

5. The temperatures at 9:00 am for seven days in January in Cairns were: 28, 30, 27, 32, 31, 29, 33.

(a) Calculate the mean temperature.

_____

_____

(b) The following week, the mean temperature was 31.5°C. In which week was it hotter on average?

_____

6. A line graph of website visitors shows an *upward trend* from Monday to Friday. Which of the following statements best describes this? Circle your answer.

A. The website had more visitors each day as the week went on.

B. The website had fewer visitors each day as the week went on.

C. The number of visitors stayed the same each day.

D. The number of visitors went up and down randomly.

7. Look at the two graphs below. Which type (bar or line) would you use for each situation? Explain your choice.

(a) Showing a school's total enrolments each year from 2015 to 2025.

_____

_____

(b) Comparing the number of students who prefer each of five subjects.

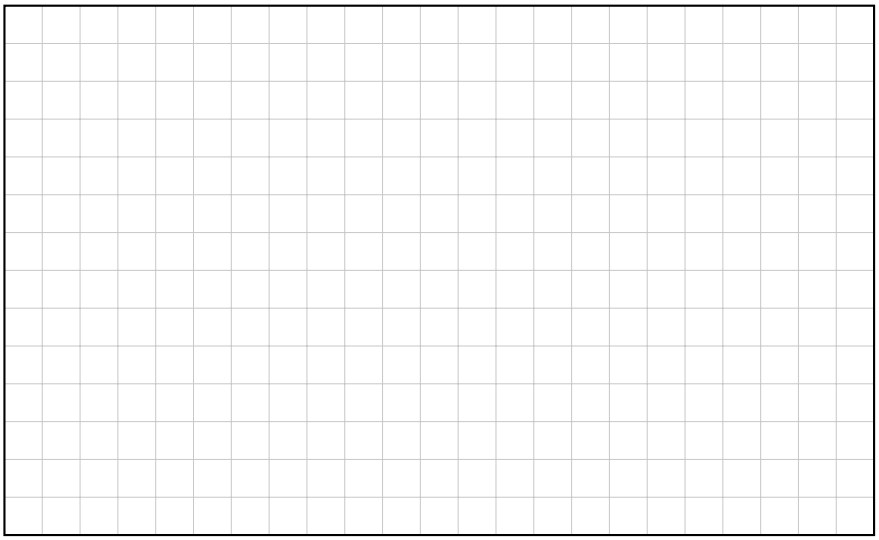
_____

_____

8. The table below shows monthly electricity usage (kWh) for a Queensland home:

Month	Jan	Feb	Mar	Apr	May	Jun
Usage (kWh)	380	360	340	300	280	265
Month	Jul	Aug	Sep	Oct	Nov	Dec
Usage (kWh)	270	285	295	315	345	375

(a) Plot this data as a line graph on the grid below.



(b) Which month had the lowest electricity usage?

(c) Describe the overall trend in electricity usage. Is there a seasonal pattern?

(d) Calculate the mean monthly usage for the year (sum = 3810 kWh).

(e) The household is charged \$0.28 per kWh. What would be the cost of electricity in January?

Multi-Step Problem 9

Springvale Junior Football Club is applying for a community grant. They need to include data showing the club's growth over the past 5 years.

Year	2021	2022	2023	2024	2025
Junior Members	64	78	85	96	110
Senior Members	42	45	48	50	55

(a) Draw a bar graph comparing junior and senior membership for each year. Use two different coloured/shaded bars side by side.

(b) Calculate the mean junior membership over the 5 years.

(c) Calculate the total number of members (junior + senior) in 2025.

(d) The grant application states: *"Our club has grown by more than 50% in junior membership since 2021."* Is this claim correct? Show your working.

(e) In the grant application, write two sentences describing what the data shows about the club's growth. Use mathematical language (e.g. trend, increase, mean, total).

Multi-Step Problem 10

Maya works at a Brisbane weather station. She recorded the daily maximum temperature (°C) for 10 days in February:

32, 35, 30, 28, 33, 36, 34, 29, 31, 32

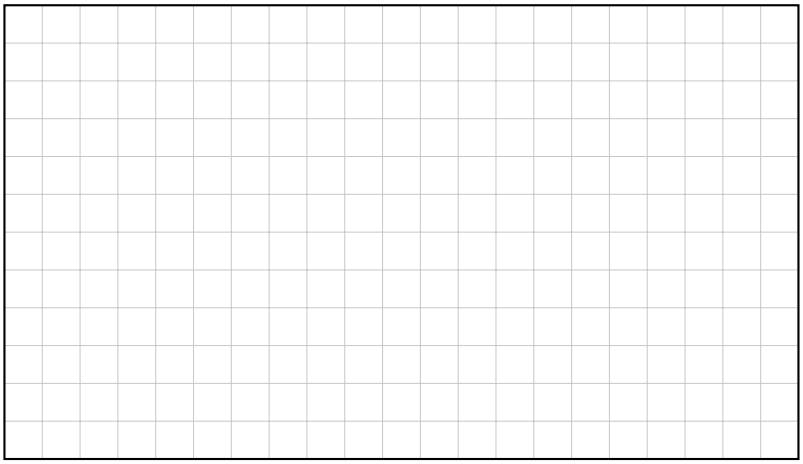
(a) Create a frequency table for the data above. Use groups: 28–30, 31–33, 34–36.

Temperature Group (°C)	Tally	Frequency
28–30		
31–33		
34–36		
Total		

(b) Calculate the mean temperature for the 10 days.

(c) The Bureau of Meteorology's long-term average for February in Brisbane is 29.8°C. Was this February above or below average? By how much?

(d) Plot the 10-day temperatures as a line graph on the grid below (Day 1 to Day 10).



(e) Looking at the line graph, would you describe the temperature pattern as an upward trend, downward trend, or fluctuating? Explain using two specific data points as evidence.

Skills Checklist

Chapter 7 — Skills Checklist

Rate your confidence for each skill. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can collect data using tally marks	⊙	⊖	⊗
I can complete a frequency table correctly	⊙	⊖	⊗
I can read and answer questions from a bar graph	⊙	⊖	⊗
I can draw a bar graph with correct labels and scale	⊙	⊖	⊗
I can read and answer questions from a line graph	⊙	⊖	⊗
I can draw a line graph with correct labels and scale	⊙	⊖	⊗
I can describe trends using words like *increasing*, *decreasing*, *fluctuating*	⊙	⊖	⊗
I can calculate the mean from a set of data	⊙	⊖	⊗
I can explain what data tells us in real-world contexts	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

Chapter 8 — Chance and Probability

Stage: Stage 4 — Student Workbook
Week: Week 8
Lessons: Lessons 29–32
Strand: Topic 1: Statistics & Probability
Sub-topic: Chance

Introduction

Probability is the mathematics of chance. We use it every time we hear a weather forecast (*"70% chance of rain"*), enter a raffle, or decide whether to carry an umbrella. Businesses use probability to manage risk; doctors use it to interpret test results; sports coaches use it to plan tactics.

In this chapter you will describe how likely events are using words and numbers, list all possible outcomes of an experiment, and calculate exact probabilities as fractions, decimals and percentages.

Learning Objectives — by the end of this chapter you will be able to:

- Use probability language to describe likelihood: impossible, unlikely, even chance, likely, certain
- Place events on a probability scale from 0 to 1
- List the sample space (all possible outcomes) of a simple experiment
- Calculate theoretical probability using the formula
- Express probability as a fraction, decimal and percentage
- Compare probabilities to determine which event is more likely
- Interpret probability statements from everyday and media contexts
- Record results from a simple experiment and calculate experimental probability

Key Terms

Term	Definition
Probability	A number between 0 and 1 that describes how likely an event is
Chance	Another word for probability
Outcome	A possible result of an experiment
Event	The specific outcome (or set of outcomes) we are interested in
Sample space	The list of *all* possible outcomes of an experiment
Favourable outcome	An outcome that matches the event we want
Certain	Will definitely happen; probability = 1
Likely	More probable than not; probability $> \frac{1}{2}$
Even chance	Equally likely to happen or not; probability $= \frac{1}{2}$
Unlikely	Less probable than not; probability $< \frac{1}{2}$
Impossible	Cannot happen; probability = 0
Theoretical probability	Expected probability calculated from equally likely outcomes
Experimental probability	Probability calculated from actual trial results
Trial	One run of an experiment (e.g. one coin flip, one dice roll)

Key Formulas and Rules

Chance and Probability — Key Formulas

Theoretical Probability:

$$\Pr(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Experimental Probability:

$$\Pr(E) = \frac{\text{number of times the event occurred}}{\text{total number of trials}}$$

Probability range	Special values
$0 \leq \Pr(E) \leq 1$	$\Pr(\text{impossible}) = 0$
All probabilities lie between 0 and 1	$\Pr(\text{certain}) = 1$
	$\Pr(\text{even chance}) = \frac{1}{2} = 0.5 = 50\%$

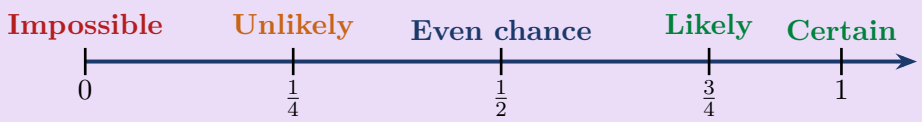
Converting between forms:

$$\textbf{Fraction} \xrightarrow{\div \text{ top by bottom}} \textbf{Decimal}$$

$$\xrightarrow{\times 100} \textbf{Percentage}$$

$$\textbf{Percentage} \xrightarrow{\div 100} \textbf{Decimal}$$

The Probability Scale:



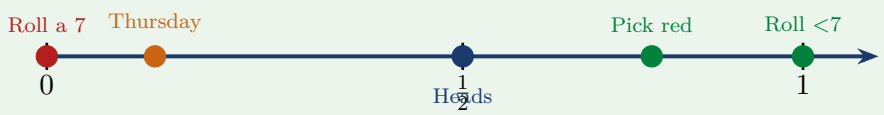
Worked Examples

Worked Example 1 — Probability Language and the Scale

Task: For each event, choose the correct probability word and give a numerical value.

Event	Probability word	Value
You roll a 7 on a standard six-sided die	Impossible	0
You roll a number less than 7 on a standard die	Certain	1
You flip a fair coin and get Heads	Even chance	$\frac{1}{2}$
A bag has 8 red and 2 blue balls; you pick red	Likely	$\frac{8}{10} = \frac{4}{5}$
You randomly pick a Thursday from a standard week	Unlikely	$\frac{1}{7}$

These events plotted on the probability scale:



Worked Example 2 — Listing the Sample Space

Task: List the sample space for each experiment.

Write **every** possible outcome inside curly brackets { }, using a systematic order so you don't miss any. The total count of outcomes becomes the denominator of the probability fraction.

(a) Flipping a fair coin once.

$$S = \{\,\text{Heads},\ \text{Tails}\,\} \qquad (2 \text{ outcomes})$$

(b) Rolling a standard six-sided die.

$$S = \{\,1,\ 2,\ 3,\ 4,\ 5,\ 6\,\} \qquad (6 \text{ outcomes})$$

(c) A spinner with equal sections: Red, Blue, Green, Yellow.

$$S = \{\,\text{Red},\ \text{Blue},\ \text{Green},\ \text{Yellow}\,\} \qquad (4 \text{ outcomes})$$

(d) Randomly choosing one day of the weekend.

$$S = \{\,\text{Saturday},\ \text{Sunday}\,\} \qquad (2 \text{ outcomes})$$

Worked Example 3 — Calculating Theoretical Probability

Situation: A bag contains 3 red marbles, 5 blue marbles and 2 green marbles. One marble is chosen at random.
Find: (a) Pr(red) (b) Pr(blue) (c) Pr(yellow) (d) Pr(not green)

Step 1: Total outcomes $= 3 + 5 + 2 = 10$.
Step 2: Count favourable outcomes for each event.
Step 3: Write the fraction and simplify. Convert to decimal and percentage.

(a) $\Pr(\text{red}) = \dfrac{3}{10} = 0.3 = 30\%$

(b) $\Pr(\text{blue}) = \dfrac{5}{10} = \dfrac{1}{2} = 0.5 = 50\%$

(c) $\Pr(\text{yellow}) = \dfrac{0}{10} = 0$ (no yellow marbles — **impossible**)

(d) "Not green" means red or blue: $3 + 5 = 8$ favourable outcomes.

$$\Pr(\text{not green}) = \frac{8}{10} = \frac{4}{5} = 0.8 = 80\%$$

Check: $\Pr(\text{green}) + \Pr(\text{not green}) = \dfrac{2}{10} + \dfrac{8}{10} = \dfrac{10}{10} = 1$ ✓

Worked Example 4 — Converting and Comparing Probabilities

Task: Order these three probabilities from least likely to most likely.

$$\Pr(A) = \frac{3}{5} \qquad \Pr(B) = 0.45 \qquad \Pr(C) = 72\%$$

Strategy: Convert everything to decimals, then compare.

- $\Pr(A) = \frac{3}{5} = 3 \div 5 = \mathbf{0.60}$
- $\Pr(B) = \mathbf{0.45}$ (already decimal)
- $\Pr(C) = 72\% \div 100 = \mathbf{0.72}$

Comparing: $0.45 < 0.60 < 0.72$

$$\text{Least likely: } B \longrightarrow A \longrightarrow C \text{ :Most likely}$$

Worked Example 5 — Interpreting Probability Statements

Interpret each statement. Is the event impossible, unlikely, about even, likely or certain?

(a) *"The Bureau of Meteorology says there is a 70% chance of rain in Brisbane tomorrow."*
$70\% = 0.7 > 0.5$ **Likely.** It is more probable that it will rain than not. Bring an umbrella.

(b) *"There is a 1 in 500 chance that a product is faulty."*
$\frac{1}{500} = 0.002 = 0.2\%$ **Very unlikely.** Only 2 in every 1000 products would be faulty.

(c) *"A competition has a probability of 0.02 of winning."*
$0.02 = 2\%$ **Unlikely.** Only 2 in every 100 entries would win.

(d) *"A footy team has won 12 of their last 15 games."*
$\Pr(\text{win}) = \frac{12}{15} = \frac{4}{5} = 0.8 = 80\%$ (based on recent form) **Likely** — but this is *experimental* probability from past results, not a guarantee.

Worked Example 6 — Experimental Probability

Situation: Mia flipped a coin 40 times and got Heads 26 times and Tails 14 times.

(a) **Experimental** $\Pr(\text{Heads}) = \frac{26}{40} = \frac{13}{20} = 0.65 = 65\%$

(b) **Theoretical** $\Pr(\text{Heads}) = \frac{1}{2} = 0.5 = 50\%$

(c) **Are the results expected?** The experimental result (65%) differs from the theoretical value (50%). This is normal with only 40 trials — random variation causes differences. If Mia repeated the experiment thousands of times, the experimental probability would get closer and closer to 50%. This is the **Law of Large Numbers**.

Practice Exercise

Practice Exercise — Chance and Probability

Part A — Probability Language

1. Write the probability word that best describes each event: **impossible / unlikely / even chance / likely / certain**.

(a) The sun rises tomorrow. _____

(b) You roll a 9 on a standard six-sided die. _____

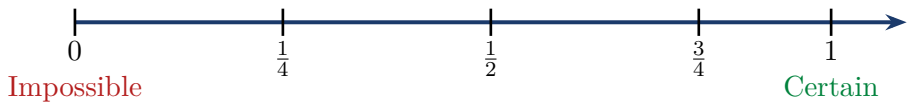
(c) You flip a fair coin and get Tails. _____

(d) A bag has 9 red and 1 blue ball. You pick a red ball. _____

(e) You randomly pick a day in July that is a Monday *and* a Friday. _____

(f) You randomly pick a vowel from the 5 vowels A, E, I, O, U. _____

2. Plot each event from Question 1 on the probability scale. Label each point with its letter (a–f).



Part B — Sample Space

3. List the sample space for each experiment.

(a) Rolling a four-sided die (faces 1–4).

_____

(b) Randomly selecting one day of the school working week.

_____

(c) Choosing a letter at random from the word MATHS.

_____

(d) Randomly picking one month from the four seasons of the year (Summer, Autumn, Winter, Spring).

4. A spinner has 8 equal sections numbered 1 to 8. How many outcomes are in the sample space?

Part C — Calculating Probability

5. A box contains 4 green counters, 6 yellow counters and 2 purple counters. One is chosen at random.

(a) Total number of outcomes:

(b) Pr(green) as a simplified fraction:

(c) Pr(yellow) as a decimal:

(d) Pr(purple) as a percentage:

(e) Pr(orange). Explain your answer.

(f) Pr(not yellow):

6. A standard six-sided die is rolled. Find each probability.

(a) Pr(rolling a 4)

(b) Pr(rolling an even number)

(c) Pr(rolling a number greater than 4)

(d) Pr(rolling a number less than 7). Explain your answer.

Part D — Converting and Comparing

7. Convert each probability to all three forms. Complete the table.

Given value	Fraction	Decimal	Percentage
$\frac{1}{4}$	$\frac{1}{4}$		
0.6		0.6	
35%			35%
$\frac{7}{20}$	$\frac{7}{20}$		
0.08		0.08	

8. Order from least likely to most likely. Convert to decimals and show your working.

$$\Pr(A) = \tfrac{2}{5}, \qquad \Pr(B) = 0.65, \qquad \Pr(C) = 48\%, \qquad \Pr(D) = \tfrac{3}{4}$$

Part E — Experimental Probability

9. A drawing pin was dropped 50 times: point-up 32 times, point-down 18 times.

(a) Experimental Pr(point-up):

(b) Experimental Pr(point-down):

(c) Do these two probabilities add to 1? Why must they?

10. James rolled a die 60 times and got a 6 exactly 8 times.

(a) Experimental Pr(rolling a 6):

(b) Theoretical Pr(rolling a 6):

(c) Is the experimental result close to the theoretical probability? Explain.

Application Exercise

Application Exercise — Probability in Australian Contexts

1. *(Weather)* The Bureau of Meteorology forecast probabilities of rain in Townsville for one week:

Day	Mon	Tue	Wed	Thu	Fri	Sat
Pr(rain)	20%	45%	60%	80%	55%	10%

(a) Which day has the highest probability of rain?

(b) On which days is rain more likely than not?

(c) Express Wednesday's probability as a fraction in simplest form.

(d) Mara plans an outdoor party on Saturday. Is rain likely or unlikely? Use the probability to support your answer.

2. *(Raffle)* A school raffle sells 200 tickets numbered 1 to 200. One ticket wins.

(a) Lachlan buys 5 tickets. What is his probability of winning? (fraction and percentage)

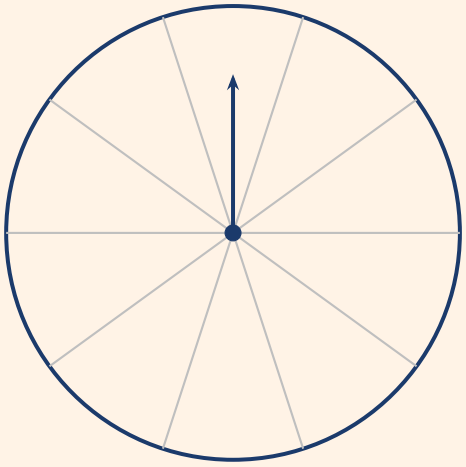
(b) Sofia buys 20 tickets. What is her probability of winning? (fraction and percentage)

(c) Who has the better chance? By how much (as a percentage difference)?

(d) Tickets cost \$2 each. How much did Sofia spend?

3. *(Spinner)* A game at a school fete uses a spinner with 10 equal sections:
4 sections Red, 3 sections Blue, 2 sections Green, 1 section Gold.
Players win a prize only for landing on Gold.

(a) In the circle below, draw and label the 10 sections of the spinner.



(b) Pr(Gold) as a fraction, decimal and percentage:

(c) Pr(Red):

(d) Pr(not Blue):

(e) If the spinner is spun 200 times, how many times would you *expect* it to land on Green?

4. *(Playing cards)* A standard deck has 52 cards: 4 suits (Hearts, Diamonds, Clubs, Spades) with 13 cards each. Hearts and Diamonds are red; Clubs and Spades are black.

(a) Pr(drawing a Heart):

(b) Pr(drawing a King):

(c) Pr(drawing a red card):

(d) Pr(drawing the Ace of Spades):

(e) Is it more likely to draw a red card or a Club? Show your working and explain.

5. *(Quality control)* A Queensland honey factory produces 400 jars per hour. On average, 6 jars per hour have faulty lids.

(a) Pr(faulty lid) as a fraction and percentage:

(b) How many faulty jars are expected in an 8-hour shift?

(c) The factory targets fewer than 2% faulty. Is the current rate within target?

_____

_____

6. *(Experimental probability)* Chloe flipped a coin 20 times (Heads: 14). Tom flipped the same coin 200 times (Heads: 128).

(a) Experimental Pr(Heads) from Chloe's results:

_____

_____

(b) Experimental Pr(Heads) from Tom's results:

_____

_____

(c) Whose results give a better estimate of the true probability? Explain why.

_____

_____

(d) Based on Tom's results, is the coin fair? Justify your answer with probability values.

_____

_____

_____

Chapter Review

Chapter 8 Review — Chance and Probability

1. Match each event with the correct probability word by writing the word next to the event.

Word bank: impossible unlikely even chance likely certain

(a) Rolling a 3 on a standard six-sided die. _____

(b) Picking a red ball from a bag containing only red balls. _____

(c) Flipping a fair coin and getting Tails. _____

(d) It snowing in Darwin in the middle of summer. _____

(e) A bag has 7 red and 3 blue balls. Picking red. _____

2. A bag contains 2 red, 4 blue, 3 green and 1 yellow marble.

(a) Total number of marbles:

_____

(b) Pr(blue) as a simplified fraction:

_____

_____

(c) Pr(red) as a decimal:

_____

_____

(d) Pr(green) as a percentage:

_____

_____

(e) Pr(not red) as a fraction:

_____

_____

3. A spinner has 5 equal sections numbered 1 to 5. List the sample space and find:

(a) Sample space:

_____

(b) Pr(spinning a 3):

_____

(c) Pr(spinning an even number):

(d) Pr(spinning a number greater than 2):

4. Convert to decimals then order from least likely to most likely.

$$45\%, \quad \frac{1}{3}, \quad 0.70, \quad \frac{9}{10}, \quad 20\%$$

5. A weather app says there is a **0.35** probability of a thunderstorm tomorrow.

(a) Write this probability as a percentage.

(b) Is a thunderstorm likely, unlikely or an even chance? Explain.

(c) What is the probability there will *not* be a thunderstorm? Show how you calculated this.

6. A student rolled a die 30 times with these results:

Number rolled	1	2	3	4	5	6
Frequency	6	3	7	4	5	5

(a) Experimental Pr(rolling a 3):

(b) Theoretical Pr(rolling a 3):

(c) Experimental Pr(rolling a number greater than 4):

(d) How many times would you *expect* to roll a 1 in 30 rolls (theoretical)?

7. A bag of mixed lollies contains: 6 snakes, 4 bears, 5 frogs and 5 worms. One lolly is picked at random.

(a) Pr(snake or worm):

(b) Pr(not a bear):

(c) Is picking a frog more or less likely than picking a bear? Show working.

Multi-Step Problem 8

Springvale Primary runs a lucky dip at the school fair. The box contains:

- 10 small prizes (pencil cases)

- 6 medium prizes (book vouchers)
- 3 large prizes (board games)
- 1 grand prize (tablet computer)

Tickets cost $3 each. Each ticket drawn wins a prize, which is then removed.

(a) How many prizes are in the box?

(b) At the start, Pr(grand prize) as a fraction and percentage (1 d.p.):

(c) Pr(large prize) at the start:

(d) Pr(small or medium prize) at the start:

(e) After 5 small prizes and 2 medium prizes are drawn, how many prizes remain?

(f) What is the new Pr(grand prize)? Has it increased or decreased? Explain why.

(g) If 200 tickets are sold, how much money does the school raise?

Multi-Step Problem 9

Experimental Investigation. Three students each flip the same thumbtack and record results:

Student	Trials	Point-Up	Point-Down
Aiden	10	6	4
Bella	50	28	22
Carlos	200	118	82

(a) Calculate the experimental Pr(point-up) for each student (decimal, 2 d.p.).

Student	Experimental Pr(point-up)
Aiden	
Bella	
Carlos	

(b) Whose result is most reliable? Why?

(c) Based on Carlos's data, estimate Pr(point-up). Is it closer to $\frac{1}{2}$ or $\frac{3}{5}$?

(d) Using Carlos's estimate, how many times would you expect point-up in 500 flips?

(e) A fair coin gives equal chance of heads/tails. Does this thumbtack behave like a fair coin? Use the data as evidence.

Skills Checklist

Chapter 8 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can use probability words (impossible → certain)	⊙	⊖	⊗
I can place events on a probability scale 0 to 1	⊙	⊖	⊗
I can list the sample space of a simple experiment	⊙	⊖	⊗
I can calculate $\Pr(E)$ using the formula	⊙	⊖	⊗
I can express probability as a fraction, decimal and %	⊙	⊖	⊗
I can compare probabilities to find the most likely event	⊙	⊖	⊗
I can interpret probability statements in everyday life	⊙	⊖	⊗
I can calculate experimental probability from trial results	⊙	⊖	⊗
I can explain the difference between theoretical and experimental probability	⊙	⊖	⊗

My goal before the next lesson:

Assessment Task 1: Project

Your Task

Students respond to a real-world scenario related to **personal identity and community** using numeracy skills from Topic 1. The scenario booklet will be provided by your teacher. Your response must clearly show your mathematical thinking, all working and calculations, and a written conclusion.

Detail	Information
Title	Assessment Task 1 — Individual Project
Topic	Topic 1: Personal Identity and Community
Type	Individual project (Option A)
When	Completed during Week 9 (Lessons 33–36) with additional personal time as needed
Submit	Submit all planning pages, working, and your final response to your teacher

Important: All work must be your own. You may use your workbook, notes and a calculator. You may not get another person to complete any part of your response.

Response Format Options

Choose **one** of the following formats for your response. Circle your choice below.

Format	Requirements
Written response	Up to 600 words, with all calculations shown clearly
Multimodal presentation	Up to 6 A4 pages or 5 minutes in length; may include diagrams, graphs and text
Spoken presentation	Up to 4 minutes; supported by notes or visual aids

My chosen format: ☐ Written response ☐ Multimodal presentation ☐ Spoken presentation

Numeracy Skills You May Use

Skills from Topic 1 (Chapters 1–8)

Your response may draw on any of the following skills. Use those that are relevant to your scenario.

- ☐ Whole numbers and operations (Ch. 1)
- ☐ Fractions, decimals and percentages (Ch. 2)
- ☐ Rates, ratios and comparisons (Ch. 3)
- ☐ Length, perimeter and area (Ch. 4)
- ☐ 2D shapes, area and maps (Ch. 5)
- ☐ Time, dates and timetables (Ch. 6)
- ☐ Data, tables and graphs (Ch. 7)
- ☐ Probability and chance (Ch. 8)

Tick the skills you actually used in your response.

Assessment Criteria

Grade	Descriptor
A	Demonstrates **comprehensive** understanding of numeracy skills; selects appropriate and efficient strategies; communicates mathematical thinking clearly and with precision
B	Demonstrates **thorough** understanding of numeracy skills; selects appropriate strategies; communicates mathematical thinking clearly
C	Demonstrates understanding of **most** numeracy skills; uses suitable strategies for most tasks; communicates mathematical thinking adequately
D	Demonstrates **some** understanding of numeracy skills; uses strategies with partial success; communicates some mathematical thinking
E	Demonstrates **limited** understanding of numeracy skills; attempts to use strategies with limited success; communicates limited mathematical thinking

Project Planning

Use the planning pages below **before** you begin writing your final response. Your teacher may collect these pages as part of your submission.

Project Planning — Page 1: Understanding the Task

What is the scenario asking me to do?

_____

_____

_____

_____

What mathematical information is given to me in the scenario?

_____

_____

_____

_____

_____

What information do I need to find or calculate?

_____

_____

_____

Project Planning — Page 2: Choosing My Approach

Which numeracy skills will I use? (List them)

_____

_____

_____

What calculations do I need to do? (List each one)

_____

_____

_____

_____

What format will I use for my response, and why?

_____

_____

_____

Project Planning — Page 3: Working Space

Show all your calculations and working below. Label each calculation clearly.

Draft Response (optional — plan here before writing your final version)

Self-Checklist Before Submitting

Pre-Submission Checklist

Check each item before handing in your project. You should be able to tick every box.

☐ I have read the scenario carefully and answered what was asked
☐ I have shown all my calculations clearly with correct units
☐ I have checked my calculations for errors
☐ My response is clearly presented in my chosen format
☐ I have written a conclusion that answers the scenario question
☐ I have included my planning pages (Pages 1, 2, and 3) in my submission
☐ My name is on every page

My response is: ☐ Written ☐ Multimodal ☐ Spoken

I have used the following chapters from my workbook:

☐ Ch. 1 ☐ Ch. 2 ☐ Ch. 3 ☐ Ch. 4 ☐ Ch. 5 ☐ Ch. 6 ☐ Ch. 7 ☐ Ch. 8

After your project is marked: Your teacher will return your project with written feedback. Read the feedback carefully and use the space below to note what you did well and what you will improve in Assessment Task 2.

After Your Project: Reflection

What I did well in this project:

_____

_____

_____

What I will focus on improving before Assessment Task 2:

_____

_____

_____

My grade: _____ **Teacher comment:**

_____

_____

TOPIC TWO

Workplace and Employment

Chapters 9–15 | Weeks 10–18

In this topic you will develop numeracy skills in the context of Australian workplaces and employment. You will apply mathematical thinking to pay calculations, measurement, data analysis, quality control and trend interpretation in realistic workplace scenarios.

CHAPTERS IN THIS TOPIC

Ch 9: Numbers at Work
Ch 10: Workplace Numbers
Ch 11: 3D Shapes and Volume
Ch 12: Mass, Capacity and Temperature
Ch 13: Workplace Measurement
Ch 14: Data at Work
Ch 15: Probability and Trends at Work

Assessment Task 2: Examination Preparation

Chapter 9 — Numbers at Work

Stage: Stage 4 — Student Workbook
Week: Week 10
Lessons: Lessons 37–40
Strand: Topic 2: Workplace and Employment
Sub-topic: Number & Algebra — Pay, Timesheets and Pay Slips

Introduction

Welcome to Topic 2. In this topic, all of the numeracy skills you have developed in Topic 1 are applied to the world of work. Whether you are working in a café, a construction site, a hospital or a retail store, you will regularly use numbers — to calculate your pay, check your hours, read invoices and manage a budget.

This chapter focuses on the most fundamental workplace numeracy skill: understanding how much you are paid. You will learn to read a roster and timesheet, calculate your gross pay, and interpret a pay slip.

Learning Objectives — by the end of this chapter you will be able to:

- Identify numeracy skills used in a range of workplaces
- Read and interpret a pay rate expressed as dollars per hour
- Calculate gross pay for a single shift and for a full week
- Complete a timesheet and calculate total hours worked
- Read and interpret a pay slip, identifying gross pay, deductions and net pay
- Calculate fortnightly pay from weekly pay
- Use a paper spreadsheet table to organise and total workplace data

Key Terms

Term	Definition
Pay rate / Hourly rate	The amount paid for each hour of work (e.g. $24.50 per hour)
Gross pay	Total pay *before* any deductions are taken out
Net pay / Take-home pay	Pay *after* deductions (what actually goes into your bank account)
Deduction	An amount subtracted from gross pay (e.g. income tax, superannuation)
Income tax (PAYG)	Tax withheld by the employer and sent to the ATO on the employee's behalf
Superannuation (Super)	A retirement savings contribution — currently 11.5% of gross pay (2025)
Timesheet	A record of hours worked each day, used to calculate pay
Pay slip	A document from your employer showing your pay details for a period
Roster	A schedule showing which employees work on which days and times
Casual	An employee with no fixed hours; paid a higher rate, no paid leave
Part-time	An employee with regular fixed hours fewer than full-time
Full-time	An employee working approximately 38 hours per week
Fortnight	A period of two weeks (14 days)
Spreadsheet	A grid of rows and columns used to organise and calculate data
SUM function	A spreadsheet formula that adds a range of cells (e.g. `=SUM(B2:B6)`)

Key Formulas and Rules

Numbers at Work — Key Formulas

Gross pay (single shift):

$$\text{Gross pay} = \text{hours worked} \times \text{hourly rate}$$

Weekly gross pay:

$$\text{Weekly pay} = \sum (\text{pay for each shift})$$

Fortnightly pay:

$$\text{Fortnightly pay} = 2 \times \text{weekly pay}$$

Net pay:

$$\text{Net pay} = \text{Gross pay} - \text{Total deductions}$$

Typical deductions (2025)	Spreadsheet basics
Income tax (PAYG): varies by income	Cells named by column + row (e.g. A1, B3)
Superannuation: 11.5% of gross pay	`=SUM(B2:B6)` adds cells B2 to B6
Union fees, salary sacrifice: optional	`=B2*C2` multiplies cell B2 by C2

Note on Australian pay rates (2025): The National Minimum Wage from 1 July 2024 is **$24.10/hour** for adult employees. Casual workers receive a 25% loading on top of their base rate. Many industries have higher award rates — always check the relevant Modern Award for your industry.

Worked Examples

Worked Example 1 — Calculating Gross Pay for a Single Shift

Situation: Emma works a casual shift at a Brisbane café. She works for 4.5 hours and is paid \$26.80 per hour.
Calculate Emma's gross pay for the shift.

Step 1: Identify the hours worked and the hourly rate.
Hours worked = 4.5 hours Hourly rate = \$26.80/hr

Step 2: Apply the formula.
Gross pay = hours worked × hourly rate

Step 3: Calculate.

$$\text{Gross pay} = 4.5 \times \$26.80 = \$120.60$$

Emma earns \$120.60 for this shift.

Quick check: $4 \times \$27 = \$108$ and $5 \times \$27 = \$135$, so \$120.60 is a reasonable answer for 4.5 hours. ✓

Worked Example 2 — Completing a Timesheet

Situation: Jake works part-time at a Woolworths supermarket. His roster for the week is shown below. Some columns have been filled in — complete the rest.

Springvale Woolworths — Weekly Timesheet

Day	Date	Start	Finish	Break (mins)	Hours worked
Mon	14 Apr	14:00	18:00	0	4 h 00 min
Tue	15 Apr	—	—	—	*not rostered*
Wed	16 Apr	09:00	13:30	30	_____
Thu	17 Apr	—	—	—	*not rostered*
Fri	18 Apr	15:00	19:00	0	_____
Sat	19 Apr	08:00	14:00	30	_____
Sun	20 Apr	—	—	—	*not rostered*
				Total hours worked:	_____

Employee signature: _____ Manager signature: _____

To calculate hours worked:
Hours = Finish time − Start time − Break time (convert break to hours or minutes as needed)

Completing the timesheet:

- **Wednesday:** 13:30 − 09:00 = 4 h 30 min; minus 30 min break = **4 h 00 min**
- **Friday:** 19:00 − 15:00 = 4 h 00 min; no break = **4 h 00 min**
- **Saturday:** 14:00 − 08:00 = 6 h 00 min; minus 30 min break = **5 h 30 min**
- **Total:** $4 + 4 + 4 + 5.5 =$ **17 h 30 min**

Worked Example 3 — Weekly Gross Pay from a Timesheet

Situation: Jake (from Worked Example 2) earns \$24.80/hour on weekdays and \$31.00/hour on Saturday (penalty rate).
Calculate Jake's total gross pay for the week.

Step 1: Separate weekday and weekend hours.
Step 2: Multiply each group by its rate.
Step 3: Add the results.

Day(s)	Hours	Rate	Pay
Monday (weekday)	4.0	\$24.80	$4.0 \times \$24.80 = \$99.20$
Wednesday (weekday)	4.0	\$24.80	$4.0 \times \$24.80 = \$99.20$
Friday (weekday)	4.0	\$24.80	$4.0 \times \$24.80 = \$99.20$
Saturday (weekend)	5.5	\$31.00	$5.5 \times \$31.00 = \$170.50$
		Total gross pay:	**\$468.10**

Jake's total gross pay for the week is \$468.10.

Worked Example 4 — Reading a Pay Slip

Situation: Jake receives the following pay slip. Read it carefully and answer the questions below.

Pay Slip — Springvale Woolworths

Employee:	Jake Nguyen	**Pay period:**	14 Apr – 20 Apr 2025
Employee ID:	WW-3847	**Pay date:**	24 Apr 2025
Position:	Casual, Grocery	**Payment method:**	Direct deposit

Earnings	**Hours**	**Amount**
Ordinary hours (weekday @ $24.80)	12.0 h	$297.60
Saturday penalty rate (@ $31.00)	5.5 h	$170.50
Gross Pay	**17.5 h**	**$468.10**
Deductions		
Income tax (PAYG withheld)		$42.00
Superannuation (11.5%)		_____
Total Deductions		_____
Net Pay (take-home pay)		_____

Superannuation is paid separately into your super fund and does not reduce your net pay.

Questions:

(a) What is Jake's gross pay? **$468.10**

(b) Calculate Jake's superannuation contribution (11.5% of gross pay).

$$11.5\% \times \$468.10 = 0.115 \times \$468.10 = \$53.83$$

(c) The only deduction from his take-home pay is PAYG tax. Calculate net pay.

$$\text{Net pay} = \$468.10 - \$42.00 = \$426.10$$

(d) What would Jake's fortnightly pay be (assuming the same hours each week)?

$$\text{Fortnightly} = 2 \times \$468.10 = \$936.20$$

Note: Super is paid by the employer on top of gross pay; it does not come out of Jake's take-home pay.

Worked Example 5 — Paper Spreadsheet (Payroll Table)

Situation: A small cafe employs four casual workers one Saturday. The manager uses a paper payroll table to calculate everyone's pay. Each worker earns $28.50/hour.
Complete the spreadsheet table below.

Row	A: Employee	B: Hours	C: Rate ($/hr)	D: Gross Pay
2	Priya	5.0	28.50	`=B2*C2` = $142.50
3	Marcus	6.5	28.50	`=B3*C3` = $185.25
4	Lily	4.0	28.50	`=B4*C4` = $114.00
5	Tom	7.0	28.50	`=B5*C5` = $199.50
6	**Total** `=SUM(B2:B5)` = 22.5 h			`=SUM(D2:D5)`

Total gross pay: $142.50 + $185.25 + $114.00 + $199.50 = **$641.25**

Key point: On paper, you calculate each cell value manually. A real spreadsheet program would calculate automatically when you enter the formula. The logic (multiply hours by rate; then SUM the column) is identical.

Practice Exercise

Practice Exercise — Numbers at Work

Part A — Single Shift Pay

1. Calculate the gross pay for each shift.

	Employee	Hours worked	Hourly rate	Gross pay
(a)	Aisha (retail casual)	3 h	$27.40/hr	
(b)	Ben (hospitality)	5 h	$26.80/hr	
(c)	Chloe (childcare)	6.5 h	$29.20/hr	
(d)	Dan (construction)	8 h	$34.50/hr	
(e)	Eva (part-time admin)	4.5 h	$24.10/hr	

2. Maya works 7 hours on Friday at $25.60/hr. Calculate her gross pay.

_____

_____

3. Liam works 3.5 hours on a public holiday and receives $49.20/hr (double time + penalty).

Calculate his gross pay.

Part B — Timesheet

4. Calculate the hours worked for each shift (subtract break from total shift time).

Day	Start	Finish	Break	Hours worked
Monday	08:00	13:00	30 min	
Tuesday	12:00	17:30	30 min	
Wednesday	09:00	14:00	0 min	
Thursday	15:00	21:00	30 min	
Friday	08:30	13:30	0 min	
			Total hours:	

5. The employee in Question 4 earns $24.80/hr. Calculate their total gross pay for the week.

6. What would their fortnightly pay be if they worked the same hours each week?

Part C — Completing a Timesheet

7. Complete the timesheet below for Zara, who works at a Springvale bakery.

Springvale Bakery — Weekly Timesheet Employee: **Zara Mitchell**

Day	Date	Start	Finish	Break (mins)	Hours worked
Mon	7 Jul	06:00	12:00	30	
Tue	8 Jul	—	—	—	not rostered
Wed	9 Jul	06:00	10:30	0	
Thu	10 Jul	06:00	12:00	30	
Fri	11 Jul	—	—	—	not rostered
Sat	12 Jul	06:00	11:00	30	
Sun	13 Jul	—	—	—	not rostered
				Total hours:	

Employee signature: _____ Manager: _____

Part D — Pay Slips

8. Use the pay slip below to answer the questions.

Pay Slip — Springvale Hardware

Employee: Rosa Gonzalez **Pay period:** 3 Feb – 9 Feb 2025
Position: Casual Sales **Pay rate:** $27.60/hr

Earnings	Hours	Amount
Ordinary hours	14.0 h	$386.40
Gross Pay	**14.0 h**	**$386.40**
Income tax (PAYG withheld)		$28.00
Total Deductions		$28.00
Net Pay		$358.40

(a) What is Rosa's gross pay for the week?

_____

(b) What is her net (take-home) pay?

(c) How much tax was withheld?

(d) Calculate Rosa's superannuation for the week (11.5% of gross pay).

(e) What would Rosa's fortnightly gross pay be if she worked the same hours each fortnight?

Part E — Spreadsheet Table

9. Complete the paper payroll spreadsheet below. Each employee earns $26.40/hr.

Row	A: Employee	B: Hours	C: Rate	D: Gross Pay
2	Hamish	4.5	$26.40	
3	Jess	6.0	$26.40	
4	Kobi	3.5	$26.40	
5	Layla	5.0	$26.40	
6	**Total (SUM)**			

10. Using your answers from Question 9, what is the total gross pay for all four employees?

Application Exercise

Application Exercise — Workplace Pay Scenarios

1. *(Retail)* Callum works at a Brisbane JB Hi-Fi store. He is a casual employee earning \$28.20/hr on weekdays and \$35.25/hr on Saturdays (time and a quarter). His roster this week is:

Day	Shift	Hours (after breaks)
Tuesday	13:00–18:00	5.0 h
Thursday	10:00–15:00	5.0 h
Saturday	09:00–15:30	6.0 h (after 30 min break)

(a) Calculate Callum's pay for each shift. Show your working.

(b) Calculate his total gross pay for the week.

(c) Tax withheld is \$52.00. What is his net pay?

(d) What is his fortnightly gross pay if he works the same hours each week?

2. *(Comparing job offers)* Mia is choosing between two part-time jobs:

	Job A — Coles checkout	Job B — Café barista
Hourly rate	\$24.80/hr	\$26.40/hr
Hours per week	20 h	16 h

(a) Calculate the weekly gross pay for each job.

_____

_____

_____

(b) Which job pays more per week? By how much?

_____

_____

(c) Which job has the higher hourly rate?

_____

(d) Mia says: *"I should take the café job because it pays more per hour."* Do you agree? Explain using the weekly pay figures.

_____

_____

_____

3. *(Hospitality)* Noah works as a casual kitchen hand at a Gold Coast resort. His timesheet for the week shows:

Day	Start	Finish	Break	Hours worked
Monday	11:00	15:00	0 min	_____
Wednesday	17:00	22:30	30 min	_____
Friday	17:00	23:00	30 min	_____
Saturday	12:00	18:00	30 min	_____
Sunday	11:00	16:00	0 min	_____
			Total hours:	_____

Noah earns \$26.80/hr on weekdays and \$33.50/hr on weekends (Sat and Sun).

(a) Complete the timesheet above.

(b) Calculate Noah's weekday gross pay (Mon, Wed, Fri).

(c) Calculate Noah's weekend gross pay (Sat, Sun).

(d) What is his total gross pay for the week?

4. *(Construction)* A construction company employs a crew of three labourers for a week. They all earn $36.80/hr. Their hours are: Riley: 38 h; Sam: 32 h; Taylor: 40 h.

(a) Complete the paper payroll spreadsheet below.

Row	A: Employee	B: Hours	C: Rate	D: Gross Pay
2	Riley	38	$36.80	
3	Sam	32	$36.80	
4	Taylor	40	$36.80	
5	**Total**			

(b) What is the total gross pay for the crew this week?

(c) What is the mean (average) gross pay per worker?

5. *(Healthcare)* Priya works as a casual aged-care support worker. Her pay slip shows gross pay of $742.80 for the fortnight.

(a) What is her weekly gross pay?

(b) Priya earns $29.10/hr. How many hours did she work in the fortnight?

(c) Tax withheld for the fortnight is $68.00. What is her fortnightly net pay?

6. *(Interpreting a payroll table)* The table below shows a week's payroll at Springvale Nursery.

Employee	Hours	Rate	Gross Pay	Tax
Finn	15	$24.80	$372.00	$30.00
Georgia	20	$27.60	$552.00	$52.00
Harry	12.5	$24.80	$310.00	$22.00
Isla	25	$27.60	$690.00	$70.00
Total				

(a) Complete the Totals row.

(b) Which employee had the highest gross pay this week?

(c) What is Finn's net pay after tax?

(d) What is the total tax withheld across all employees?

(e) Explain why Georgia and Isla earn a higher rate than Finn and Harry.

Chapter Review

Chapter 9 Review — Numbers at Work

1. Calculate the gross pay for each scenario.

(a) 6 hours at $25.40/hr.

(b) 4.5 hours at $32.00/hr.

(c) 8 hours at $24.10/hr (national minimum wage).

2. Calculate the hours worked for each shift.

Start	Finish	Break	Hours worked
07:00	13:30	30 min	
14:00	20:00	0 min	
09:30	16:00	30 min	

3. Sam works the following shifts at a Brisbane café this week. His rate is $27.20/hr on weekdays and $34.00/hr on Sunday.

Day	Hours (after break)	Rate	Pay
Tuesday	5.0 h	$27.20	
Thursday	4.5 h	$27.20	
Saturday	6.0 h	$27.20	
Sunday	5.0 h	$34.00	
		Total gross pay:	

4. Sam's tax for the week is $56.00. What is his net (take-home) pay?

5. What is Sam's fortnightly gross pay if he works the same hours each week?

6. Calculate Sam's superannuation for the week (11.5% of gross pay).

7. A pay slip shows: Gross pay = $534.00, Tax withheld = $47.00. What is the net pay?

8. An employee earns $31.50/hr. How many hours did they work if their gross pay was $504.00?

9. Explain in your own words the difference between gross pay and net pay.

Multi-Step Problem 10

Ava's Week at Work. Ava is a Year 10 student working a casual job at a local pharmacy after school and on weekends.

Her pay rates: weekday = \$24.80/hr; Saturday = \$31.00/hr; Sunday = \$37.20/hr (double time).

Springvale Pharmacy — Weekly Timesheet Employee: **Ava Chen**

Day	Start	Finish	Break	Hours worked
Mon	15:30	19:00	0 min	
Tue	—	—	—	not rostered
Wed	15:30	19:00	0 min	
Thu	—	—	—	not rostered
Fri	15:30	19:30	0 min	
Sat	09:00	14:00	30 min	
Sun	10:00	14:00	0 min	
			Total hours:	

Employee signature: _____ Manager: _____

(a) Complete the timesheet. Calculate hours worked each day and the total.

(b) Calculate Ava's pay for each group of days. Show your working.

(c) What is Ava's total gross pay for the week?

(d) Tax withheld is \$38.00. Calculate Ava's net pay.

(e) If Ava works the same roster every week, what will her annual gross pay be? (There are 52 weeks in a year.)

(f) Ava wants to save 30% of her weekly net pay. How much will she save per week?

(g) After 10 weeks of saving 30% of her net pay, will Ava have enough to buy a $500 laptop? Show your calculation.

Multi-Step Problem 11

Team Payroll. The manager at Springvale Café needs to calculate the weekly payroll for three employees. Complete the payroll table and answer the questions.

Springvale Café — Weekly Payroll Summary

Employee	Wkday hrs	Sat hrs	Wkday pay	Sat pay	Gross pay
Ben	16	5			
Cleo	20	6			
Dom	12	0			
				Total gross pay:	

Weekday rate: $26.80/hr Saturday rate: $33.50/hr

(a) Complete the payroll table.

(b) The café must also pay superannuation (11.5% of each worker's gross pay). Calculate the super contribution for each worker and the total.

(c) What is the total weekly cost to the café (gross pay + total super)?

(d) Cleo is offered the opportunity to work 4 extra weekday hours next week at the same rate. How much extra would she earn?

(e) Dom asks for a pay rise to $28.50/hr. If approved, how much more would Dom earn in a 12-week weekday roster (12 h/week)?

Skills Checklist

Chapter 9 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can calculate gross pay for a single shift	⊙	⊖	⊗
I can calculate hours worked from start and finish times	⊙	⊖	⊗
I can complete a timesheet including breaks	⊙	⊖	⊗
I can calculate total weekly gross pay from a timesheet	⊙	⊖	⊗
I can calculate fortnightly pay from weekly pay	⊙	⊖	⊗
I can read a pay slip and identify gross pay, tax and net pay	⊙	⊖	⊗
I can calculate superannuation (11.5% of gross pay)	⊙	⊖	⊗
I can complete a paper payroll spreadsheet table	⊙	⊖	⊗
I can compare two job offers using weekly gross pay	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

Chapter 10 — Workplace Numbers

Stage: Stage 4 — Student Workbook
Week: Week 11
Lessons: Lessons 41–44
Strand: Topic 2: Workplace and Employment
Sub-topic: Number & Algebra — GST, Ratios, Invoices and Budgets

Introduction

Numbers appear constantly in the workplace — on price tags, invoices, supply orders and budgets. Whether you are working in a trade, a retail store, a kitchen or an office, you need to calculate GST, work with mixing ratios, apply discounts, and stay within a budget.

This chapter brings together Number and Algebra skills from Topic 1 and applies them directly to real workplace situations. The skills you practise here appear regularly in the QCAA Numeracy examination.

Learning Objectives — by the end of this chapter you will be able to:

- Mix and scale quantities using ratios in workplace contexts
- Calculate GST (10%) on a price and find the total including GST
- Find the pre-GST price from a GST-inclusive total
- Apply percentage markups and discounts in retail and business contexts
- Complete a workplace invoice including line items, subtotal, GST and total
- Build a purchase list and check it stays within a given budget

Key Terms

Term	Definition
GST (Goods and Services Tax)	A 10% tax added to most goods and services sold in Australia
Price excluding GST	The price before GST is added (the base price)
Price including GST	The price after GST has been added (what the customer pays)
Markup	An increase in price above the cost price (to make a profit)
Markdown / Discount	A reduction in the selling price
Invoice	A document from a supplier listing items sold and amounts owed
Subtotal	The total of all items before GST is added
Budget	A plan for spending that sets a maximum total amount
Purchase list	A list of items to buy with quantities and unit prices
Ratio	A comparison showing how to mix two or more quantities
Concentrate	A strong mixture intended to be diluted before use
Profit	Income minus costs (what is left over after expenses)

Key Formulas and Rules

Workplace Numbers — Key Formulas

GST calculations:

$$\text{GST amount} = \text{price} \times 0.10$$
$$\text{Price incl. GST} = \text{price} \times 1.10$$
$$\text{Price excl. GST} = \text{total} \div 1.10$$

Markup and discount:

$$\text{New price (markup)} = \text{original} \times (1 + r)$$
$$\text{Sale price (disc.)} = \text{original} \times (1 - r)$$

where r is the rate as a decimal.

Side-by-side GST summary:

Ratio mixing:

$$\text{Total parts} = \text{part A} + \text{part B} + \ldots$$

$$\text{Quantity per part} = \frac{\text{total volume}}{\text{total parts}}$$

$$\text{Each ingredient} = \text{parts} \times \text{qty per part}$$

Scaling a ratio:

Find the multiplier: $\text{multiplier} = \frac{\text{new total}}{\text{original total}}$

Multiply every part by the multiplier.

Adding GST (excl. → incl.)	Removing GST (incl. → excl.)
$80 × 1.10 = $88	$88 ÷ 1.10 = $80
$150 × 1.10 = $165	$165 ÷ 1.10 = $150
$245 × 1.10 = $269.50	$269.50 ÷ 1.10 = $245

Worked Examples

Worked Example 1 — Mixing a Solution Using a Ratio

Situation: A cleaner at a Brisbane office mixes a disinfectant solution using a ratio of **1 part concentrate : 4 parts water**. She needs to make **10 litres** of solution in total.
How many litres of concentrate and how many litres of water does she need?

Step 1: Find the total number of parts: $1 + 4 = 5$ parts total.
Step 2: Find the quantity per part: $10 \div 5 = 2$ litres per part.
Step 3: Multiply each ingredient by the quantity per part.

- Concentrate: $1 \times 2 =$ **2 litres**
- Water: $4 \times 2 =$ **8 litres**

Check: $2 + 8 = 10$ litres ✓ Ratio $2 : 8 = 1 : 4$ ✓

Worked Example 2 — Scaling a Workplace Ratio

Situation: A concreting team uses a cement mix ratio of **1 part cement : 2 parts sand : 4 parts gravel**. They originally planned to mix **7 bags** ($1 + 2 + 4 = 7$ parts for a small pour), but now need to scale up to **35 bags** total.
How many bags of each ingredient do they need for the larger pour?

Step 1: Find the multiplier: multiplier $= 35 \div 7 = 5$
Step 2: Multiply each part of the original ratio by the multiplier.

Ingredient	Original parts	Multiplier	New quantity
Cement	1	×5	5 bags
Sand	2	×5	10 bags
Gravel	4	×5	20 bags
Total	7		**35 bags**

Check: $5 : 10 : 20 = 1 : 2 : 4$ ✓ Total = 35 bags ✓

Worked Example 3 — GST: Adding and Removing

(a) Adding GST: A plumber charges $320 (excluding GST) for a repair job. What is the total price including GST?

Method 1 (two steps): GST $= \$320 \times 0.10 = \$32.00$ Total $= \$320 + \$32 = \$352.00$
Method 2 (one step): Total $= \$320 \times 1.10 = \$352.00$

The customer pays $352.00.

(b) Removing GST: A receipt shows a total of $176.00 including GST. What is the pre-GST price, and how much GST was charged?

Pre-GST price $= \$176.00 \div 1.10 = \$160.00$
GST amount $= \$176.00 - \$160.00 = \$16.00$
Check: $\$160 \times 0.10 = \$16$ ✓

Pre-GST price = $160.00 GST = $16.00

Worked Example 4 — Markup and Discount

(a) Markup: A hardware store buys a drill for $85 and sells it with a 40% markup. What is the selling price?

Method 1: Markup amount $= \$85 \times 0.40 = \$34.00$ Selling price $= \$85 + \$34 = \$119.00$
Method 2: Selling price $= \$85 \times 1.40 = \$119.00$

The selling price is $119.00.

(b) Discount: A jacket has a recommended retail price of $180. The store offers a 25% discount. What is the sale price?

Method 1: Discount $= \$180 \times 0.25 = \$45.00$ Sale price $= \$180 - \$45 = \$135.00$
Method 2: Sale price $= \$180 \times 0.75 = \$135.00$

The sale price is $135.00.

Worked Example 5 — Completing an Invoice

Situation: Springvale Plumbing Supplies sends an invoice to a customer. Complete all missing values.

TAX INVOICE — Springvale Plumbing Supplies Pty Ltd

ABN: 45 678 901 234 **Invoice No.:** INV-0814
Date: 18 March 2025 **Due date:** 1 April 2025
To: Riverside Renovations

Description	Qty	Unit price (excl. GST)	Line total
Copper pipe (6 m lengths)	4	$48.50	$194.00
Pipe fittings (pack of 10)	3	$22.00	$66.00
Plumber's tape	5	$4.80	$24.00
Sealant (tube)	2	$14.50	$29.00
		Subtotal (excl. GST)	**$313.00**
		GST (10%)	$31.30
		TOTAL (incl. GST)	**$344.30**

Payment by EFT to: BSB 014-456 Account: 1234 5678

Key steps:

- Line total = Qty × unit price
- Subtotal = sum of all line totals
- GST = subtotal × 0.10
- Total = subtotal + GST = subtotal × 1.10

Worked Example 6 — Purchase List and Budget

Situation: A school canteen manager has a budget of $200 (excluding GST) to restock supplies. She selects items from the price list below.

Price List — Springvale Wholesale Foods (prices exclude GST)

Bread rolls (bag of 12)	$3.80	Tomato sauce (2 L bottle)	$6.40
Sliced cheese (500 g)	$5.60	Butter (500 g)	$4.20
Ham (200 g pack)	$4.90	Napkins (pack of 200)	$3.50
Tuna (canned, 4-pack)	$7.20	Takeaway containers (50-pk)	$8.80

Her order:

Item	Qty	Unit price	Line total
Bread rolls (bag of 12)	8	\$3.80	\$30.40
Ham (200 g pack)	6	\$4.90	\$29.40
Sliced cheese (500 g)	4	\$5.60	\$22.40
Tomato sauce (2 L bottle)	3	\$6.40	\$19.20
Takeaway containers (50-pk)	4	\$8.80	\$35.20
Napkins (pack of 200)	2	\$3.50	\$7.00
		Subtotal (excl. GST)	**\$143.60**
		GST (10%)	\$14.36
		Total (incl. GST)	**\$157.96**

Is the order within budget?
Budget (excl. GST) = \$200.00. Subtotal = \$143.60 < \$200.00 **Yes, within budget.**
Remaining budget: \$200.00 − \$143.60 = \$56.40 (excl. GST)

Practice Exercise

Practice Exercise — Workplace Numbers

Part A — Ratio Mixing and Scaling

1. A gardener mixes fertiliser using a ratio of **1 part fertiliser : 5 parts water**. She needs to make 12 litres of solution.

(a) How many total parts are there?

(b) How many litres per part?

(c) How many litres of fertiliser does she need?

(d) How many litres of water does she need?

(e) Check your answer: do the litres add up to 12?

2. A kitchen uses a marinade ratio of **2 parts soy sauce : 3 parts honey : 1 part vinegar**. Complete the table for different batch sizes.

Total volume	Soy sauce	Honey	Vinegar
6 cups			
12 cups			
18 cups			

3. A paint is mixed in the ratio **3 parts white : 1 part blue**. A painter needs 20 litres total. How many litres of each colour?

Part B — GST Calculations

4. Calculate the GST (10%) and total price including GST for each item.

Item	Price excl. GST	Price incl. GST
Toolbox	$65.00	
Safety boots	$148.00	
Work gloves	$22.50	
First aid kit	$84.00	
Hi-vis vest	$36.80	

5. A receipt shows the following GST-inclusive prices. Find the pre-GST price and the GST amount for each.

Price incl. GST	Price excl. GST	GST amount
$44.00		
$99.00		
$132.00		
$275.00		

Part C — Markup and Discount

6. Calculate the new price after each markup or discount.

Original price	Change	New price
\$60.00	20% markup	
\$250.00	30% discount	
\$45.00	15% markup	
\$180.00	10% discount	
\$320.00	50% discount	

7. A retailer buys a pair of headphones for \$54 and marks them up by 60%.

(a) What is the selling price (excl. GST)?

_____

_____

(b) What is the selling price including GST?

_____

_____

Part D — Completing Invoices

8. Complete the invoice below by filling in all blank cells.

TAX INVOICE — Springvale Office Supplies

Date: 5 May 2025 **Invoice No.:** INV-0291

To: Springvale Community Centre

Description	Qty	Unit price (excl. GST)	Line total
A4 copy paper (ream)	10	$6.80	$_____
Ballpoint pens (box 12)	4	$9.50	$_____
Sticky notes (pack)	6	$3.40	$_____
Whiteboard markers (set)	3	$12.60	$_____
Stapler	2	$8.90	$_____
		Subtotal (excl. GST)	$_____
		GST (10%)	$_____
		TOTAL (incl. GST)	$_____

Part E — Budget Planning

9. A tradie has a budget of $150 (excl. GST) for supplies. Use the price list and complete the purchase list.

Ace Hardware — Prices exclude GST

Sandpaper (pack)	$4.80	Wood screws (box)	$7.20
Paintbrush set	$14.50	Masking tape (roll)	$3.90
Filler (tube)	$6.40	Safety glasses	$11.00
Drop sheet	$9.80	Sponge (pack of 4)	$5.50

Complete the table for the tradie's order. Choose at least 4 items and stay within the $150 budget (excl. GST).

Item	Qty	Unit price	Line total
		Subtotal (excl. GST)	$_____
		GST (10%)	$_____
		Total (incl. GST)	$_____

Is the order within the $150 budget (excl. GST)?

10. A business has a $500 catering budget (excl. GST). They spend $312.40 (excl. GST). How much of the budget remains?

Application Exercise

Application Exercise — Workplace Scenarios

1. *(Trade ratios)* A painter uses an exterior paint that requires a ratio of **4 parts paint : 1 part water (thinner)** for the first coat.

(a) The painter needs 15 litres of mixture. How many litres of paint and water are needed?

(b) For the second coat, the ratio changes to **9 parts paint : 1 part water**. For 20 litres of mixture, how much of each?

(c) The paint costs \$28.50/litre (excl. GST). How much does the paint for the second coat cost (excl. GST)?

2. *(Cleaning products)* A commercial cleaner uses three different solutions:

Solution	Mixing ratio (conc:water)	Total needed	Concentrate needed
Floor cleaner	1 : 7	16 L	
Window spray	1 : 3	8 L	
Sanitiser	1 : 9	20 L	

(a) Complete the table — calculate the concentrate needed for each solution.

(b) The concentrate costs \$18.40 per litre (excl. GST). What is the total cost of concentrate for all three solutions (excl. GST)?

(c) What is the total cost including GST?

3. *(Retail invoice)* Complete the invoice below.

TAX INVOICE — Springvale Sports & Outdoors

Date: 12 June 2025 **Invoice No.:** INV-1148

To: Springvale Junior Football Club

Description	Qty	Unit price (excl. GST)	Line total
Football (size 5)	12	$28.00	$_____
Training bibs (pack of 10)	3	$22.50	$_____
Drink bottles (750 mL)	20	$6.80	$_____
Mesh bag (for balls)	2	$18.40	$_____
First aid kit (junior)	1	$54.00	$_____
		Subtotal (excl. GST)	$_____
		GST (10%)	$_____
		TOTAL (incl. GST)	$_____

(a) The club has a budget of $600 (incl. GST). Is the invoice within budget?

_____

_____

(b) The club decides to reduce the number of drink bottles to 10. What is the new total (incl. GST)?

_____

_____

_____

4. *(Comparing quotes)* Springvale Community Hall needs 50 folding chairs. Two suppliers quote:

	Supplier A	Supplier B
Price per chair	$38.00 excl. GST	$44.00 incl. GST
Delivery	$25.00 excl. GST	Free

(a) Calculate Supplier A's total (incl. GST) for 50 chairs plus delivery.

_____

(b) Calculate Supplier B's total (incl. GST) for 50 chairs including delivery.

(c) Which supplier is cheaper, and by how much (incl. GST)?

5. *(Retail markup and discount)* A boutique buys clothing at wholesale prices and applies a 60% markup before GST. They then run a 20% off sale.

Item	Wholesale cost	Price after markup	Sale price (20% off)
Dress	$45.00		
Jeans	$38.00		
Jacket	$72.00		

(a) Complete the table.

(b) For the jacket, what is the final sale price including GST?

(c) Does the boutique still make a profit on the jacket during the sale? Show your working.

6. *(Catering budget)* A company is organising a staff morning tea for 20 people. Their catering budget is $120 (excl. GST). Use the price list below.

Springvale Bakery Wholesale — Prices exclude GST

Sausage rolls (12-pack)	$8.40	Juice bottles (6-pack)	$7.20
Mini quiches (12-pack)	$9.60	Coffee pods (10-pack)	$6.80
Fruit platter (serves 10)	$18.50	Tea bags (25-pack)	$4.20
Assorted pastries (8-pk)	$10.40	Disposable cups (50-pk)	$5.60

(a) Choose items to feed 20 people and stay within the $120 budget (excl. GST). Write your purchase list below.

Item	Qty	Unit price	Line total
		Subtotal	
		GST (10%)	
		Total incl. GST	

(b) Is your order within the $120 budget (excl. GST)?

_____

(c) What is the total cost per person (incl. GST)? (Total incl. GST ÷ 20)

_____

_____

Chapter Review

Chapter 10 Review — Workplace Numbers

1. Mix a solution using ratio **1 part concentrate : 3 parts water** to make 8 litres total. How much of each ingredient?

2. A concrete mix uses ratio **1 : 3 : 5** (cement : sand : gravel). Scale up from 9 bags to 45 bags. How many bags of each?

3. Calculate the GST-inclusive price for each.

(a) \$75.00 (excl. GST) _____

(b) \$240.00 (excl. GST) _____

(c) \$18.50 (excl. GST) _____

4. Find the pre-GST price and GST amount for each receipt total.

(a) Total \$55.00 (incl. GST)

(b) Total \$363.00 (incl. GST)

5. A camera costs \$320.00 (excl. GST). It is marked up by 35% before being put on sale at 10% off. Find:

(a) The price after markup (excl. GST).

(b) The sale price after the 10% discount (excl. GST).

_____

_____

(c) The final price including GST.

_____

_____

6. Complete the invoice.

TAX INVOICE — Springvale Garden Centre

Description	Qty	Unit price (excl. GST)	Line total
Potting mix (25 L bag)	6	$12.80	$_____
Fertiliser (1 kg)	4	$9.40	$_____
Garden gloves (pair)	3	$8.60	$_____
Seedling trays (4-pack)	5	$6.20	$_____
		Subtotal (excl. GST)	$_____
		GST (10%)	$_____
		TOTAL (incl. GST)	$_____

7. A receipt shows a total of $407.00 (incl. GST). A colleague says the GST was $40.00. Is this correct? Show your calculation.

_____

_____

_____

Multi-Step Problem 8

Trade job. Flynn is a tiler completing a bathroom renovation. He needs to tile a 12 m^2 floor and two feature walls (6 m^2 each). Tiles come in boxes covering 1.5 m^2 per box.

He also needs to mix tile adhesive. The adhesive comes as a powder mixed with water in a ratio of **3 parts powder : 1 part water**, and he needs 8 litres of adhesive mix.

His supplier, Springvale Tiles, provides the following quote:

Item	Unit price (excl. GST)
Floor tiles (box, covers 1.5 m^2)	\$48.00/box
Feature wall tiles (box, covers 1.5 m^2)	\$62.00/box
Tile adhesive powder (per litre equiv.)	\$4.50/litre
Tile grout (per bag)	\$18.00/bag

Flynn needs 3 bags of grout. He always buys 10% extra tiles to allow for cuts and breakage.

(a) How many boxes of floor tiles does Flynn need (including 10% extra)? Round up to the nearest whole box.

(b) How many boxes of feature wall tiles does he need (10% extra, two walls of 6 m^2 each)?

(c) How many litres of adhesive powder does he need for 8 litres of mix?

(d) Complete the invoice below.

TAX INVOICE — Springvale Tiles

Description	Qty	Unit price (excl. GST)	Line total
Floor tiles (boxes)		$48.00	$_____
Feature wall tiles (boxes)		$62.00	$_____
Tile adhesive powder (litres)		$4.50	$_____
Tile grout (bags)	3	$18.00	$_____
		Subtotal (excl. GST)	$_____
		GST (10%)	$_____
		TOTAL (incl. GST)	$_____

(e) Flynn's budget for materials (incl. GST) is $900. Is the total within budget?

_____

_____

Multi-Step Problem 9

School canteen restock. The canteen manager has a $400 budget (excl. GST) and needs to restock for the week. She receives two competing quotes for the same items.

Item (and qty needed)	Quote A (excl. GST)	Quote B (excl. GST)
Bread rolls, 10 bags of 12	$3.60/bag	$3.90/bag
Cheese slices, 8 packs	$5.40/pack	$4.80/pack
Ham, 10 packs	$4.80/pack	$5.20/pack
Juice bottles, 5 six-packs	$6.80/pack	$6.40/pack
Takeaway containers, 4 packs	$8.60/pack	$9.20/pack

(a) Calculate the subtotal (excl. GST) for each quote for all items at the quantities listed.

_____

_____

_____

(b) Which quote is cheaper overall (excl. GST)? By how much?

(c) Add GST (10%) to the cheaper quote. What is the total (incl. GST)?

(d) Is the cheaper quote within the $400 budget (excl. GST)?

(e) Using the cheaper quote, the manager wants to add 2 extra bags of bread rolls. What is the new subtotal and new total (incl. GST)?

Skills Checklist

Chapter 10 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can mix quantities using a given ratio	⊙	⊖	⊗
I can scale a ratio up or down for a new total	⊙	⊖	⊗
I can calculate GST (10%) and the total including GST	⊙	⊖	⊗
I can find the pre-GST price from a GST-inclusive total	⊙	⊖	⊗
I can apply a percentage markup to find a selling price	⊙	⊖	⊗
I can apply a percentage discount to find a sale price	⊙	⊖	⊗
I can complete a workplace invoice (line totals, subtotal, GST, total)	⊙	⊖	⊗
I can build a purchase list and check it is within a budget	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

Numeracy Short Course — Stage 4 Student Workbook
Chapter 10: Workplace Numbers | Topic 2: Workplace and Employment | Week 11, Lessons 41–44
Aligned to QCAA 2025 Senior Short Course Numeracy Syllabus v1.2

Chapter 11 — 3D Shapes and Volume

Stage: Stage 4 — Student Workbook
Week: Week 12
Lessons: Lessons 45–48
Strand: Topic 2: Workplace and Employment
Sub-topic: Measurement & Space — 3D Shapes and Volume

Introduction

Three-dimensional shapes are everywhere in the workplace. Builders calculate the volume of concrete needed for a slab. Warehouse managers work out how many boxes fit in a storage room. Manufacturers design packaging to minimise waste. Delivery drivers load trucks to maximise space.

In this chapter you will identify 3D shapes, describe their properties, and calculate the volume of rectangular and triangular prisms using a two-step method.

Learning Objectives — by the end of this chapter you will be able to:

- Identify and name prisms and cylinders from diagrams and descriptions
- Describe 3D shapes using faces, edges and vertices
- Calculate the volume of a rectangular prism using $V = L \times W \times H$
- Calculate the volume of a triangular prism using the two-step method
- Apply volume calculations to practical workplace problems
- Convert between volume units (cm^3, m^3)

Key Terms

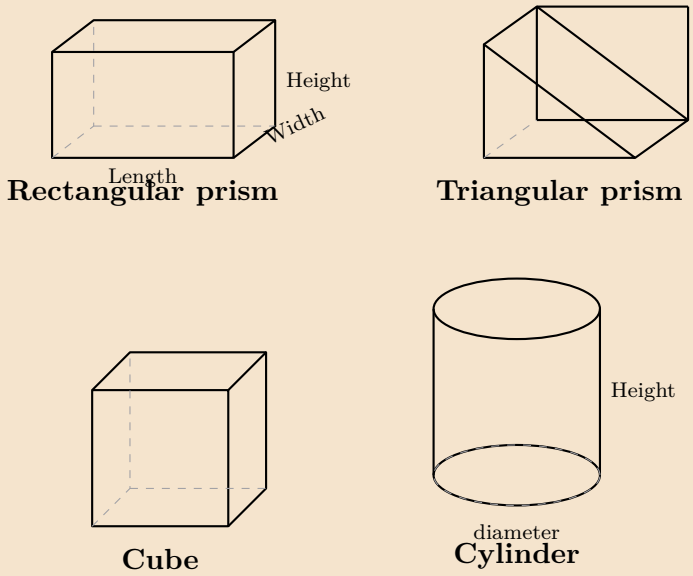
Term	Definition
Prism	A 3D shape with two identical parallel bases and rectangular side faces
Rectangular prism	A prism with a rectangular base (also called a cuboid or box shape)
Triangular prism	A prism with a triangular base
Square prism / cube	A prism where all faces are squares (length = width = height)
Cylinder	A 3D shape with two circular bases (not a prism, but similar in volume method)
Face	A flat surface of a 3D shape
Edge	A line where two faces meet
Vertex (vertices)	A corner where three or more edges meet
Cross-section	The 2D shape you see when you slice straight through a 3D shape
Base	The face used as the foundation for volume calculations
Height	The perpendicular distance between the two bases of a prism
Volume	The amount of 3D space inside a shape (measured in cubic units)
Capacity	The amount a container can hold (1 cm^3 = 1 mL; 1 L = 1000 cm^3)

3D Shape Reference

3D Shape Properties — Reference Table

Shape	Faces	Edges	Vertices	Base shape
Rectangular prism (cuboid)	6	12	8	Rectangle
Triangular prism	5	9	6	Triangle
Square prism (cube)	6	12	8	Square
Pentagonal prism	7	15	10	Pentagon
Cylinder	2 flat + 1 curved	—	—	Circle

Diagrams of common 3D shapes:



Key Formulas

Volume Formulas

Volume of any prism:

$$V = A_{\text{base}} \times H$$

Step 1: Find the area of the base.
Step 2: Multiply by the height (length) of the prism.

Rectangular prism:

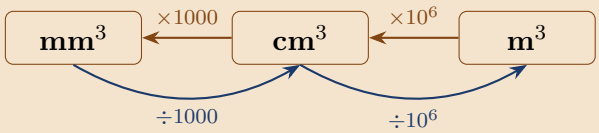
$$V = L \times W \times H$$

Triangular prism:

$$V = \frac{b \times h}{2} \times L$$

where b = base of triangle, h = height of triangle, L = length of prism.

Volume unit conversions:



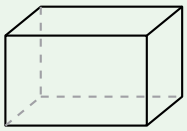
Capacity link:
$1\ \text{cm}^3 = 1\ \text{mL}$
$1000\ \text{cm}^3 = 1\ \text{L}$
$1\ \text{m}^3 = 1000\ \text{L}$

Worked Examples

Worked Example 1 — Identifying 3D Shapes and Their Properties

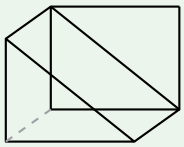
Task: Name each shape and state the number of faces, edges and vertices.

Shape A



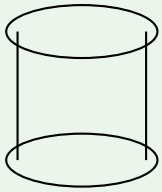
Rectangular prism

Shape B



Triangular prism

Shape C



Cylinder

Shape	Faces	Edges	Vertices	Base shape
A — Rectangular prism	6	12	8	Rectangle
B — Triangular prism	5	9	6	Triangle
C — Cylinder	3 (2 flat + 1 curved)	2 (curved)	0	Circle

Key point: A prism always has the same cross-section along its entire length.

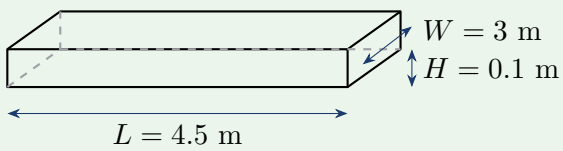
Worked Example 2 — Why 3D Shapes Are Used at Work

Task: Match each workplace item to its 3D shape and explain why that shape is used.

Workplace item	Shape	Reason
Cardboard shipping box	Rectangular prism	Rectangular faces stack easily; no wasted space when packed together
Water pipe / drainage pipe	Cylinder	Circular cross-section allows fluid flow with no sharp corners to block
Wedge door stopper	Triangular prism	Triangular profile grips the floor; thin end slides under the door easily
Storage container (freight)	Rectangular prism	Maximises volume for a given footprint; stacks securely
Tin of paint	Cylinder	Strong under pressure; easy to manufacture; round lid seals evenly

Worked Example 3 — Volume of a Rectangular Prism

Situation: A concrete slab for a garden shed floor measures 4.5 m long, 3 m wide and 0.1 m thick. How many cubic metres of concrete are needed?



Step 1: Find the area of the base (rectangle).
$A = L \times W = 4.5 \times 3 = 13.5 \text{ m}^2$

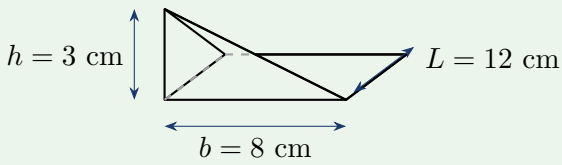
Step 2: Multiply the base area by the height.
$V = A \times H = 13.5 \times 0.1 = 1.35 \text{ m}^3$

The slab requires 1.35 m^3 of concrete.

Alternative (direct formula): $V = L \times W \times H = 4.5 \times 3 \times 0.1 = 1.35 \text{ m}^3$ ✓

Worked Example 4 — Volume of a Triangular Prism

Situation: A wedge-shaped door stopper is a triangular prism. It has a triangular face with a base of 8 cm and a height of 3 cm. The stopper is 12 cm long. What is its volume?



Step 1: Find the area of the triangular base.
$A_\triangle = \frac{b \times h}{2} = \frac{8 \times 3}{2} = \frac{24}{2} = 12 \text{ cm}^2$

Step 2: Multiply the base area by the length of the prism.
$V = A_\triangle \times L = 12 \times 12 = 144 \text{ cm}^3$

The volume of the door stopper is 144 cm^3.

Worked Example 5 — How Many Boxes Fit in a Storage Room?

Situation: A storeroom at a warehouse is 6 m long, 4 m wide and 3 m high. Each cardboard box is 60 cm long, 40 cm wide and 30 cm tall. How many boxes fit in the storeroom?

Step 1: Convert all measurements to the same unit.
Storeroom: $6\,\text{m} = 600\,\text{cm}$; $4\,\text{m} = 400\,\text{cm}$; $3\,\text{m} = 300\,\text{cm}$

Step 2: Calculate the volume of the storeroom.
$V_{\text{room}} = 600 \times 400 \times 300 = 72\,000\,000\,\text{cm}^3$

Step 3: Calculate the volume of one box.
$V_{\text{box}} = 60 \times 40 \times 30 = 72\,000\,\text{cm}^3$

Step 4: Divide room volume by box volume.
Number of boxes $= 72\,000\,000 \div 72\,000 = 1000$ boxes

Up to 1000 boxes fit in the storeroom.

Practical note: In real life, you also need to consider access aisles, weight limits and stacking height. The volume method gives the theoretical maximum.

Practice Exercise

Practice Exercise — 3D Shapes and Volume

Part A — Shape Identification and Properties

1. Name each 3D shape described below.

(a) A shape with 6 rectangular faces, 12 edges and 8 vertices. _____

(b) A shape with 5 faces (2 triangles and 3 rectangles), 9 edges and 6 vertices. _____

(c) A shape with 2 circular faces and 1 curved surface. _____

(d) A special rectangular prism where all 6 faces are identical squares. _____

2. Complete the properties table for each shape.

Shape	Faces	Edges	Vertices
Rectangular prism			
Triangular prism			
Cube			

3. A prism has a hexagonal base. How many faces, edges and vertices does it have? *(Hint: a hexagon has 6 sides.)*

Part B — Volume of Rectangular Prisms

4. Calculate the volume of each rectangular prism. Show your working and include units.

	Length	Width	Height	Volume
(a)	5 cm	4 cm	3 cm	
(b)	8 m	3 m	2 m	
(c)	12 cm	7 cm	5 cm	
(d)	2.5 m	1.5 m	0.2 m	
(e)	40 cm	25 cm	15 cm	

5. A storage box has a volume of 3000 cm^3. It is 25 cm long and 12 cm wide. How tall is the box?

Part C — Volume of Triangular Prisms

6. For each triangular prism, calculate the base area first, then the volume.

(a) Triangle base = 6 cm, triangle height = 4 cm, prism length = 10 cm.

Step 1: Base area $= \dfrac{b \times h}{2} = \dfrac{\quad \times \quad}{2} =$_____ cm^2

Step 2: $V = A \times L =$_____$\times$_____$=$_____ cm^3

(b) Triangle base = 9 m, triangle height = 5 m, prism length = 8 m.

Step 1: Base area $= \dfrac{b \times h}{2} = \dfrac{\quad \times \quad}{2} =$_____ m^2

Step 2: $V = A \times L =$_____$\times$_____$=$_____ m^3

(c) Triangle base = 14 cm, triangle height = 8 cm, prism length = 20 cm.

_____

_____

_____

Part D — Volume Unit Conversions

7. Convert each volume.

(a) 5 m^3 = _____cm^3

(b) 8000 cm^3 = _____L

(c) 2500 cm^3 = _____mL

(d) 3 m^3 = _____L

(e) 450 000 cm^3 = _____m^3

8. A rectangular tank is 80 cm long, 50 cm wide and 40 cm high. What is its capacity in litres?

_____

_____

_____

Application Exercise

Application Exercise — Workplace Volume Problems

1. *(Concrete slab)* A driveway slab is 12 m long, 3.5 m wide and 0.12 m thick.

(a) Calculate the volume of concrete needed (m^3).

(b) Concrete costs \$185 per m^3 (excl. GST). What is the cost of concrete for the driveway (excl. GST)?

(c) What is the total cost including GST (10%)?

2. *(Shipping containers)* A standard 20-foot shipping container has internal dimensions of approximately 5.9 m long, 2.35 m wide and 2.39 m high.

(a) Calculate the internal volume of the container (m^3, to 2 decimal places).

(b) A pallet of goods measures 1.2 m $\times$ 1.0 m $\times$ 1.8 m (L $\times$ W $\times$ H). What is the volume of one pallet?

(c) How many pallets fit in the container? (Use the volume method; round down.)

3. *(Triangular ramp)* A builder constructs a concrete access ramp with a triangular cross-section. The triangle has a base of 2.4 m and a height of 0.3 m. The ramp is 1.5 m wide.

(a) Draw a labelled diagram of the ramp shape.

(b) Calculate the volume of concrete needed for the ramp.

(c) Concrete is sold in 0.2 m^3 bags. How many bags are needed? (Round up.)

4. *(Packaging)* A gift box manufacturer makes rectangular boxes in two sizes:

Box	Length	Width	Height
Small	20 cm	15 cm	10 cm
Large	35 cm	25 cm	15 cm

(a) Calculate the volume of each box.

(b) How many small boxes fit inside one large box? (Use volumes; round down.)

(c) A customer needs to store 500 small boxes. The storage room is 4 m × 3 m × 2.4 m. Will all 500 boxes fit? Show your working.

5. *(Water tank)* A rectangular water tank on a farm is 3 m long, 2 m wide and 1.5 m tall.

(a) Calculate the volume of the tank in m^3.

(b) Convert the volume to litres ($1\ m^3 = 1000$ L).

(c) The tank is currently 40% full. How many litres of water are in the tank?

(d) Water is pumped in at 50 litres per minute. How many minutes will it take to fill the tank from its current level?

6. *(Comparing storage options)* A business is choosing between two storage containers:

Container	Length	Width	Height	Weekly rental
Option A	6 m	3 m	2.5 m	\$85/week
Option B	5 m	4 m	2.8 m	\$90/week

(a) Calculate the volume of each container.

(b) Which container has the greater volume?

(c) Calculate the cost per cubic metre per week for each option.

(d) Which container offers better value for money? Justify your answer.

Chapter Review

Chapter 11 Review — 3D Shapes and Volume

1. A shape has 5 faces, 9 edges and 6 vertices. Name the shape and state the shape of its base.

2. How many faces does a prism with an octagonal base have? Show your reasoning.

3. Calculate the volume of each shape. Include units in your answer.

(a) Rectangular prism: $L = 9$ cm, $W = 6$ cm, $H = 4$ cm.

(b) Rectangular prism: $L = 5$ m, $W = 2.5$ m, $H = 0.15$ m.

(c) Triangular prism: triangle base = 10 cm, triangle height = 6 cm, prism length = 15 cm.

(d) Triangular prism: triangle base = 4 m, triangle height = 3 m, prism length = 7 m.

_____

_____

_____

4. Convert: 2.4 m^3 = _____ cm^3 15 000 cm^3 = _____ L

5. A rectangular fish tank is 90 cm long, 40 cm wide and 45 cm deep.

(a) Calculate the volume in cm^3.

_____

_____

(b) Convert the volume to litres.

_____

(c) The tank should only be filled to 80% of its capacity. How many litres is that?

_____

_____

6. Explain the difference between *volume* and *capacity*. Give a workplace example of each.

_____

_____

_____

Multi-Step Problem 7

Concrete pour. A construction crew is pouring concrete for a warehouse floor. The floor is a rectangle 20 m long and 15 m wide. The concrete slab must be 120 mm thick.

(a) Convert 120 mm to metres.

_____

(b) Calculate the volume of concrete needed (m^3).

_____

(c) Concrete is delivered in trucks holding 6 m^3 per load. How many truck loads are needed? (Round up to the nearest whole truck.)

(d) Concrete costs \$195 per m^3 (excl. GST). Calculate the total cost of concrete (excl. GST).

(e) What is the total cost including GST?

(f) The crew pours concrete at a rate of 3 m^3 per hour. How many hours will it take to pour the slab?

(g) Workers are paid \$38/hr each. There are 4 workers. What is the total labour cost for the pour?

Multi-Step Problem 8

Warehouse loading. A warehouse has a storage space measuring 12 m long, 8 m wide and 4 m high. Two types of boxes are being stored:

Box type	L	W	H	Contents
Type X	60 cm	50 cm	40 cm	Electronic parts
Type Y	80 cm	60 cm	50 cm	Clothing items

(a) Calculate the volume of the storage space (m^3; convert all to metres first).

(b) Calculate the volume of one Type X box (cm^3, then convert to m^3).

(c) Calculate the volume of one Type Y box (m^3).

(d) Using the volume method, how many Type X boxes could fit in the storage space?

(e) Using the volume method, how many Type Y boxes could fit?

(f) The warehouse manager decides to fill half the storage space with Type X boxes and the other half with Type Y boxes. How many of each type will fit?

(g) Type X boxes are worth \$240 each and Type Y boxes are worth \$185 each. What is the total value of stock in the mixed-storage scenario?

Skills Checklist

Chapter 11 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can name common 3D shapes and describe their faces, edges and vertices	⊙	⊖	⊗
I can calculate the volume of a rectangular prism	⊙	⊖	⊗
I can find the area of a triangular base (Step 1)	⊙	⊖	⊗
I can calculate the volume of a triangular prism (Step 2)	⊙	⊖	⊗
I can convert between cm^3, m^3 and litres	⊙	⊖	⊗
I can solve multi-step workplace problems involving volume	⊙	⊖	⊗
I can explain why certain 3D shapes are used in the workplace	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

Chapter 12 — Mass, Capacity and Temperature

Stage: Stage 4 — Student Workbook
Week: Week 13
Lessons: Lessons 49–52
Strand: Topic 2: Workplace and Employment
Sub-topic: Measurement & Space — Mass, Capacity and Temperature

Introduction

From weighing ingredients in a kitchen to managing chemical stock in a warehouse, measurement is a core workplace skill. In hospitality, getting a recipe ratio right means measuring mass accurately. In logistics, loading a truck safely means understanding weight limits. In food service, checking that food is stored at the right temperature protects customers from illness.

This chapter covers metric mass, capacity and temperature — the three measurement areas most commonly tested in the QCAA Numeracy examination.

Learning Objectives — by the end of this chapter you will be able to:

- Identify and use metric mass units: mg, g, kg, t
- Convert between metric mass units in both directions
- Identify and use metric capacity units: mL, L, kL, ML
- Convert between metric capacity units in both directions
- Read temperatures from thermometer diagrams
- Apply Australian food safety temperature guidelines to workplace scenarios
- Solve multi-step measurement problems in workplace contexts

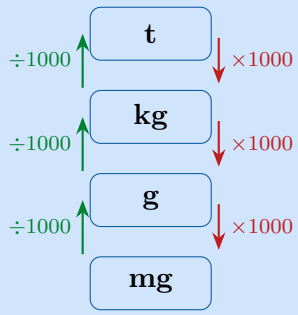
Key Terms

Term	Definition
Mass	The amount of matter in an object (commonly called weight in everyday use)
Milligram (mg)	The smallest common mass unit; used for medicines and small doses
Gram (g)	Used for food ingredients, small packages
Kilogram (kg)	Used for most everyday goods, produce, body weight
Tonne (t)	Used for heavy loads: vehicles, freight, bulk materials
Capacity	The amount of liquid or gas a container can hold
Millilitre (mL)	Very small liquid amounts; used for medicines, beverages
Litre (L)	Standard unit for beverages, fuel, recipes
Kilolitre (kL)	1000 litres; used for water tanks, pool volumes
Megalitre (ML)	1 000 000 litres; used for reservoirs, irrigation
Temperature	A measure of how hot or cold something is, in degrees Celsius (°C)
Danger zone	The temperature range (5°C – 60°C) where bacteria grow rapidly in food
Cold chain	The process of keeping food cold throughout transport and storage

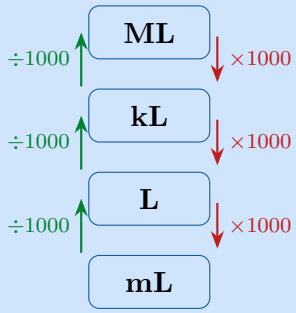
Key Formulas and Conversion Ladders

Mass, Capacity and Temperature — Key Conversions

Mass conversion ladder:

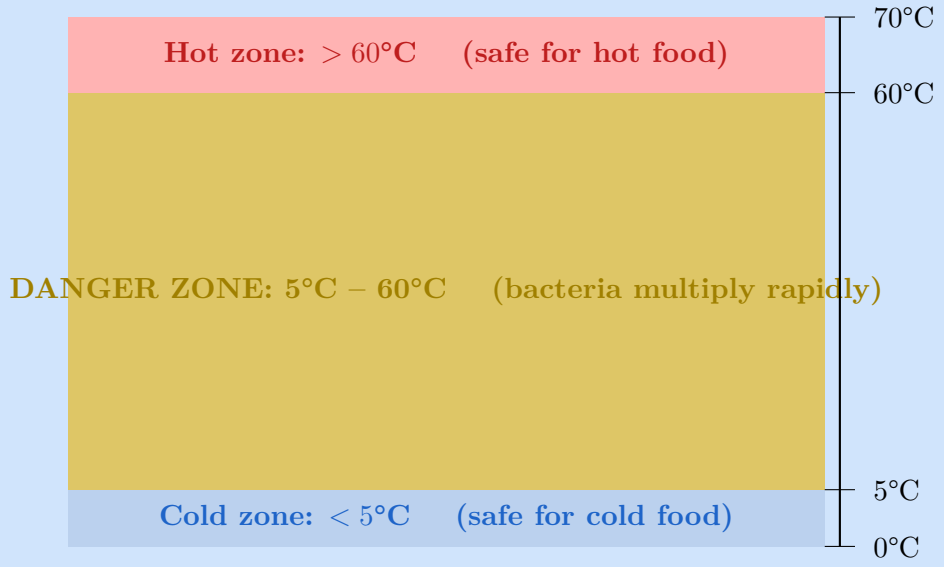


Capacity conversion ladder:



Rule: Moving **down** the ladder (to a smaller unit) ⇒ multiply by 1000.
Moving **up** the ladder (to a larger unit) ⇒ divide by 1000.

Australian Food Safety Temperature Guidelines:



Worked Examples

Worked Example 1 — Converting Mass Units

Convert each mass as directed.

Use the ladder: moving **down** (to smaller unit) $\times 1000$; moving **up** (to larger unit) $\div 1000$.

(a) Convert 2.5 kg to grams.

$$2.5 \text{ kg} \xrightarrow{\times 1000} 2500 \text{ g}$$

(b) Convert 3500 g to kilograms.

$$3500 \text{ g} \xrightarrow{\div 1000} 3.5 \text{ kg}$$

(c) Convert 4.2 t to kilograms.

$$4.2 \text{ t} \xrightarrow{\times 1000} 4200 \text{ kg}$$

(d) Convert 750 mg to grams.

$$750 \text{ mg} \xrightarrow{\div 1000} 0.75 \text{ g}$$

(e) Convert 3.8 kg to milligrams (two steps: kg $\rightarrow$ g $\rightarrow$ mg).

$$3.8 \text{ kg} \xrightarrow{\times 1000} 3800 \text{ g} \xrightarrow{\times 1000} 3\,800\,000 \text{ mg}$$

Worked Example 2 — Converting Capacity Units

Convert each capacity as directed.

(a) Convert 1.5 L to millilitres.

$$1.5 \text{ L} \xrightarrow{\times 1000} 1500 \text{ mL}$$

(b) Convert 2500 mL to litres.

$$2500 \text{ mL} \xrightarrow{\div 1000} 2.5 \text{ L}$$

(c) Convert 3.2 kL to litres.

$$3.2 \text{ kL} \xrightarrow{\times 1000} 3200 \text{ L}$$

(d) Convert 850 000 L to megalitres.

$$850\,000 \text{ L} \xrightarrow{\div 1000} 850 \text{ kL} \xrightarrow{\div 1000} 0.85 \text{ ML}$$

(e) Convert 0.045 kL to millilitres (two steps: kL → L → mL).

$$0.045 \text{ kL} \xrightarrow{\times 1000} 45 \text{ L} \xrightarrow{\times 1000} 45\,000 \text{ mL}$$

Worked Example 3 — Capacity Word Problem

Situation: A café uses 220 mL of milk per coffee drink. On a busy Saturday they make 280 coffees.

(a) **How many millilitres of milk are used in total?**

$$280 \times 220 = 61\,600 \text{ mL}$$

(b) **Convert this to litres.**

$$61\,600 \div 1000 = 61.6 \text{ L}$$

(c) **Milk is ordered in 2-litre cartons. How many cartons are needed?**

$$61.6 \div 2 = 30.8 \text{ cartons}$$

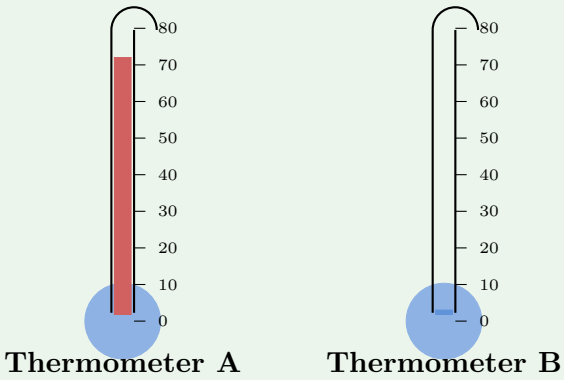
Round **up**: they need **31 cartons** (you can't order a fraction of a carton).

(d) **Each carton costs \$3.20. What is the total milk cost?**

$$31 \times \$3.20 = \$99.20$$

Worked Example 4 — Reading Thermometers

Task: Read the temperature shown on each thermometer, then answer the questions.



Thermometer	Reading	Zone	Food safety status
A	72°C	Hot zone (> 60°C)	**Safe** — hot food is safe above 60°C
B	3°C	Cold zone (< 5°C)	**Safe** — cold food is safe below 5°C

Worked Example 5 — Food Safety Temperature Scenario

Situation: A hospitality student checks food temperatures at a café kitchen. Here are the readings.

Food item	Temp.	Required	Safe?
Chicken soup (on bain-marie)	68°C	> 60°C (hot)	**Yes**
Caesar salad (in display fridge)	8°C	< 5°C (cold)	**No — in danger zone**
Beef stew (in warming oven)	55°C	> 60°C (hot)	**No — in danger zone**
Cheesecake (in cold display)	4°C	< 5°C (cold)	**Yes**

Action required for unsafe items:
Caesar salad (8°C): Return to fridge immediately. Do not serve until temperature drops below 5°C.
Beef stew (55°C): Increase oven temperature. Do not serve until temperature reaches at least 60°C.
Any food left in the danger zone for more than 2 hours must be discarded (Australian Food Standards Code).

Practice Exercise

Practice Exercise — Mass, Capacity and Temperature

Part A — Mass Conversions

1. Complete the mass conversion table.

	mg	g	kg	t
(a)		500		
(b)			3.2	
(c)	2000			
(d)				0.5
(e)		8500		
(f)			0.25	

2. Convert each mass.

(a) 4.7 kg = _____ g

(b) 12 500 g = _____ kg

(c) 3 t = _____ kg

(d) 650 mg = _____ g

(e) 2.1 t = _____ g (two steps)

(f) 0.008 kg = _____ mg (two steps)

Part B — Capacity Conversions

3. Complete the capacity conversion table.

	mL	L	kL	ML
(a)		4.5		
(b)	750			
(c)			2.8	
(d)				0.003
(e)		12 000		
(f)	6 500 000			

4. Convert each capacity.

(a) 3.6 L = _____mL

(b) 4200 mL = _____L

(c) 5 kL = _____L

(d) 0.75 L = _____mL

(e) 8 ML = _____kL

(f) 3 500 L = _____kL

Part C — Choosing Appropriate Units

5. Write the most appropriate unit (mg, g, kg, t, mL, L, kL, ML) for each measurement.

(a) The mass of a tablet of paracetamol. _____

(b) The fuel tank capacity of a family car. _____

(c) The mass of a loaded semi-trailer truck. _____

(d) The volume of water in a swimming pool. _____

(e) The mass of a bag of potatoes at the supermarket. _____

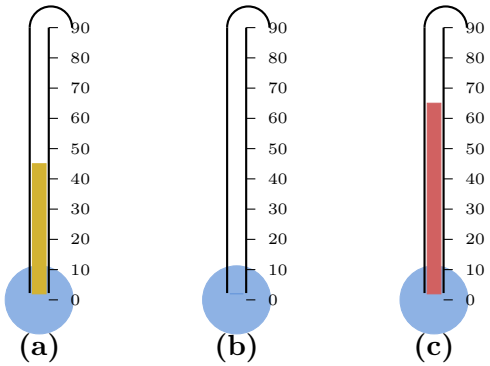
(f) The amount of medicine in a syringe. _____

(g) The water storage in a large dam. _____

(h) The mass of a single grain of rice. _____

Part D — Reading Thermometers

6. Read the temperature on each thermometer.



Temperature (a): _____°C Temperature (b): _____°C Temperature (c): _____°C

7. For each temperature reading in Question 6, state whether it is in the cold zone, danger zone or hot zone.

(a) _____

(b) _____

(c) _____

Part E — Food Safety

8. A food safety inspector records these temperatures. Complete the table.

Food item	Recorded temp.	Required temp.	Safe? (Y/N)
Hot chips (takeaway display)	58°C	> 60°C	
Yoghurt (fridge display)	4°C	< 5°C	
Roast chicken (warming oven)	65°C	> 60°C	
Cream cake (counter display)	12°C	< 5°C	

9. For each unsafe food item in Question 8, describe the action that should be taken.

_____

_____

_____

10. Explain in your own words what the "danger zone" means and why food should not stay in it for long periods.

_____

_____

_____

Application Exercise

Application Exercise — Measurement in the Workplace

1. *(Pharmacy — dosage)* A patient is prescribed 500 mg of a medication twice a day for 10 days.

(a) How many milligrams does the patient take per day?

(b) How many milligrams over the full 10-day course?

(c) Convert the total 10-day dose to grams.

(d) Each tablet contains 250 mg. How many tablets are needed for the full course?

2. *(Freight)* A delivery truck has a maximum payload of 4.5 t.

(a) Convert 4.5 t to kilograms.

(b) The truck is loaded with 36 boxes each weighing 85 kg. What is the total load in kg?

(c) Is the total load within the maximum payload? By how much (kg)?

(d) The driver wants to add more boxes of 65 kg each. How many additional boxes can be loaded without exceeding the payload?

3. *(Catering — scaling a recipe)* A caterer is preparing pumpkin soup for an event. The original recipe makes 4 servings and uses 600 g of pumpkin and 800 mL of stock. She needs 60 servings.

(a) What is the scaling factor? (60 ÷ 4)

(b) How many grams of pumpkin does she need?

(c) Convert this to kilograms.

(d) How many millilitres of stock does she need?

(e) Convert this to litres.

(f) Stock is sold in 2-litre cartons. How many cartons are needed? (Round up.)

4. *(Irrigation)* A farm irrigates paddocks using a system that pumps 850 L per minute.

(a) How many litres are pumped in one hour (60 minutes)?

(b) Convert this to kilolitres.

(c) The paddock needs 12 kL of water per irrigation. How many minutes does the pump need to run?

_____

_____

(d) A storage dam holds 45 ML. Express this in litres.

_____

5. *(Food safety inspection)* A health inspector visits a restaurant and records the following temperatures. For each item:

- State whether it is safe or unsafe.
- If unsafe, state what action must be taken.

Item	Temp.	Assessment and action
Prawn cocktail (display)	9°C	
Beef stew (bain-marie)	63°C	
Chocolate mousse (fridge)	3°C	
Fried rice (warming lamp)	52°C	
Grilled fish (service pass)	74°C	

6. *(Cold chain)* A supermarket receives a delivery of frozen goods. The driver records the truck temperature log below.

Time	Truck temperature	Within safe cold range (< 5°C)?
06:00 (departure)	2°C	
08:30	4°C	
10:15	7°C	
11:45	6°C	
13:00 (arrival)	4°C	

(a) Complete the table.

(b) At what times was the temperature outside the safe cold range?

_____

(c) The goods were in the unsafe temperature range for approximately 2.5 hours. Under Australian Food Standards, goods left in the danger zone for more than 2 hours must be discarded. What should happen to this delivery?

_____

_____

_____

Chapter Review

Chapter 12 Review — Mass, Capacity and Temperature

1. Convert each mass.

(a) 6.4 kg = _____g

(b) 900 mg = _____g

(c) 7500 kg = _____t

(d) 0.35 t = _____kg

(e) 2.5 kg = _____mg (two steps)

2. Convert each capacity.

(a) 2.8 L = _____mL

(b) 5600 mL = _____L

(c) 4 kL = _____L

(d) 0.6 ML = _____kL

(e) 3200 L = _____kL

3. Write the correct unit for each situation.

(a) Volume of a household water tank. _____

(b) Mass of a vitamin C tablet. _____

(c) Capacity of an IV drip bag in hospital. _____

(d) Mass of a shipping container of steel. _____

4. A temperature probe reads 14°C inside a coolroom storing cooked chicken.

(a) Is this temperature in the danger zone?

_____

(b) What should the temperature be for safe cold food storage?

_____

(c) What action should be taken?

_____

_____

5. A baker uses 750 g of flour per loaf of bread.

(a) How many grams of flour are needed for 24 loaves?

_____

_____

(b) Convert this to kilograms.

_____

(c) Flour is sold in 5 kg bags. How many bags are needed?

_____

_____

6. A sports drink bottle holds 750 mL. A team of 18 players each drink one bottle.

(a) Total mL consumed:

_____

(b) Convert to litres:

_____

(c) Drinks are purchased in 1.5 L bottles. How many bottles are needed?

_____

_____

Multi-Step Problem 7

Hospital kitchen. A hospital kitchen manager is planning meals for 120 patients over one day. For the morning tea, each patient receives a 200 mL cup of soup and a 150 g bread roll.

(a) Calculate the total volume of soup needed in mL.

(b) Convert the total soup volume to litres.

(c) Soup is made in pots holding 8 L each. How many full pots are needed?

(d) Calculate the total mass of bread rolls needed in grams.

(e) Convert the total mass to kilograms.

(f) Bread rolls are delivered in trays of 12. How many trays are needed?

(g) The soup is kept warm at 68°C on the bain-marie. Is this temperature safe for hot food? Explain.

(h) After the service, leftover soup measures 3.5 L at 45°C. Is this soup safe to refrigerate and reuse? What must happen to it first according to food safety guidelines?

Multi-Step Problem 8

Water tank management. A Queensland farm has two rectangular water tanks:

- **Tank A:** 4 m long, 2.5 m wide, 2 m high
- **Tank B:** 3 m long, 3 m wide, 1.8 m high

(a) Calculate the volume of each tank in m^3.

(b) Convert each volume to litres ($1\ m^3 = 1000$ L).

(c) What is the combined capacity of both tanks in kL?

(d) Tank A is currently 60% full and Tank B is 35% full. How many litres are in each tank?

(e) Water is pumped from a bore at 120 L per minute. How long (in hours and minutes) would it take to fill both tanks from their current levels?

(f) During a drought, the farm uses 4500 L per day from the combined tanks. How many days will the current water supply last? (Use the current totals from part d.)

Skills Checklist

Chapter 12 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can convert between mg, g, kg and t	⊙	⊖	⊗
I can convert between mL, L, kL and ML	⊙	⊖	⊗
I can select the appropriate unit for a given context	⊙	⊖	⊗
I can read a temperature from a thermometer diagram	⊙	⊖	⊗
I can identify the danger zone and explain why it matters	⊙	⊖	⊗
I can apply food safety guidelines to workplace scenarios	⊙	⊖	⊗
I can solve multi-step measurement problems	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

Chapter 13 — Workplace Measurement

Stage: Stage 4 — Student Workbook
Week: Week 14
Lessons: Lessons 53–56
Strand: Topic 2: Workplace and Employment
Sub-topic: Measurement & Space — Consolidation (Volume units, Floor plans, Maps)

Introduction

This chapter consolidates all Topic 2 Measurement and Space skills. You will extend your volume unit knowledge, work with a real workplace floor plan, and plan delivery routes on a map. These skills are used every day by warehouse staff, delivery drivers, builders, retailers and tradies across Queensland.

By the end of this chapter you will have covered all Measurement and Space content for Topic 2, and the Chapter Review revisits Chapters 11–13 together.

Learning Objectives — by the end of this chapter you will be able to:

- Convert between volume units (mm^3, cm^3, m^3)
- Read and interpret a workplace floor plan using the scale and legend
- Use a scale to calculate real dimensions and areas from a plan
- Identify locations on a delivery map using grid references and compass directions
- Plan and describe an efficient route using compass directions and distances

Key Terms

Term	Definition
Floor plan	A top-down (bird's eye) drawing of a building showing rooms and features
Scale	The ratio between a measurement on a drawing and the real measurement
Scale factor	The number you multiply by to convert plan measurements to real measurements
Legend / Key	A list explaining the symbols used on a plan or map
Loading dock	The area of a warehouse where goods are loaded and unloaded from trucks
Route	A path taken from one place to another
Efficient route	A route that minimises distance or travel time
Bearing / direction	A compass direction (N, NE, E, SE, S, SW, W, NW) used to describe movement
Grid reference	A letter–number code used to identify a location on a map (e.g. B4)

Key Formulas

Workplace Measurement — Key Formulas

Volume unit conversions:

$$1\ \text{cm}^3 = 1000\ \text{mm}^3$$
$$1\ \text{m}^3 = 1\,000\,000\ \text{cm}^3$$
$$1\ \text{m}^3 = 1000\ \text{L} \quad \text{(capacity link)}$$

Floor plan scale:

$$\text{Real length} = \text{plan length} \times \text{scale factor}$$

$$\text{Plan length} = \frac{\text{real length}}{\text{scale factor}}$$

Example: Scale 1:100 means
1 cm on plan = 100 cm = 1 m in reality.

Common scales	What they mean
1:50	1 cm on plan = 50 cm (0.5 m) real
1:100	1 cm on plan = 100 cm (1 m) real
1:200	1 cm on plan = 200 cm (2 m) real
1:400	1 cm on plan = 400 cm (4 m) real
1:20 000	1 cm on map = 200 m real

Worked Examples

Worked Example 1 — Volume Unit Conversions

Convert each volume as directed.

$cm^3 \to mm^3$: $\times 1000$ $mm^3 \to cm^3$: $\div 1000$ $m^3 \to cm^3$: $\times 1\,000\,000$ $cm^3 \to m^3$: $\div 1\,000\,000$

(a) Convert $0.005\,m^3$ to cm^3.

$$0.005 \times 1\,000\,000 = 5000\ cm^3$$

(b) Convert $5\,000\,000\,cm^3$ to m^3.

$$5\,000\,000 \div 1\,000\,000 = 5\ m^3$$

(c) Convert $8.4\,cm^3$ to mm^3.

$$8.4 \times 1000 = 8400\ mm^3$$

(d) Convert $250\,000\,mm^3$ to cm^3, then to m^3.

$$250\,000 \div 1000 = 250\ cm^3 \quad \longrightarrow \quad 250 \div 1\,000\,000 = 0.00025\ m^3$$

Worked Example 2 — Reading the Warehouse Floor Plan

Refer to the Springvale Warehouse floor plan on the next page (Scale: 1 cm = 4 m).
Task: Answer the questions using the floor plan.

(a) **Name the room in the south-east corner of the building.**

Looking at the floor plan, the south-east corner contains the **WC / Amenities** area.

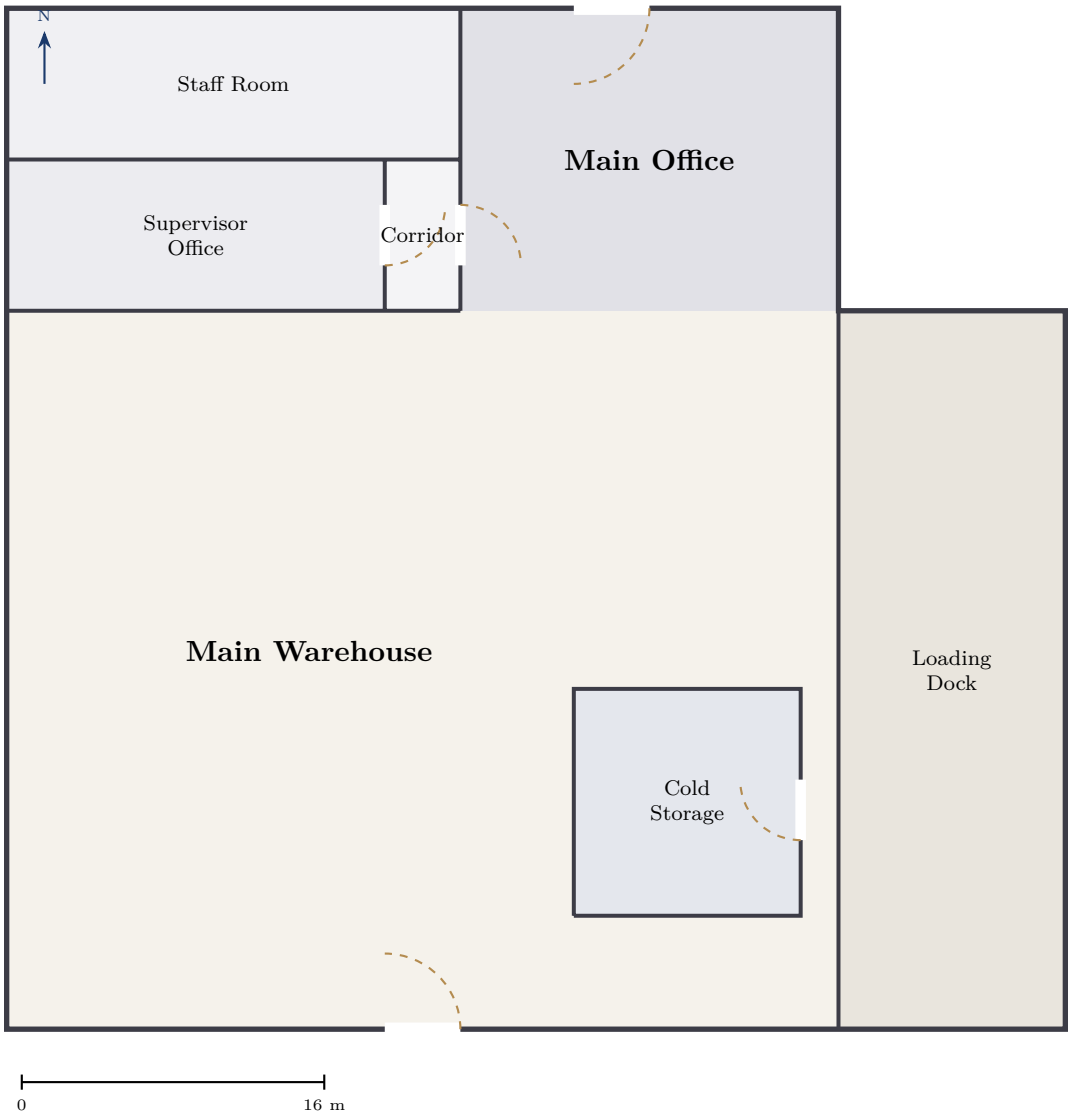
(b) **How many separate rooms/areas does the building have?**
Main Warehouse, Cold Store, Loading Dock area, Main Office, Supervisor Office, Staff Room, WC/Amenities — **7 areas**.

(c) **Where is the main entrance?**
The main entrance is in the **south wall**, roughly in the centre of the building.

(d) **What is the purpose of the loading dock?**
The loading dock is on the south side of the warehouse where trucks park to load and unload goods.

Springvale Warehouse — Floor Plan

Scale: 1 cm = 4 m (1:400)



Floor Plan Legend

- Walls
- Doors (opening arc shown)
- Loading dock / road surface
- Interior room floor
- Reference point
- North arrow

Worked Example 3 — Using Scale to Find Real Dimensions

Situation: On the Springvale Warehouse floor plan (Scale: 1 cm = 4 m), the Main Warehouse measures approximately 11 cm wide and 9.5 cm tall on the plan.
Find: (a) the real width, (b) the real depth, (c) the real floor area of the Main Warehouse.

Step 1: Multiply the plan measurement by the scale factor.
Step 2: Include units (scale factor = 4 m per cm, so answer is in metres).
Step 3: For area, multiply real length × real width.

(a) Real width = 11 cm × 4 = 44 m

(b) Real depth = 9.5 cm × 4 = 38 m

(c) Real floor area = 44 × 38 = 1672 m^2

Note: To measure from the plan, you would use a ruler on the printed page. The values above assume the plan printed at the intended scale.

Worked Example 4 — Reading the Delivery Map

Refer to the Springvale Delivery Map on the following page (Scale: 1 cm = 200 m).
Task: Answer the questions using the map.

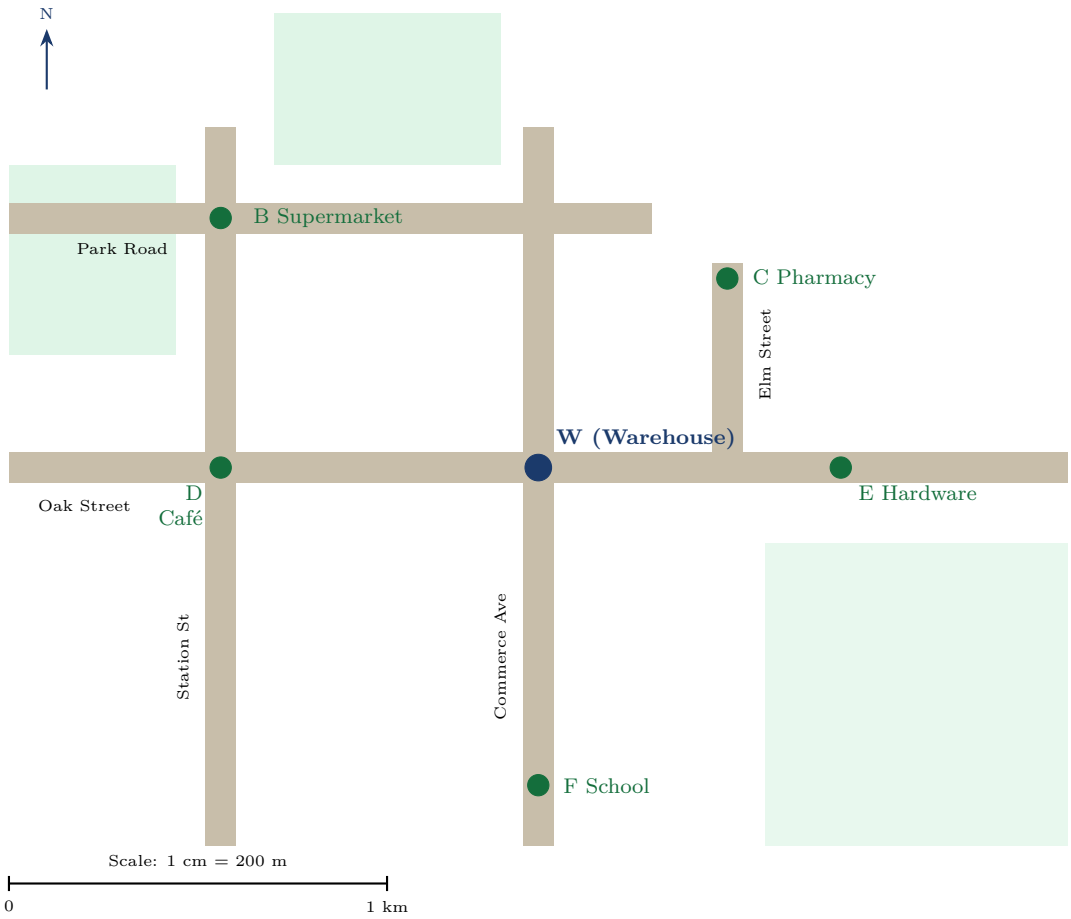
(a) **Which delivery stop is directly north of the Warehouse?**
Looking at the map, the Supermarket (B) and Pharmacy (C) are both north of the Warehouse (W) along Park Road. The Supermarket is to the north-west; the Pharmacy is to the north-east.

(b) **Estimate the straight-line distance on the map from the Warehouse (W) to the Café (D).**
The Warehouse is at approximately (7, 5) and the Café is at (2.8, 5.0) on the map grid.
Map distance ≈ 7.0 − 2.8 = 4.2 cm
Real distance = 4.2 × 200 = 840 m

(c) **In which direction would you travel from the Warehouse to the Hardware Store (E)?**
Hardware Store is to the **east** of the Warehouse (same horizontal level, further right).

Springvale — Local Delivery Map

Scale: 1 cm = 200 m (1:20 000)



Delivery Map Legend

- Warehouse / Depot (W)
- Delivery stop (B–F)
- Road
- Parks / open space

Worked Example 5 — Planning an Efficient Delivery Route

Situation: A driver leaves the Warehouse (W) and must deliver to three stops: the Café (D), the Supermarket (B) and the School (F). They must return to the Warehouse at the end.

Task: Plan the route and calculate the total distance.

Strategy: Choose a route that avoids backtracking. Look for stops that are close to each other or in the same direction. Measure each leg on the map, then multiply by the scale factor.

Efficient route: W → D → B → W → F → W

Better route (no backtrack): W → D → B → F → W

Map measurements and real distances:

Leg	Map dist. (cm)	Real distance (m)	Direction
W → D (Café)	≈ 4.2 cm	$4.2 \times 200 = 840$ m	West along Oak St
D → B (Supermarket)	≈ 3.3 cm	$3.3 \times 200 = 660$ m	North along Station St
B → F (School)	≈ 8.0 cm	$8.0 \times 200 = 1600$ m	South-east
F → W (Warehouse)	≈ 4.2 cm	$4.2 \times 200 = 840$ m	North along Commerce Ave
Total	≈ 19.7 cm	≈ 3940 m ≈ **3.9 km**	

Practice Exercise

Practice Exercise — Workplace Measurement

Part A — Volume Unit Conversions

1. Convert each volume.

(a) 3 cm^3 = _____ mm^3

(b) 12 000 mm^3 = _____ cm^3

(c) 2.5 m^3 = _____ cm^3

(d) 3 400 000 cm^3 = _____ m^3

(e) 0.008 m^3 = _____ cm^3

(f) 750 cm^3 = _____ mm^3

2. A small plastic component measures 25 mm × 15 mm × 8 mm.

(a) Calculate its volume in mm^3.

_____

_____

(b) Convert to cm^3.

_____

_____

Part B — Floor Plan Reading

Use the Springvale Warehouse floor plan to answer Questions 3–7.

3. Name the room located in the north-east section of the building.

4. How many doors are shown on the floor plan?

5. Which room is completely enclosed within the Main Warehouse?

6. The Supervisor Office measures approximately 5 cm × 2 cm on the plan. Using the scale (1 cm = 4 m), calculate its real dimensions and floor area.

7. The Main Office measures approximately 5 cm × 4 cm on the plan. Calculate its real floor area.

Part C — Delivery Map

Use the Springvale Delivery Map to answer Questions 8–12.

8. Which delivery stop is furthest south?

9. State the compass direction from the Warehouse (W) to each stop.

(a) W → Supermarket (B): _____

(b) W → Pharmacy (C): _____

(c) W → School (F): _____

10. Using the map scale (1 cm = 200 m), estimate the straight-line map distance and real distance from:

From → To	Map distance (cm)	Real distance (m)
W → Pharmacy (C)		
W → Hardware (E)		
Supermarket (B) → Pharmacy (C)		

11. A driver must visit the Pharmacy (C) and the Hardware Store (E) before returning to the Warehouse. Write the most efficient route and the total estimated distance.

12. Describe the route from the Café (D) to the School (F) using street names and compass directions. Use the map.

Application Exercise

Application Exercise — Workplace Measurement in Context

1. *(Volume conversions)* A construction engineer is specifying drainage pipes.

(a) A small fitting has a volume of 4800 mm^3. Convert to cm^3.

(b) A concrete block has a volume of 0.015 m^3. Convert to cm^3.

(c) A water tank holds 3.6 m^3. Convert to litres.

(d) A gravel delivery fills 750 000 cm^3. Convert to m^3.

2. *(Floor plan — material quantities)* Use the Springvale Warehouse floor plan (Scale: 1 cm = 4 m).

(a) Measure the Cold Store on the plan (approximately 4 cm × 4 cm). Calculate its real floor area.

(b) The Cold Store needs new floor tiles (30 cm × 30 cm each). How many tiles are needed? (Convert the real area to cm^2 first.)

(c) The Main Warehouse (approximately 11 cm × 9.5 cm on plan) is to have new racking installed. Each rack section is 2.4 m wide and runs along the full depth of the warehouse. How many rack sections can fit across the width?

(d) The entire building perimeter (outer walls) needs a weatherproofing strip. Estimate the building's perimeter from the plan (approximately 16 cm × 12 cm) and calculate the real perimeter.

3. *(Delivery map — route planning)* Use the Springvale Delivery Map (Scale: 1 cm = 200 m).
A driver starts at the Warehouse (W) and must visit all five delivery stops (B, C, D, E, F) before returning.

(a) Plan an efficient route that minimises backtracking. Write your route.

(b) Estimate the total distance in metres for your route (measure each leg on the map).

(c) Convert the total distance to kilometres.

(d) The driver averages 30 km/h in the local area. Estimate the total driving time (in minutes) for the route.

(e) Each delivery stop takes 15 minutes. Including driving time, how long does the full run take?

4. *(Scale drawing — office layout)* A new office space is 8 m long and 5 m wide. The architect's plan uses a scale of 1:50.

(a) What length (in cm) would the 8 m wall be on the plan?

(b) What length would the 5 m wall be on the plan?

(c) The real area of the office is:

(d) On the plan, a desk measures 4 cm × 2 cm. What are the real dimensions of the desk?

(e) The desk occupies what percentage of the total office floor area?

5. *(Warehouse storage planning)* The Main Warehouse at Springvale has real floor dimensions of approximately 44 m × 38 m. Pallets are stored in rows. Each pallet occupies a space of 1.2 m × 1.0 m.

(a) How many pallets fit along the 44 m length (each pallet is 1.2 m wide)?

(b) How many pallets fit along the 38 m depth (each pallet is 1.0 m deep)?

(c) What is the maximum number of pallets on the ground floor?

(d) In practice, 25% of the floor space is kept clear for access aisles and safety. How many pallets can actually be stored?

_____

_____

Chapter Review

Chapter 13 Review — Topic 2 Measurement Consolidation (Chapters 11–13)

Section A — 3D Shapes and Volume (Chapter 11 revision)

1. Calculate the volume of each shape.

(a) Rectangular prism: $L = 7$ m, $W = 4$ m, $H = 2.5$ m.

_____

_____

(b) Triangular prism: triangle base = 6 cm, height = 4 cm, prism length = 9 cm.

_____

_____

_____

2. A rectangular shipping box is 0.8 m × 0.6 m × 0.4 m. Convert its volume to cm^3.

_____

_____

_____

Section B — Mass, Capacity and Temperature (Chapter 12 revision)

3. Convert each measurement.

(a) 3.5 kg = _____g

(b) 2800 mL = _____L

(c) 4 kL = _____L

(d) 600 mg = _____g

(e) 1.2 t = _____kg

4. A food temperature probe reads 28°C inside a coolroom storing cooked ham. Is this safe? What must be done?

_____

_____

_____

Section C — Volume Unit Conversions

5. Convert each volume.

(a) 4 cm^3 = _____mm^3

(b) 9 m^3 = _____cm^3

(c) 500 000 cm^3 = _____m^3

Section D — Floor Plan and Map

Use the Springvale Warehouse floor plan and Delivery Map for Questions 6–9.

6. On the Warehouse floor plan, the Staff Room measures approximately 2 cm × 6 cm. What is its real area?

_____

_____

7. A vehicle enters the warehouse through the loading dock. Which wall is the loading dock on?

_____

8. On the Delivery Map, estimate the real straight-line distance from the Café (D) to the Supermarket (B).

_____

_____

9. Describe the direction of travel for each leg of this route: W → F → E → C → W.

_____

_____

_____

Multi-Step Problem 10

Warehouse Cold Store fit-out. The Cold Store in the Springvale Warehouse (floor dimensions approximately 16 m × 16 m real) is being fitted out with new shelving.

(a) Calculate the floor area of the Cold Store.

(b) Each shelving unit is 1.5 m wide and takes up a floor depth of 0.6 m. A 0.9 m access aisle is required in front of each row of shelving. How many shelving units fit along the 16 m wall if access aisles are included?

(c) Each shelving unit holds 8 shelves. Each shelf holds 12 boxes. Each box has dimensions 40 cm × 30 cm × 25 cm. Calculate the volume of one box in cm^3.

(d) How many boxes can the Cold Store hold in total (using your answer from part b)?

(e) The Cold Store must be kept below 5°C. The temperature log shows: 08:00 = 3°C; 10:00 = 4.5°C; 12:00 = 6°C; 14:00 = 5.5°C. At what time(s) was the temperature in the danger zone? What action should have been taken?

Multi-Step Problem 11

Full delivery run. A driver at Springvale Warehouse needs to deliver to all 5 stops (B–F) and return.

(a) Using the Delivery Map, write your planned route and estimate the total distance (in metres and km).

(b) Each stop has a delivery of boxes. The cargo weights are listed below. Does the total load exceed 1 tonne?

Stop	Boxes	Mass per box
B — Supermarket	24	18 kg
C — Pharmacy	6	8 kg
D — Café	10	12 kg
E — Hardware	15	22 kg
F — School	8	6 kg

(c) The driver leaves at 08:00 and drives at an average of 25 km/h. Each stop takes 20 minutes to unload. What time does the driver return to the warehouse?

Skills Checklist

Chapter 13 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can convert between mm^3, cm^3 and m^3	⊙	⊖	⊗
I can read a floor plan and identify rooms using the legend	⊙	⊖	⊗
I can use a scale to find real dimensions from a floor plan	⊙	⊖	⊗
I can calculate real areas from scaled plan measurements	⊙	⊖	⊗
I can identify locations on a map using compass directions	⊙	⊖	⊗
I can estimate distances using a map scale	⊙	⊖	⊗
I can plan an efficient route between multiple locations	⊙	⊖	⊗
I can solve multi-step problems combining measurement skills	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

Numeracy Short Course — Stage 4 Student Workbook
Chapter 13: Workplace Measurement | Topic 2: Workplace and Employment | Week 14, Lessons 53–56
Aligned to QCAA 2025 Senior Short Course Numeracy Syllabus v1.2

Chapter 14 — Data at Work

Stage: Stage 4 — Student Workbook
Week: Week 15
Lessons: Lessons 57–60
Strand: Topic 2: Workplace and Employment
Sub-topic: Statistics & Probability — Data Recording and Graphs

Introduction

Data is everywhere in the workplace. A retail manager tracks daily sales to order stock. A café owner monitors which items sell most. A logistics company records delivery times to improve efficiency. Being able to organise, display and interpret workplace data is an essential numeracy skill.

In this chapter you will record data in structured tables, draw bar graphs from workplace data, compare tables to make decisions, and set up a paper spreadsheet. These skills are applied directly in most Australian workplaces.

Learning Objectives — by the end of this chapter you will be able to:

- Record workplace data accurately in a correctly structured two-column table
- Construct a bar graph from a data table, including all required features (SALT)
- Read and compare information from two-column workplace tables
- Compare two supplier price tables and make a purchasing recommendation
- Set up a paper spreadsheet with headers, data, formulas and totals
- Make data-based decisions and write a short justification

Key Terms

Term	Definition
Workplace record	A document that captures data from a business activity (e.g. sales, stock, time)
Stock list	A table recording quantities and details of items held in a business
Sales record	A table showing sales data (items sold, quantities, revenue) by period
Temperature log	A record of temperature readings taken at regular intervals
Two-column table	A simple table with two columns — often category and value
Column header	The label at the top of a column that describes the data in that column
Data entry	Recording a value into the correct cell of a table
Spreadsheet	A grid of rows and columns used to record and calculate data
Cell reference	The address of a cell in a spreadsheet (e.g. B3 = column B, row 3)
SUM formula	A spreadsheet formula that adds a range of cells (`=SUM(B2:B8)`)
SALT	A graph checklist: **S**cale, **A**xes labelled, **L**egend, **T**itle

Key Rules

Data at Work — Key Rules and Formulas

Parts of a good data table:

- A clear title above the table
- Column headers that name the data
- Units shown in headers (e.g. *Sales ($)*)
- Data entered accurately in the correct row/column
- Totals row at the bottom where needed

Spreadsheet basics (on paper):

- Columns are labelled A, B, C...
- Rows are numbered 1, 2, 3...
- A cell is named column + row (e.g. C4)
- Multiply: `=B2*C2`
- Add a range: `=SUM(D2:D8)`
- Total cost = qty × unit price

Graph checklist — SALT:

Scale	**A**xes labelled	**L**egend	**T**itle
Consistent numbers on both axes	Both horizontal and vertical axes named with units	Included when more than one data set	Describes what the graph shows

Comparing two tables — steps:

1. Read the question carefully — which column/row are you comparing?
2. Extract the relevant values from each table.
3. Calculate totals or differences as needed.
4. Write a conclusion sentence: *"Based on the data, [Option A/B] is better because…"*

Worked Examples

Worked Example 1 — Recording Data in a Table

Situation: A retail assistant at Springvale Newsagency records the number of each item sold on a Monday morning.
Raw data: Newspapers: 42, Magazines: 18, Lottery tickets: 67, Stationery: 24, Confectionery: 35, Drinks: 51.
Task: Record this data in a correctly structured two-column table.

Step 1: Give the table a title.
Step 2: Create column headers — one for the category, one for the value (with units).
Step 3: Enter each data value in the correct row.
Step 4: Add a Total row and calculate the sum.

Springvale Newsagency — Monday Morning Sales

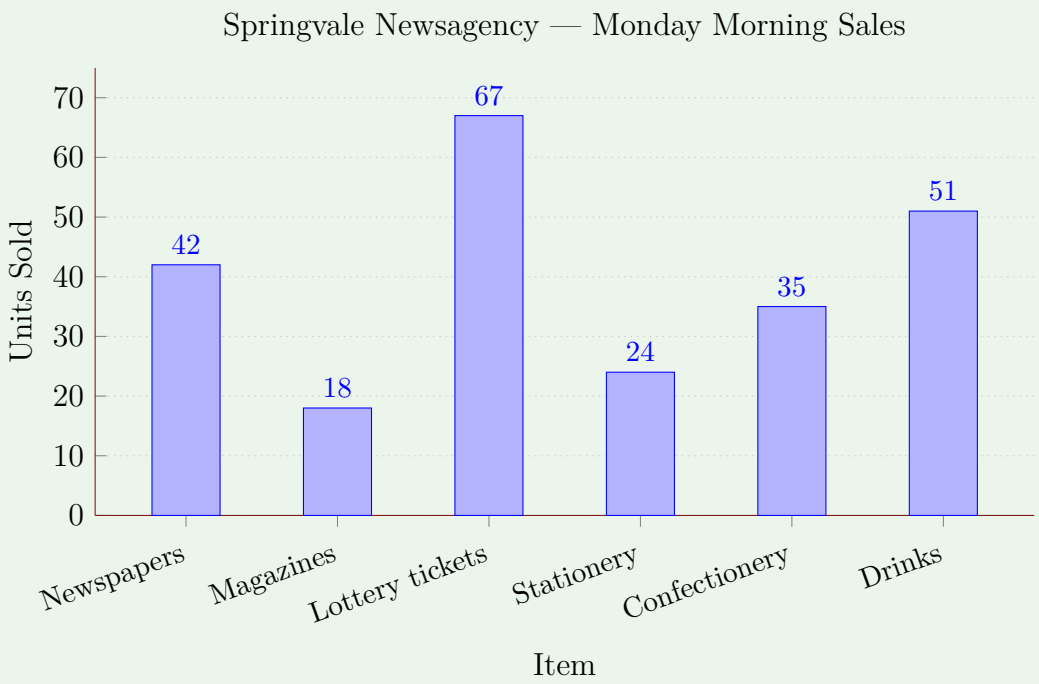
Item	Units sold
Newspapers	42
Magazines	18
Lottery tickets	67
Stationery	24
Confectionery	35
Drinks	51
Total	**237**

Key point: The table has a title, labelled columns, accurate data, and a total row.
The most popular item was **Lottery tickets** (67 units); the least popular was **Magazines** (18 units).

Worked Example 2 — Constructing a Bar Graph from Workplace Data

Situation: Use the newsagency sales data from Worked Example 1 to draw a bar graph.

Step 1: Choose the axes — categories on horizontal; units sold on vertical.
Step 2: Choose a scale that fits the data (max = 67, so count by 10s up to 70).
Step 3: Draw bars of equal width with equal gaps between them.
Step 4: Check SALT: Scale, Axes labelled, Legend (not needed here), Title.



SALT check: Scale (0–70, intervals of 10) ✓ Axes labelled ✓ Legend (not needed) ✓ Title ✓

Worked Example 3 — Reading and Comparing a Workplace Table

Situation: The table below shows weekly revenue for a Springvale café over five weeks.

Week	**1**	**2**	**3**	**4**	**5**
Food sales ($)	2840	3120	2980	3350	3200
Drink sales ($)	1250	1380	1190	1520	1440
Total ($)	4090	4500	4170	4870	4640

Questions and answers:

(a) **Which week had the highest total revenue?**
Week 4 had the highest total: $4870.

(b) **What was the mean weekly total revenue?**
$(4090 + 4500 + 4170 + 4870 + 4640) \div 5 = 22\,270 \div 5 = \$4454$

(c) **In which weeks did food sales exceed $3000?**
Weeks 2, 4 and 5 (food sales were $3120, $3350 and $3200 respectively).

(d) **What fraction of total Week 1 revenue came from drink sales?**
$\frac{1250}{4090} \approx 0.306 = 30.6\%$

Worked Example 4 — Comparing Two Supplier Tables

Situation: A café owner compares two coffee suppliers for the same weekly order.

Supplier A — Springvale Roasters

Item	Qty	Price/unit
Espresso blend (1 kg)	5	$28.40
Almond milk (L)	10	$3.20
Oat milk (L)	6	$3.80
Takeaway cups (50-pk)	4	$9.60

Supplier B — Metro Coffee Co.

Item	Qty	Price/unit
Espresso blend (1 kg)	5	$26.80
Almond milk (L)	10	$3.50
Oat milk (L)	6	$4.10
Takeaway cups (50-pk)	4	$10.20

Step 1: Calculate total cost for each item (qty × price) for each supplier.
Step 2: Sum all item totals to get the overall order total.
Step 3: Compare and write a recommendation.

Item	Supplier A total	Supplier B total
Espresso blend	$5 \times \$28.40 = \$142.00$	$5 \times \$26.80 = \$134.00$
Almond milk	$10 \times \$3.20 = \$32.00$	$10 \times \$3.50 = \$35.00$
Oat milk	$6 \times \$3.80 = \$22.80$	$6 \times \$4.10 = \$24.60$
Takeaway cups	$4 \times \$9.60 = \$38.40$	$4 \times \$10.20 = \$40.80$
Total	**$235.20**	**$234.40**

Recommendation: Supplier B is cheaper overall by $\$235.20 - \$234.40 = \$0.80$ per week. Over 52 weeks, this saves $52 \times \$0.80 = \$41.60$ per year. However, the difference is very small — the owner might prefer Supplier A for other reasons (e.g. quality, delivery reliability).

Worked Example 5 — Paper Spreadsheet Setup

Situation: Set up a stock-value spreadsheet for a small hardware store. Complete all calculations.

Springvale Hardware — Stock Value Spreadsheet

Row	A: Item	B: Units in stock	C: Unit price ($)	D: Total value ($)
1	*(Header row)*			
2	Hammers	12	24.99	`=B2*C2` = $299.88
3	Screwdrivers	30	12.50	`=B3*C3` = $375.00
4	Drill bits	50	8.40	`=B4*C4` = $420.00
5	Paint brushes	20	7.80	`=B5*C5` = $156.00
6	Sandpaper	45	3.20	`=B6*C6` = $144.00
7	**TOTAL**	`=SUM(B2:B6)` = 157	—	`=SUM(D2:D6)` = $1394.88

Key points:

- Row 1 is the header row — column letters A, B, C, D label the columns.
- Each data cell in column D uses `=B#*C#` to multiply quantity by unit price.
- The total row uses `=SUM(...)` to add all values in a column.
- On paper, you calculate the formula values manually — on a computer, the software calculates automatically.

Practice Exercise

Practice Exercise — Data at Work

Part A — Completing Tables

1. A supermarket checkout operator records the number of customers served each hour. Enter the data correctly in the table, then answer the questions.
Raw data (customers per hour, 8 am to 3 pm):
8 am: 24, 9 am: 38, 10 am: 45, 11 am: 52, 12 pm: 61, 1 pm: 48, 2 pm: 36.

Springvale Supermarket — Customers Served per Hour

Hour	Customers served
8 am	
9 am	
10 am	
11 am	
12 pm	
1 pm	
2 pm	
Total	

(a) Which hour was the busiest?

_____

(b) Which hour was the quietest?

_____

(c) How many more customers were served at 12 pm than at 8 am?

_____

(d) What was the mean (average) number of customers per hour?

_____

_____

2. A Springvale bakery records daily sales revenue for one week. Complete the table and answer the questions.

Springvale Bakery — Weekly Sales Revenue

Day	Food sales ($)	Drink sales ($)
Monday	620	185
Tuesday	540	162
Wednesday	680	204
Thursday	595	178
Friday	820	246
Saturday	1140	342
Sunday	980	294
Total		

(a) Complete the Total row.

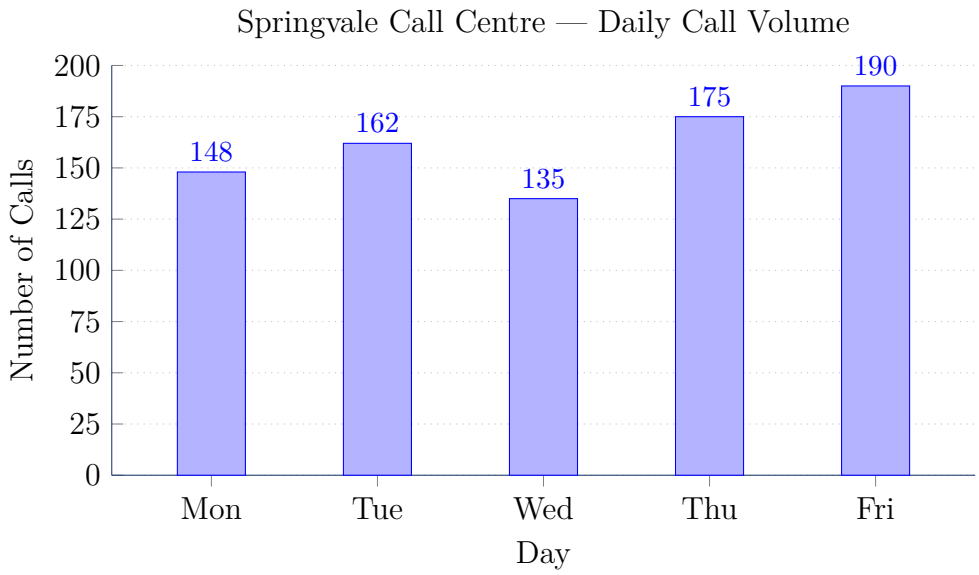
(b) Which day had the highest food sales?

(c) What fraction of Saturday's food sales came from drink sales? Simplify.

(d) What percentage of the total weekly revenue came from drinks? (Round to 1 decimal place.)

Part B — Reading a Bar Graph

Questions 3–7 refer to the bar graph below showing call centre call volumes.



3. Which day had the most calls?

4. How many calls were received in total across the week?

5. How many more calls were received on Friday than on Wednesday?

6. What was the mean number of calls per day?

7. On the busiest day, each call takes an average of 4 minutes. How many total minutes of call time was that?

Part C — Drawing a Bar Graph

8. Draw a bar graph for the bakery drink sales from Question 2 on the grid below. Include all SALT features.

Part D — Spreadsheet

9. Complete the paper spreadsheet below for a café weekly stock order.

Row	A: Item	B: Qty	C: Unit price ($)	D: Line total ($)
2	Coffee beans (kg)	4	28.50	
3	Full cream milk (2L)	15	3.40	
4	Almond milk (L)	8	3.80	
5	Sugar (kg)	3	2.60	
6	Paper cups (50-pk)	6	9.20	
7	**TOTAL**		—	

10. Using your spreadsheet totals, what is the cost per cup of coffee if the café makes 300 coffees from 4 kg of beans? (Use only the coffee bean cost.)

Application Exercise

Application Exercise — Data in Workplace Contexts

1. *(Stock record)* A Springvale pharmacy manager does a weekly stock count. Record the data in the table and answer the questions.
Stock count: Paracetamol 500mg (boxes): 48; Ibuprofen 200mg (boxes): 32; Antihistamine tablets (boxes): 19; Antiseptic cream (tubes): 27; Bandages (rolls): 55; Hand sanitiser (bottles): 63.

Springvale Pharmacy — Weekly Stock Count

Item	Units in stock
Total	

(a) Which item has the lowest stock?

(b) The reorder threshold is 20 units. Which items need to be reordered?

(c) Hand sanitiser costs $6.80 each (excl. GST). What is the current stock value (excl. GST)?

2. *(Sales bar graph)* The table shows daily sandwich sales at a school canteen for one week. Construct a bar graph on the grid provided.

Day	Sandwiches sold
Monday	38
Tuesday	44
Wednesday	52
Thursday	48
Friday	65

(a) What is the total number of sandwiches sold for the week?

(b) What is the mean daily sales figure?

(c) On which day would you recommend the canteen prepare the most sandwiches? Justify using the data.

3. *(Comparing supplier tables)* A restaurant owner compares two suppliers for a weekly produce order.

Supplier A — Fresh Direct QLD

Item	Qty	Price
Tomatoes (kg)	10	$3.20/kg
Lettuce (head)	8	$2.80 ea
Capsicum (kg)	5	$4.50/kg
Mushrooms (kg)	4	$8.40/kg
Avocados (ea)	12	$1.90 ea

Supplier B — Metro Produce Co.

Item	Qty	Price
Tomatoes (kg)	10	$3.60/kg
Lettuce (head)	8	$2.50 ea
Capsicum (kg)	5	$4.20/kg
Mushrooms (kg)	4	$7.80/kg
Avocados (ea)	12	$2.10 ea

(a) Calculate the total cost from each supplier. Show all working.

(b) Which supplier is cheaper overall, and by how much?

(c) Explain one reason the restaurant owner might still choose the more expensive supplier.

4. *(Staffing data)* The table shows staff hours worked and hourly rates at Springvale Café for one week.

Row	A: Staff member	B: Hours	C: Rate ($/hr)	D: Gross pay ($)
2	Anika	18	26.80	
3	Brennan	22	26.80	
4	Carla	15	24.80	
5	Diego	25	24.80	
6	Elena	12	26.80	
7	**TOTAL**	`=SUM(B2:B6)`	—	`=SUM(D2:D6)`

(a) Complete column D by calculating each staff member's gross pay.

(b) Complete the totals row (B7 and D7).

(c) Who worked the most hours?

_____

(d) What is the total weekly wages bill?

_____

(e) Calculate the mean hours worked per staff member this week.

_____

_____

5. *(Temperature log)* A kitchen supervisor records the coolroom temperature every two hours.

Springvale Restaurant — Coolroom Temperature Log

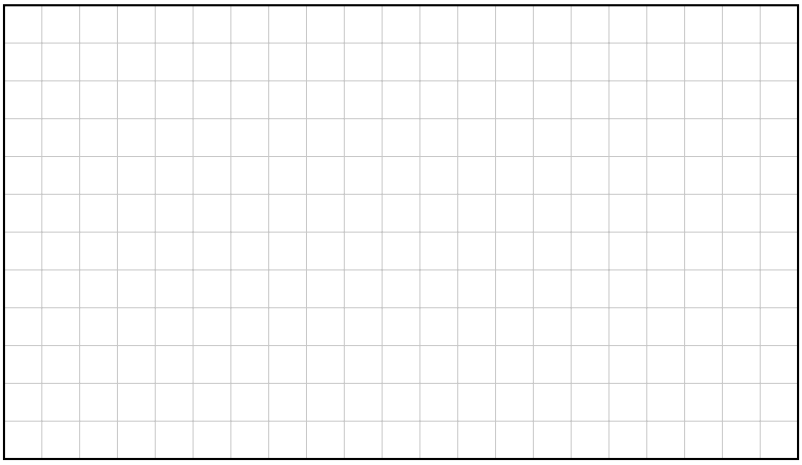
Time	Temperature (°C)	Within safe range (<5°C)?
06:00	3	
08:00	4	
10:00	6	
12:00	5	
14:00	7	
16:00	4	
18:00	3	

(a) Complete the "Within safe range?" column.

(b) At what times was the coolroom outside the safe range?

(c) What action should the supervisor take at those times?

(d) Construct a line graph of the temperature data on the grid below. Mark the 5°C safety line.



6. *(Identifying trends)* A retail manager reviews monthly sales data for two products.

Month	Jan	Feb	Mar	Apr	May	Jun
Product X sales	240	255	270	285	298	310
Product Y sales	180	165	148	132	120	105

(a) Describe the trend for Product X.

(b) Describe the trend for Product Y.

(c) Based on the data, which product should the manager order more of in July? Justify your answer.

Chapter Review

Chapter 14 Review — Data at Work

1. A building manager logs the number of maintenance requests received each day. Raw data (Mon–Fri): Mon: 8, Tue: 12, Wed: 6, Thu: 15, Fri: 11.

(a) Record the data in the table below.

Day	**Requests**
Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Total	

(b) Which day had the most requests?

(c) Calculate the mean number of requests per day.

2. Draw a bar graph for the maintenance data from Question 1 on the grid below.

3. Complete the stock value spreadsheet below.

Row	A: Product	B: Units	C: Unit price ($)	D: Total value ($)
2	Safety helmets	25	18.50	
3	Hi-vis vests	40	12.80	
4	Safety boots	15	85.00	
5	Work gloves	60	9.40	
6	**TOTAL**		—	

4. What is the total stock value from Question 3?

5. Which item has the highest total stock value?

6. Two office supply companies quote for the same order:

Item (Qty)	Quote A	Quote B
A4 paper, 5 reams	$6.40/ream	$5.90/ream
Pens, 3 boxes	$8.80/box	$9.60/box
Folders, 10	$2.20 ea	$2.50 ea
Staples, 4 boxes	$4.60/box	$4.10/box

(a) Calculate the total for each quote.

(b) Which quote is cheaper? By how much?

Multi-Step Problem 7

Retail store performance. Springvale Sports Store records daily revenue for two weeks.

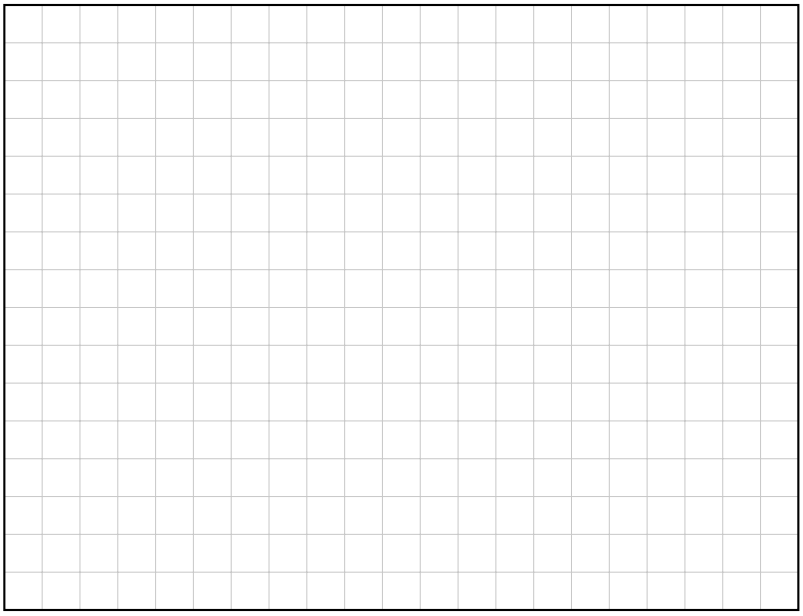
Day	Week 1 ($)	Week 2 ($)
Monday	1240	1380
Tuesday	980	1050
Wednesday	1120	1290
Thursday	1350	1420
Friday	1680	1750
Saturday	2450	2680
Sunday	2180	2340
Total		

(a) Complete the Total row for both weeks.

(b) Which week had higher total revenue? By how much?

(c) Calculate the mean daily revenue for each week.

(d) On the grid below, draw a grouped bar graph comparing the two weeks. Use different shading for Week 1 and Week 2 bars, and include a legend.



(e) Describe the overall trend visible in the grouped bar graph.

(f) The store manager says: *"Revenue is growing week on week."* Does the data support this? Write 2–3 sentences using specific figures.

Multi-Step Problem 8

Café monthly report. A Springvale café manager prepares a monthly performance report using two data sources.

Table 1 — Revenue by category:

Category	Week 1	Week 2	Week 3	Week 4
Hot drinks	1240	1380	1190	1450
Cold drinks	620	750	580	820
Food	2840	3120	2680	3380
Weekly total				

Table 2 — Staff hours and wages:

Staff	Hours/week	Rate ($/hr)	Weekly wage ($)
Manager	38	34.50	
Barista	30	27.80	
Wait staff (3 people)	20 each	24.80	
Total weekly wages			

(a) Complete both tables (calculate weekly totals in Table 1; calculate wages in Table 2).

(b) What is the total monthly revenue across all 4 weeks?

(c) What is the total monthly wages bill (4 weeks)?

(d) Calculate the monthly profit if costs (wages + other expenses of $3200/month) are subtracted from total revenue.

(e) In which week was food revenue highest? What might explain this?

(f) Write a 3–4 sentence summary for the manager describing the month's performance, using figures from both tables.

Skills Checklist

Chapter 14 — Skills Checklist

Rate your confidence. Circle the face that best describes how you feel.

Skill	Got it!	Nearly	Need help
I can record data accurately in a structured two-column table	⊙	⊖	⊗
I can construct a bar graph with correct SALT features	⊙	⊖	⊗
I can read a data table and answer comparison questions	⊙	⊖	⊗
I can identify the highest and lowest values in a data table	⊙	⊖	⊗
I can compare two supplier tables and determine the better value	⊙	⊖	⊗
I can complete a paper spreadsheet using multiply and SUM	⊙	⊖	⊗
I can describe a trend in workplace data using mathematical language	⊙	⊖	⊗
I can write a data-based recommendation or conclusion	⊙	⊖	⊗

My goal before the next lesson:

_____

_____

15. Probability and Trends at Work

Chapter 15 Information

Chapter:	15 — Probability and Trends at Work
Week:	Week 16
Topic:	Topic 2 — Workplace and Employment
Skills Area:	S&P — Statistics and Probability
Lesson Reference:	Lessons 61–64

Introduction

In the workplace, two key mathematical skills help workers and managers make better decisions: **probability** and **trend analysis**. Probability tells us how likely something is to happen — such as the chance of a faulty product on a production line. Trend analysis means reading graphs over time to see whether something is increasing, decreasing, or staying the same. Together, these skills help businesses plan ahead, manage risk, and make evidence-based recommendations. In this final chapter you will apply both skills in realistic Australian workplace contexts.

Learning Objectives

By the end of this chapter you will be able to:

- Identify situations in the workplace where probability applies
- Calculate probability for workplace events (defect rates, random selection)
- Express probability as a fraction, decimal and percentage
- Identify and describe trends in line graphs (increasing, decreasing, stable, seasonal)
- Write conclusions from trend data using the sentence stem guide
- Make basic workplace recommendations based on data

Key Terms

Key Vocabulary

Probability	How likely an event is to occur, written as a fraction, decimal or percentage.
Defect rate	The proportion of products that fail quality control.
Quality control	The process of checking products to ensure they meet required standards.
Random selection	Choosing an item or person where every choice has an equal chance of being selected.
Trend	The general direction in which data is moving over time (up, down or stable).
Seasonal variation	A regular pattern that repeats at certain times of year (e.g. summer peaks).
Forecast	A prediction about future values based on a current trend.
Conclusion	A statement that summarises what the data shows and what it means for the business.
Recommendation	An action suggested based on the evidence in the data.

Key Formulas

- $\Pr(E) = \dfrac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ (revised from Chapter 8)
- **Defect rate:** $\Pr(\text{defect}) = \dfrac{\text{number of defective items}}{\text{total items checked}}$
- **Expected defects:** Expected $= \Pr(\text{defect}) \times$ total items produced
- **Converting probability:**
  - Fraction $\rightarrow$ Decimal: divide the top by the bottom (e.g. $\frac{3}{8} = 3 \div 8 = 0.375$)
  - Decimal $\rightarrow$ Percentage: multiply by 100 (e.g. $0.375 \times 100 = 37.5\%$)
- **Reading a trend:** Compare the start point and end point of a line graph. Identify the highest and lowest points. Describe the overall direction.

Writing Conclusions: Sentence Stem Guide

Use these sentence starters when writing a professional conclusion from a graph or table:

- **The data shows that...** *(describe the trend you can see)*
- **Between [period] and [period], [quantity] increased/decreased by...**
- **The highest/lowest point was in [month/period], when [value]...**
- **This suggests that...** *(explain what the trend means for the business)*
- **I recommend that...** *(state an action based on what the data shows)*

A good conclusion uses **3–4 sentences** and refers to specific values from the graph.

Worked Examples

Worked Example 1 — Calculating a Defect Rate Probability

Worked Example 1

Context: A quality control officer at Springvale Food Packaging inspects a batch of 200 biscuit packets. She finds that 6 packets are defective (torn seal or damaged packaging).

(a) What is the probability that a randomly selected packet is defective?
(b) Write this probability as a fraction (simplified), decimal and percentage.

Step 1 — Identify the values

- Number of defective packets (favourable outcomes) = 6
- Total packets inspected (total outcomes) = 200

Step 2 — Write as a fraction and simplify

$$\Pr(\text{defect}) = \frac{6}{200} = \frac{3}{100}$$

Divide numerator and denominator by 2 to simplify.

Step 3 — Convert to a decimal

$$\frac{3}{100} = 3 \div 100 = 0.03$$

Step 4 — Convert to a percentage

$$0.03 \times 100 = 3\%$$

Answer: $\Pr(\text{defect}) = \frac{3}{100} = 0.03 = 3\%$
The defect rate is **3%**, meaning approximately 3 in every 100 packets will be defective.

Worked Example 2 — Calculating the Expected Number of Defects

Worked Example 2

Context: Using the defect rate from Worked Example 1 ($\Pr(\text{defect}) = 0.03$), the factory runs a production batch of 1 500 packets.

(a) How many packets are expected to be defective in this production run?
(b) Each defective packet costs \$2.40 to replace. What is the total expected replacement cost?

Step 1 — Write the formula

$$\text{Expected defects} = \Pr(\text{defect}) \times \text{total items}$$

Step 2 — Substitute and calculate

$$\text{Expected defects} = 0.03 \times 1{,}500 = 45 \text{ packets}$$

Step 3 — Calculate the replacement cost

$$\text{Replacement cost} = 45 \times \$2.40 = \$108.00$$

Answer: 45 packets are expected to be defective, costing **$108.00** in replacements.

Key Insight

Knowing the probability (defect rate) lets you predict how many problems to expect in *any* batch size. This helps factories plan for waste, set budgets, and decide whether a process needs improvement.

Worked Example 3 — Expressing Probability as a Fraction, Decimal and Percentage

Worked Example 3

Context: A warehouse team at Northgate Logistics has 8 workers. Three workers hold a forklift licence. The team leader randomly selects one worker to operate the forklift for the afternoon.

(a) What is the probability the selected worker holds a forklift licence?
(b) What is the probability the selected worker does **not** hold a forklift licence?
(c) Express both probabilities as fractions, decimals and percentages.

Step 1 — Identify the outcomes

- Total workers = 8
- Workers with forklift licence = 3
- Workers without forklift licence = $8 - 3 = 5$

Step 2 — Write both probabilities as fractions

$$\Pr(\text{has licence}) = \frac{3}{8} \qquad \Pr(\text{no licence}) = \frac{5}{8}$$

Check: $\frac{3}{8} + \frac{5}{8} = \frac{8}{8} = 1$ ✓ (probabilities always add to 1)

Step 3 — Convert to decimals and percentages

Event	Fraction	Decimal	Percentage
Has forklift licence	$\frac{3}{8}$	0.375	37.5%
No forklift licence	$\frac{5}{8}$	0.625	62.5%

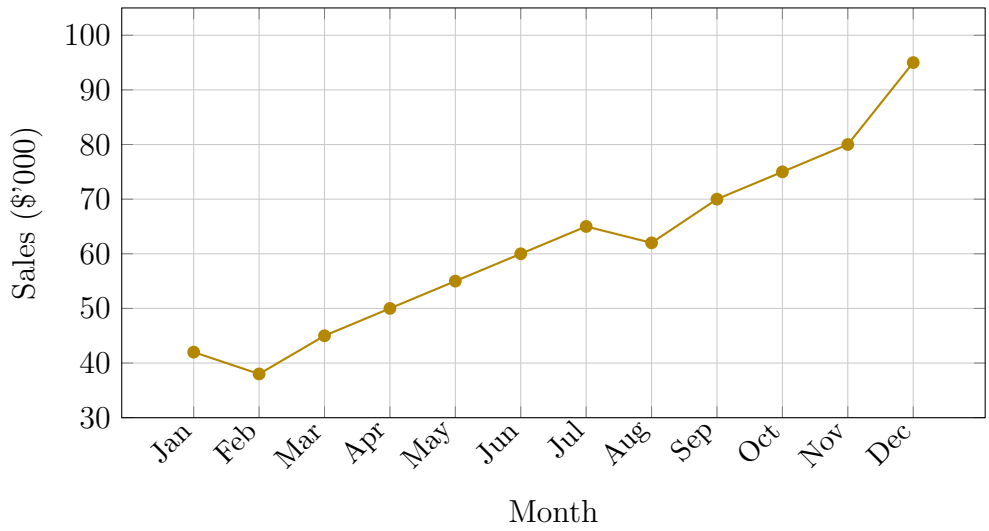
Answer: There is a **37.5%** chance the randomly selected worker holds a forklift licence.

Worked Example 4 — Reading a Line Graph and Identifying Trends

Worked Example 4

Context: The graph below shows monthly sales figures ($'000) for Springvale Hardware Store during 2024. Study the graph carefully and answer the questions.

Springvale Hardware Store — Monthly Sales 2024



(a) In which month were sales highest? _____

(b) In which month were sales lowest? _____

(c) Describe the overall trend in sales during 2024.

_____

_____

(d) Between which two consecutive months did sales increase by the greatest amount? State the increase in $'000.

_____

Answers

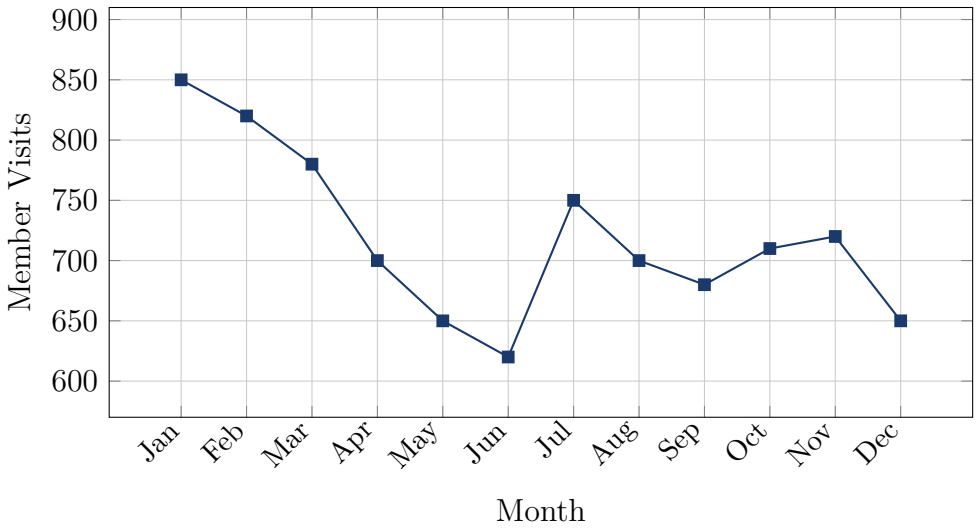
- **(a)** December ($95 000)
- **(b)** February ($38 000)
- **(c)** Sales show a **generally increasing trend** throughout 2024. There is a slight dip in February and again in August, but the overall direction is strongly upward, with the fastest growth in the final quarter of the year.
- **(d)** November to December: $95,000 − $80,000 = $15,000 increase.

Worked Example 5 — Writing a Conclusion from a Trend Graph

Worked Example 5

Context: The graph below shows monthly member visits to BrightFit Gym in Brisbane during 2024. Use the sentence stem guide (page 1) to write a 3–4 sentence conclusion about the trend.

BrightFit Gym, Brisbane — Monthly Member Visits 2024



Write a 3–4 sentence conclusion about this trend.

_____

_____

_____

_____

Model Answer (using sentence stems)

"The data shows that member visits to BrightFit Gym generally decreased throughout 2024, falling from 850 visits in January to 650 visits in December. The highest number of visits occurred in January, which likely reflects New Year's fitness resolutions. There was a notable increase in July (750 visits), which may be related to school holiday activities, before visits declined again in the second half of the year. This suggests that the gym experiences lower demand from August onwards, and I recommend that management introduce promotional membership offers in October and November to boost attendance before the Christmas period."

Worked Example 6 — Making a Recommendation Based on Trend Data

Worked Example 6

Context: The manager of Springvale Hardware Store has reviewed the 2024 monthly sales data from Worked Example 4. She needs to write a staffing and stock ordering recommendation for the store owner.

Using the Springvale Hardware Store data, write a recommendation covering:

- When the store is busiest and what the data shows
- What action should be taken before the busy period
- What staffing changes might be needed

Step 1 — Identify key facts from the data

- Sales increase significantly from September onward
- Strongest growth: November to December (from $80k to $95k)
- Lowest sales in January–February ($38k–$42k)
- Clear pattern: the store is busiest in spring/summer and quietest in late summer

Step 2 — Write the recommendation using sentence stems

"The data shows that Springvale Hardware Store experiences its highest sales in the final quarter of 2024, peaking at $95 000 in December. This suggests that demand is strongest during the lead-up to summer and the Christmas holiday period. I recommend that the store significantly increase its stock orders in October and November to ensure adequate inventory through the peak period. Additionally, I recommend rostering additional casual staff in November and December to manage the higher customer demand and maintain service quality."

Practice Exercise

Practice Exercise — Probability and Trends at Work

Part A: Probability Calculations

1. A packing line at Coastal Cannery checks 250 tins of tuna. 10 are found to have faulty seals.

(a) What is the probability that a randomly selected tin has a faulty seal?

Fraction (simplified): _____ Decimal: _____ Percentage: _____

(b) Describe in words what this probability means.

_____

2. Using the defect rate from Question 1, if the factory produces 2 000 tins in a shift, how many faulty tins are expected?

Working:

_____

Expected faulty tins: _____

3. A call centre in Brisbane has 12 staff members: 5 full-time and 7 part-time. One person is randomly selected to attend a training session.

(a) What is the probability of selecting a full-time staff member?

Fraction: _____ Decimal (2 d.p.): _____ Percentage: _____

(b) What is the probability of *not* selecting a full-time staff member?

Fraction: _____ Decimal (2 d.p.): _____ Percentage: _____

(c) Do the two probabilities from (a) and (b) add to 1? Show this check.

_____

4. A café roster covers shifts across 5 weekdays. If a casual worker is randomly assigned to one day, what is the probability they work on Wednesday?

Fraction: _____ Decimal: _____ Percentage: _____

5. A quality audit bag contains 20 items: 8 pass inspection, 7 need minor repair, and 5 are scrapped. If one item is drawn at random, find:

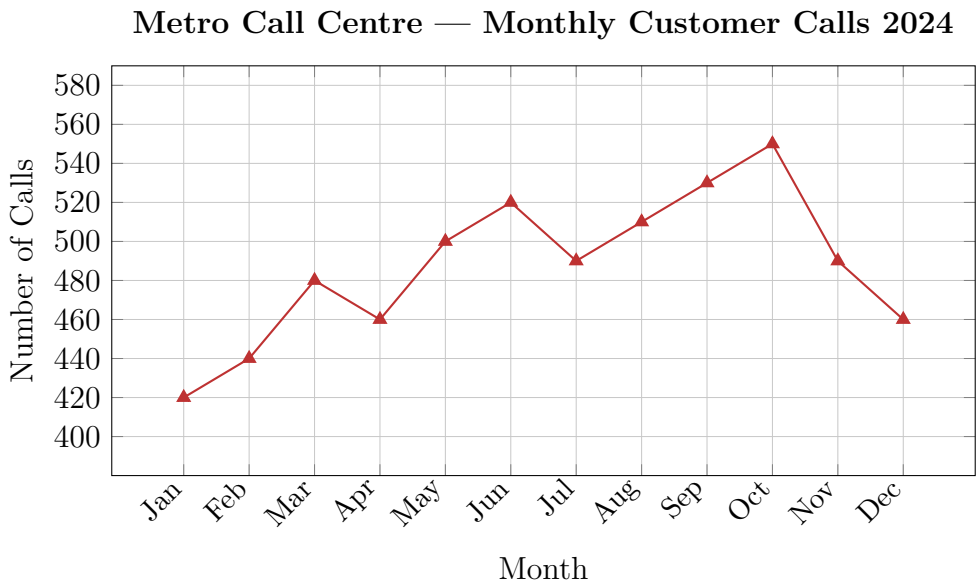
(a) Pr(pass) as a fraction: _____

(b) Pr(scrap) as a fraction and percentage: Fraction: _____ %: _____

(c) Pr(item is NOT scrapped) as a fraction: _____

Part B: Reading a Line Graph

The graph below shows the number of customer calls received by Metro Call Centre each month during 2024.



6. In which month were call volumes highest? _____

7. In which month were call volumes lowest? _____

8. How many calls were received in March? _____

9. By how much did call volumes increase from January to October?

Working: _____ Increase: _____

10. Complete the sentence using one of the following words:

increasing *decreasing* *stable* *fluctuating*

"Overall, the number of customer calls at Metro Call Centre during 2024 was _____, with some variation from month to month."

Application Exercise

Application Exercise — Workplace Probability and Trends

Question 1: Quality Control at Northside Bakery

A bakery produces 480 loaves of bread per day. Quality control checks find that 12 loaves are undercooked or misshapen and must be discarded.

(a) Calculate the defect rate as a fraction (simplified), decimal and percentage.

_____

Fraction: _____ Decimal: _____ Percentage: _____

(b) If the bakery increases production to 720 loaves one day, how many defective loaves should they expect?

Working:

_____

Expected defective loaves: _____

(c) Each defective loaf costs the bakery $3.50 in ingredients. What is the total expected cost of defective loaves over a 5-day working week (at normal production of 480 loaves/day)?

Working:

_____

_____

Total weekly cost: _____

Question 2: Random Rostering

Sunrise Café has 15 casual staff available for weekend shifts: 6 on Friday night, 5 on Saturday, and 4 on Sunday. The manager randomly selects one staff member from the full casual pool to receive a bonus shift.

(a) What is the probability of selecting a Saturday staff member?

Fraction: _____ Decimal (2 d.p.): _____ Percentage: _____

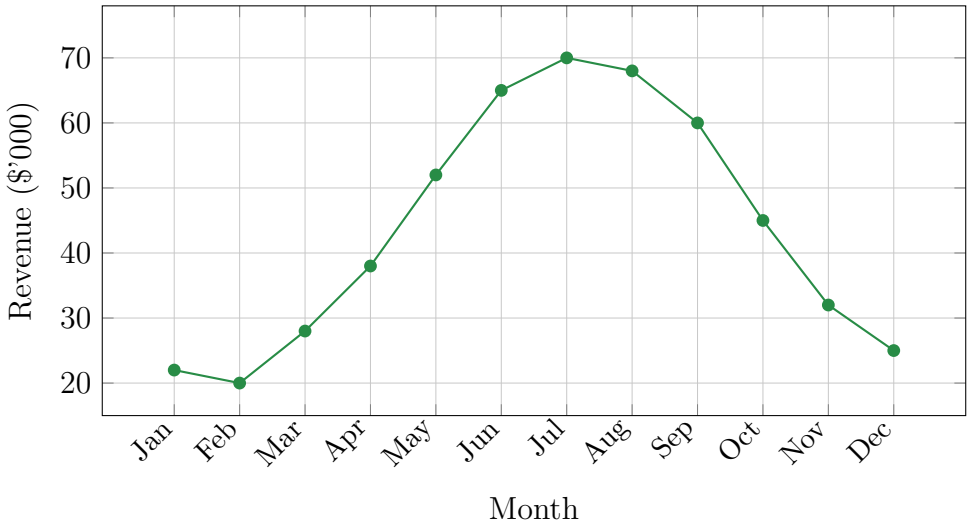
(b) What is the probability of selecting a staff member who is **not** rostered on Sunday?

Fraction: _____ Decimal (2 d.p.): _____ Percentage: _____

Question 3: Garden Centre Seasonal Sales Trend

The graph below shows monthly revenue ($'000) for Greenthumb Garden Centre in Queensland during 2024.

Greenthumb Garden Centre — Monthly Revenue 2024 ($'000)



(a) Identify the peak revenue month and state the revenue.

Month: _____ Revenue: _____

(b) Identify the lowest revenue month and state the revenue.

Month: _____ Revenue: _____

(c) Describe the seasonal trend shown in the graph. When does the garden centre perform best?

_____

_____

(d) Write a recommendation for stock ordering and staffing based on this seasonal trend. Use at least one sentence stem.

_____

_____

_____

Question 4: Drawing and Interpreting a Trend Graph

The table below shows the average customer waiting time (in minutes) at Eastside Medical Centre over 6 months after a new appointment system was introduced.

Month	Jan	Feb	Mar	Apr	May	Jun
Wait time (min)	24	22	19	15	13	11

(a) Plot the data on the blank grid below. Draw a line graph. Label both axes with units and give the graph a title.

(b) Describe the trend in waiting times.

(c) Write a 3–4 sentence professional conclusion using the sentence stem guide.

Question 5: Stock Ordering from a Trend

A hardware store manager notices that sales of garden hoses increase each year from August to November. The table shows sales data for two years:

Month	Aug	Sep	Oct	Nov
2023 Sales (units)	45	78	120	145
2024 Sales (units)	52	90	138	162

(a) By how many units did November sales increase from 2023 to 2024?

(b) Based on the two-year trend, predict approximate November 2025 sales. Explain your reasoning.

(c) Write one recommendation for the manager about placing a stock order before November.

Question 6: Improving Quality Control

An electronics factory produces circuit boards. The quality control supervisor has recorded defect data over three months. Complete the table, then answer the questions.

Month	Total Produced	Defective	Defect Rate (fraction)	Defect %
July	1 200	36	_____	_____
August	1 500	30	_____	_____
September	1 800	18	_____	_____

(a) Complete the Defect Rate column (as a simplified fraction) and the Defect % column.

(b) Describe the trend in defect rates across the three months.

(c) If this trend continues, predict the approximate defect rate for October.

(d) Write one recommendation for the factory manager based on this data.

Chapter Review

Chapter Review — Chapters 14 and 15: Statistics and Probability

Part A: Data and Tables (Chapter 14 Revision)

1. A retail assistant records sales during a morning shift. Complete the table and answer the questions.

Product Category	Number Sold
Clothing	24
Footwear	12
Accessories	18
Electronics	6
TOTAL	_____

(a) Total items sold: _____

(b) Which category had the highest sales? _____

(c) What fraction of total sales were Electronics? Write as a simplified fraction.

_____

2. A supervisor records the number of workplace incidents per week over 6 weeks: 5, 3, 4, 2, 4, 3.

(a) Calculate the mean number of incidents per week.

_____

Mean = _____ incidents per week.

(b) In Week 7, there are 0 incidents. Recalculate the mean over all 7 weeks.

_____

New mean = _____ incidents per week.

Part B: Probability (Chapter 15)

3. A grocery distribution centre processes 600 delivery orders per day. From past records, 18 orders contain picking errors.

(a) Calculate the probability of a picking error. Write as a fraction (simplified), decimal and percentage.

Fraction: _____ Decimal: _____ Percentage: _____

(b) If the centre processes 1 000 orders during a busy period, how many picking errors are expected?

Working:

_____

Expected errors: _____

4. A café team has 10 employees: 4 baristas, 3 wait staff and 3 kitchen hands. One employee is randomly selected to attend a food safety refresher course.

(a) Find Pr(barista selected) as a fraction, decimal and percentage.

Fraction: _____ Decimal: _____ Percentage: _____

(b) Find Pr(kitchen hand is NOT selected) as a fraction.

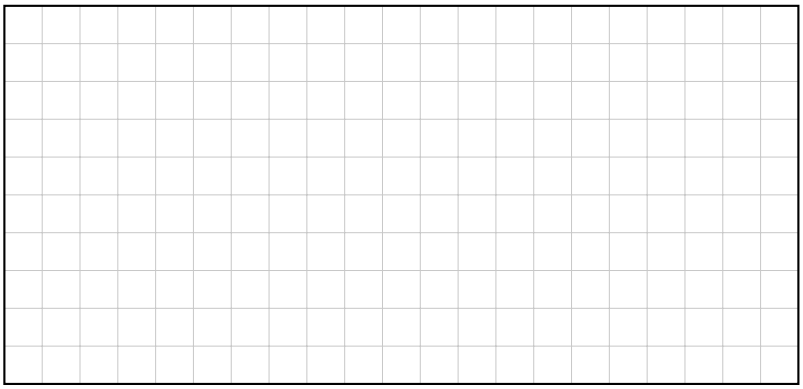
_____

Part C: Line Graph Trends (Chapter 15)

5. The table shows monthly units sold at a sporting goods store from July to December.

Month	Jul	Aug	Sep	Oct	Nov	Dec
Units sold	120	145	175	200	230	290

(a) Plot this data on the blank grid below and draw a line graph. Label axes with units and give the graph a title.



(b) Describe the trend.

_____

(c) By how much did units sold increase from July to December?

_____

6. A manager says: *"Our defect rate is 3%. In a batch of 300 items, we will get exactly 9 defects."* Is this statement correct? Explain your answer.

_____

_____

7. Why is it more useful for a workplace report to express probability as a percentage rather than a fraction? Give one example.

_____

_____

Multi-Step Problems

Multi-Step Problem 1 — Retail Store Two-Year Trend Analysis

A retail store manager provides revenue data for two years (January to June):

Month	Jan	Feb	Mar	Apr	May	Jun
2023 Revenue ($'000)	48	45	52	58	60	62
2024 Revenue ($'000)	55	52	60	66	70	75

(a) Calculate the total revenue for each year (January to June only).

Working:

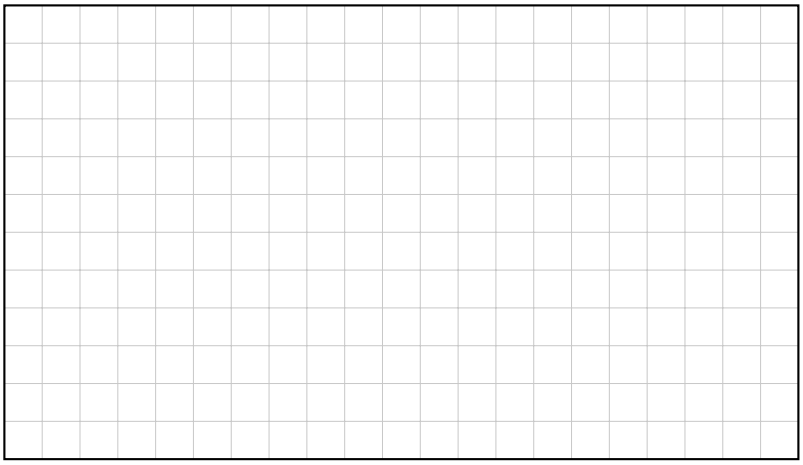
_____

2023 Total: _____ 2024 Total: _____

(b) By how much ($'000) did the January–June total revenue increase from 2023 to 2024?

_____

(c) Plot **both** years of data on the blank grid below as two separate line graphs. Use a different colour or line style for each year. Label both axes with units, include a legend, and give the graph a title.



(d) Are the trends in 2023 and 2024 similar or different? Describe what you see.

_____

(e) Write a 3–4 sentence manager's recommendation based on this two-year comparison. Use the sentence stem guide.

Multi-Step Problem 2 — Quality Control and Cost Analysis

A manufacturing company produces sports drink bottles. A quality audit finds 15 defective bottles in a sample of 500.

(a) Calculate the defect rate as a fraction (simplified), decimal and percentage.

Fraction: _____ Decimal: _____ Percentage: _____

(b) The factory runs three production shifts per day, each producing 1 200 bottles. How many defective bottles are expected across all three shifts in one day?

Working:

Expected defects per day: _____

(c) Each defective bottle costs \$1.85 to recycle or dispose of. Calculate the total expected daily disposal cost.

Working:

Disposal cost per day: _____

(d) The company introduces a new quality control process. After one week, the defect rate drops to 1.5%. How many fewer defective bottles should be expected per day? (Keep the same 3-shift production of 3 600 bottles/day.)

Working:

Reduction in defects: _____ bottles per day.

Daily cost saving: _____

(e) Write a 3–4 sentence conclusion about the impact of the new quality control process. Include: the change in defect rate, the reduction in defective bottles, the cost saving per day, and a recommendation.

_____

_____

_____

_____

Chapter 15 Skills Checklist

Chapter 15 Skills Checklist

Rate your confidence for each skill by ticking the column that best describes you:

Skill	Not yet	Getting there	Got it!
I can calculate the probability of a workplace event (e.g. defect rate) as a fraction	☐	☐	☐
I can express probability as a fraction, decimal and percentage	☐	☐	☐
I can calculate the expected number of defects in a production run	☐	☐	☐
I can read a line graph and identify the peak, lowest point and trend	☐	☐	☐
I can describe a trend using correct language (increasing, decreasing, stable, seasonal)	☐	☐	☐
I can write a 3–4 sentence conclusion using the sentence stem guide	☐	☐	☐
I can make a workplace recommendation based on trend data	☐	☐	☐
I can combine probability and trend skills in a multi-step problem	☐	☐	☐

My learning goal for revision:

_____

_____

Congratulations — you have completed all 15 chapters of Numeracy!

Assessment Task 2: Examination Preparation

Examination Details

Type:	Individual, supervised, open-book examination
Perusal time:	5 minutes (reading only — no writing permitted)
Working time:	45 minutes
Permitted materials:	This workbook, your notes, exercise books, the QCAA Formula Book, and a calculator
Question style:	Short-response questions set in real-world contexts from both Topic 1 and Topic 2
When:	Weeks 17–18 (date to be advised by your teacher)

Open-book means: You may use your workbook and notes during the exam. However, you must still understand the material — you will not have time to re-learn skills during the exam. Preparation is essential.

Grade Cut-Offs

Grade	Mark Range
A	Greater than 80%
B	Greater than 60%
C	Greater than 40%
D	Greater than 20%
E	0% or above

Assessment Criteria

Grade	Descriptor
A	Demonstrates comprehensive understanding of numeracy skills in both topics; selects appropriate and efficient strategies consistently; communicates mathematical thinking clearly and with precision
B	Demonstrates thorough understanding of numeracy skills; selects appropriate strategies for most problems; communicates mathematical thinking clearly
C	Demonstrates understanding of most numeracy skills; uses suitable strategies for most tasks; communicates mathematical thinking adequately
D	Demonstrates some understanding of numeracy skills; uses strategies with partial success; communicates some mathematical thinking
E	Demonstrates limited understanding of numeracy skills; attempts to use strategies with limited success; communicates limited mathematical thinking

Open-Book Exam Tips

7 Strategies for Open-Book Exam Success

1. **Tab your workbook before the exam.** Use sticky notes or tabs to mark each chapter so you can find skills instantly. Know exactly where your Key Formulas boxes are.
2. **Use perusal time wisely.** You may not write during perusal, but you can read every question and mentally locate where the relevant skills appear in your workbook.
3. **Read each question carefully.** Identify the numeracy skill required before you start calculating. Underline key information and circle the question being asked.
4. **Always show your working.** Method marks may be awarded even if the final answer is incorrect. Write each step clearly, with units.
5. **Estimate first, then calculate.** Use rounding to check your answer makes sense before writing it. A price should never be negative; a probability should never exceed 1.
6. **Check your answer in context.** Ask: *Does this answer make sense in the real world?* A gross pay of \$5 per day or a mass of -3 kg would both indicate an error.
7. **Manage your time.** If you are stuck on a question, move on and return to it later. Do not spend more than 2–3 minutes on any single question before moving forward.

Suggested Revision Plan

Use this table to plan your revision across the weeks before the examination. Tick each area as you revise it.

Revision Area	Chapters	Revised	Confident
Numbers and operations	1	☐	☐
Fractions, decimals, %	2	☐	☐
Rates and ratios	3	☐	☐
Length, perimeter and area	4	☐	☐
Shapes and maps	5	☐	☐
Time and angles	6	☐	☐
Data and graphs	7	☐	☐
Probability (basics)	8	☐	☐
Pay and timesheets	9	☐	☐
Workplace numbers	10	☐	☐
3D shapes and volume	11	☐	☐
Mass, capacity, temperature	12	☐	☐
Workplace measurement	13	☐	☐
Data at work	14	☐	☐
Probability and trends	15	☐	☐

Skills Checklist — Full Course

Use this checklist to track your readiness for the examination. Review any skills marked “I need to revise” before the exam.

Topic 1 Skills: Personal Identity and Community (Ch. 1–8)

Skill	I can do this	I need to revise
Read, write and order whole numbers	☐	☐
Identify the place value of digits	☐	☐
Add, subtract, multiply and divide whole numbers	☐	☐
Estimate answers and use a calculator effectively	☐	☐
Simplify fractions and find equivalent fractions	☐	☐
Convert between fractions, decimals and percentages	☐	☐
Calculate a percentage of an amount	☐	☐
Find discounted prices and apply 10% GST	☐	☐
Calculate rates and unit prices	☐	☐
Write, simplify and scale ratios	☐	☐
Convert between metric length units (mm, cm, m, km)	☐	☐
Calculate perimeter of rectangles, squares and triangles	☐	☐
Calculate area of rectangles, squares and triangles	☐	☐
Identify 2D shapes and describe their properties	☐	☐
Convert between area units (mm^2, cm^2, m^2, ha, km^2)	☐	☐
Read a map using a scale, key and compass directions	☐	☐
Convert between 12-hour and 24-hour time	☐	☐
Calculate elapsed time and read timetables	☐	☐
Identify, classify and estimate angles	☐	☐
Construct and read tally charts and frequency tables	☐	☐
Draw and interpret bar graphs and line graphs	☐	☐
Calculate probability and express it as a fraction, decimal and %	☐	☐

Topic 2 Skills: Workplace and Employment (Chapters 9–15)

Skill	I can do this	I need to revise
Calculate gross pay from hours worked and an hourly pay rate	☐	☐
Complete a timesheet and interpret a pay slip	☐	☐
Calculate superannuation and net pay from a pay slip	☐	☐
Apply GST, markup and discount in workplace contexts	☐	☐
Read and complete a workplace invoice	☐	☐
Mix and scale ratios in workplace contexts	☐	☐
Identify 3D shapes and describe faces, edges and vertices	☐	☐
Calculate the volume of rectangular and triangular prisms	☐	☐
Convert between mass units (mg, g, kg, t)	☐	☐
Convert between capacity units (mL, L, kL, ML)	☐	☐
Apply food safety temperature guidelines (danger zone 5°C–60°C)	☐	☐
Convert between volume units (mm^3, cm^3, m^3)	☐	☐
Read and interpret a workplace floor plan with a scale	☐	☐
Read and interpret a delivery/road map with a scale	☐	☐
Record data in a two-column frequency table	☐	☐
Construct a bar graph from workplace data (SALT checklist)	☐	☐
Compare two data sets using tables and graphs	☐	☐
Set up a paper spreadsheet with multiply and SUM formulas	☐	☐
Calculate defect rates using the probability formula	☐	☐
Calculate the expected number of defects in a production run	☐	☐
Identify and describe trends in line graphs	☐	☐
Write a 3–4 sentence conclusion using sentence stems	☐	☐
Make a workplace recommendation based on data	☐	☐

Note: Practice Examination questions will be provided by your teacher in Weeks 17–18. Complete the practice examination under timed conditions to build confidence.

Key Formula Quick-Reference

Use this page to write your own summary of the most important formulas before the exam. Your teacher may also direct you to the QCAA Formula Book during the examination.

N&A — Number and Algebra

Gross pay	= hours worked × hourly rate
GST (adding)	= original price × 1.10
GST amount only	= original price × 0.10
GST (removing)	= GST-inclusive price ÷ 1.10
Markup price	= cost price × (1 + markup%)
Discount price	= original price × (1 − discount%)
Unit price	= total price ÷ number of units
Superannuation	= gross pay × 0.115 (11.5% rate, 2025)

M&S — Measurement and Space

Perimeter of rectangle	$= 2(l + w)$
Area of rectangle	$= l \times w$
Area of triangle	$= \frac{1}{2} \times b \times h$
Volume of rect. prism	$= l \times w \times h$ (or base area × height)
Volume of tri. prism	$= \frac{1}{2} \times b \times h \times$ length
Length conversions	×1000: km → m → mm; ÷1000 to go back
Mass conversions	×1000: t → kg → g → mg; ÷1000 to go back
Capacity conversions	×1000: ML → kL → L → mL; ÷1000 to go back
Volume conversions	×1,000,000: $m^3 \to cm^3$; ×1000: $cm^3 \to mm^3$
Scale (maps)	Map distance × scale factor = real distance
Food safety danger zone	5 °C to 60 °C (Australian Food Standards Code)

S&P — Statistics and Probability

Probability	$\Pr(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$
Defect rate	$\Pr(\text{defect}) = \dfrac{\text{defective items}}{\text{total items}}$
Expected defects	$= \Pr(\text{defect}) \times \text{total produced}$
Fraction → Decimal	Divide top by bottom
Decimal → Percentage	Multiply by 100
Mean (average)	$= \dfrac{\text{sum of all values}}{\text{number of values}}$
SALT checklist (graphs)	**S**cale | **A**xes labelled | **L**egend | **T**itle
Trend language	Increasing / Decreasing / Stable / Seasonal

My Exam Preparation Notes

My Exam Preparation Notes

Use this page to record key formulas, strategies and things to remember. Tab the most important pages in your workbook and write the chapter or page reference here.

Before You Leave for the Exam — Final Checklist

- ☐ My workbook is tabbed and I know where each Key Formulas box is
- ☐ I have a working calculator (with charged batteries or new batteries)
- ☐ I have reviewed all skills I marked as "I need to revise"
- ☐ I have completed at least one full practice examination under timed conditions
- ☐ I know to show all working, even for simple calculations
- ☐ I understand how to read exam questions carefully and identify what is being asked
- ☐ I am ready. I have prepared well and I know my material.

Key Formulas Reference Sheet

Topic 1 — Personal Identity and Community (Chapters 1–8)

Chapter 1 — Numbers in Everyday Life

BODMAS: Brackets → Orders → Division/Multiplication (L to R) → Addition/Subtraction (L to R)

Rounding: Look at the digit to the *right*. 5 or more → round up. 4 or less → round down.

Chapter 2 — Fractions, Decimals and Percentages

Fraction → decimal: divide top by bottom
Decimal → %: $\times 100$ % → decimal: $\div 100$

$$\text{Percentage of amount} = \text{Amount} \times \frac{\%}{100}$$

$$\text{Sale price} = \text{Original} \times \left(1 - \frac{\text{Disc. }\%}{100}\right)$$

Chapter 3 — Rates, Ratios and Comparisons

$$\text{Rate} = \frac{\text{Quantity}}{\text{Unit}} \qquad \text{Unit price} = \frac{\text{Total cost}}{\text{Qty}}$$

$$D = S \times T \qquad S = \frac{D}{T} \qquad T = \frac{D}{S}$$

Simplify ratio: Divide both parts by HCF.

Share in a ratio: (1) Find total parts. (2) Total ÷ parts = value of 1 part. (3) Multiply each share.

Chapter 4 — Length, Perimeter and Area

Length: mm $\xrightarrow{\div 10}$ cm $\xrightarrow{\div 100}$ m $\xrightarrow{\div 1000}$ km

$$P_{\text{rect}} = 2L + 2W \quad P_{\text{sq}} = 4S \quad P_{\triangle} = a + b + c$$

$$A_{\text{rect}} = L \times W \quad A_{\text{sq}} = S^2 \quad A_{\triangle} = \tfrac{B \times H}{2}$$

Chapter 5 — Shapes, Area and Maps

Area conversions:
$1\,\text{cm}^2 = 100\,\text{mm}^2 \qquad 1\,\text{m}^2 = 10{,}000\,\text{cm}^2$

Scale / maps:

$$\text{Actual} = \text{map length} \times \text{scale factor}$$

Chapter 6 — Time and Angles

12-hr → 24-hr: am = same (4 digits); pm = add 12; midnight = 00:00
Elapsed time = End time − Start time

Angle types:
Acute 0°–90° Right = 90° Obtuse 90°–180°
Straight = 180° Reflex 180°–360°

Chapter 7 — Data and Graphs

Graph checklist — SALT:
Scale **A**xes labelled **L**egend **T**itle

Count tallies → frequency → total must equal all items.

Chapter 8 — Chance and Probability

$$\Pr(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

$$\Pr(E) = \frac{\text{times event occurred}}{\text{total trials}}$$

$0 \le \Pr(E) \le 1 \qquad \Pr(\text{even chance}) = 0.5 = 50\%$

Topic 2 — Workplace and Employment (Chapters 9–15)

Chapter 9 — Numbers at Work

$$\text{Gross pay} = \text{hours} \times \text{hourly rate}$$

$$\text{Net pay} = \text{Gross pay} - \text{Total deductions}$$

$$\text{Fortnightly pay} = 2 \times \text{weekly pay}$$

Key deductions (2025): Super = 11.5% of gross; Casual loading = 25% on base rate.

Chapter 10 — Workplace Numbers

GST: incl. = price × 1.10 excl. = total ÷ 1.10

Markup/discount:

$$\text{New price} = \text{original} \times (1 \pm r)$$

where r is the rate as a decimal.

Ratio mixing: qty/part $= \dfrac{\text{total volume}}{\text{total parts}}$

Chapter 11 — 3D Shapes and Volume

$$V_{\text{prism}} = A_{\text{base}} \times H \qquad V_{\text{rect}} = L \times W \times H$$

$$V_{\triangle\text{-prism}} = \tfrac{b \times h}{2} \times L$$

Conversions: $1\,\text{cm}^3 = 1000\,\text{mm}^3$ $1\,\text{m}^3 = 10^6\,\text{cm}^3$
$1\,\text{cm}^3 = 1\,\text{mL}$ $1\,\text{m}^3 = 1000\,\text{L}$

Chapter 12 — Mass, Capacity and Temperature

Mass: mg $\xrightarrow{\times 1000}$ g $\xrightarrow{\times 1000}$ kg $\xrightarrow{\times 1000}$ t (reverse: ÷1000)

Capacity: mL $\xrightarrow{\times 1000}$ L $\xrightarrow{\times 1000}$ kL $\xrightarrow{\times 1000}$ ML

Food safety: Cold < 5°C Danger 5°–60°C Hot > 60°C

Chapter 13 — Workplace Measurement

$$\text{Real length} = \text{plan length} \times \text{scale factor}$$

Common scales: 1:50 = 1 cm → 0.5 m; 1:100 = 1 cm → 1 m; 1:200 = 1 cm → 2 m

Chapter 14 — Data at Work

SALT: **S**cale **A**xes labelled **L**egend **T**itle

Spreadsheet: `=B2*C2` multiply `=SUM(D2:D8)` add range

Compare tables: Extract values → calculate totals/differences → write conclusion.

Chapter 15 — Probability and Trends at Work

$$\Pr(\text{defect}) = \frac{\text{defective items}}{\text{total items checked}}$$

$$\text{Expected defects} = \Pr(\text{defect}) \times \text{total produced}$$

Trend reading: Compare start and end points. Identify highest/lowest. Describe overall direction (increasing, decreasing, stable).

Conclusion stem: *"The data shows that... Between [period] and [period], [value] increased/decreased by..."*

Answer Section

Notes: [**timetable**] = refer to Route 72 timetable in workbook. [**map**] = measure from map/floor plan in workbook. [**graph**] = refer to graph in workbook. Questions requiring graph construction, map measurement, or extended written responses are noted as *Student construction* or *Accept reasonable response*.

Topic 1 — Personal Identity and Community (Chapters 1–8)

Chapter 1 — Numbers in Everyday Life

Practice Exercise

1. Four thousand, five hundred and seven

2. One million, thirty thousand and sixty

3. 62 315 **4.** 9 400 007

5. Ten thousands **6.** 300 000

7. (a) 2 (b) 8 (c) 5

8. $45\,087 < 45\,678 < 45\,768 < 45\,870$

9. $1\,200\,200 > 1\,200\,000 > 1\,020\,000 > 1\,002\,000$

10. 6 353 **11.** 7 664 **12.** 8 760 **13.** 234

14. Estimate: $500 + 300 = 800$; Exact: 801

15. Estimate: $70 \times 50 = 3\,500$; Exact: 3 216

16. $(12 + 8) \times 5 - 14 \div 2 = 20 \times 5 - 7 = 93$

Application Exercise

1. \$190 **2.** \$367 **3.** 2 344 000

4. 1 754 km **5.** \$4 752

6. (a) 28 (b) 31

7. 2 220 MB

8. (a) 28 per group (b) 336 total

9. \$1 000 **10.** 400 m^2

11. (a) 100 000 (b) 100 000

12. (a) 14 190 MB (b) Yes each month; $11\,350 < 15\,000$ and $2\,840 < 15\,000$

Chapter Review

1. One million, two hundred and four thousand, seven hundred and fifty

2. 600 000 **3.** $330\,400 > 303\,400 > 303\,040 > 300\,340$

4. 9 523 **5.** 11 625 **6.** 6 000 **7.** 333

8. Estimate: $900 + 500 = 1\,400$; Exact: 1 401

9. 6 054 km

10. (a) \$4 275 (b) \$775 more needed

11. (a) \$260 left/week (b) \$3 120

12. \$1 675

Chapter 2 — Fractions, Decimals and Percentages

Practice Exercise

1. $\frac{2}{3}$ (HCF = 3) **2.** $\frac{3}{5}$ **3.** $\frac{3}{4}$

4. 12 **5.** 21

6. Yes: $\frac{4}{10} = \frac{2}{5} = 0.4$ and $\frac{6}{15} = \frac{2}{5} = 0.4$

7. $\frac{1}{4}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ (0.25, 0.5, 0.667, 0.75)

8. Student conversion table

9. \$30 **10.** \$20 **11.** 140 items
12. $\frac{3}{5}$; 60%
13. 0.85; $\frac{17}{20}$
14. 7 absent
15. $\frac{2}{3} \approx 0.667 > 65\% = 0.65$; $\frac{2}{3}$ is greater

Application Exercise

1. 25% off \$48 = \$12 off; sale price \$36
2. 450 mL drunk; 300 mL remaining
3. 85%
4. (a) $\frac{13}{20}$ (b) 65% (c) 35%
5. \$1 039.20
6. 18.5%
7. (a) \$34 (b) \$374
8. Milo: 90%; Maths: 95%; maths test is higher
9. (a) 90 students (b) 62.5%
10. (a) \$20 (b) \$2 GST
11. (a) \$25.50 (b) \$36 (c) \$143.50
12. (a) 40% (b) 25% (c) $\frac{1}{4}$

Chapter Review

1. $\frac{2}{3}$ **2.** 0.375; 37.5%
3. 0.45; $\frac{9}{20}$ **4.** 25
5. $\frac{1}{3}$, $\frac{3}{8}$, 0.4, 42% (0.333, 0.375, 0.4, 0.42)
6. \$91 **7.** 90% **8.** GST \$34; total \$374
9. \$55.25
10. (a) \$135 (b) \$13.50 (c) \$148.50
11. (a) 500 mL (b) 375 mL (c) 375 mL
12. Store 2: \$849×0.90 = \$764.10 (better)

Chapter 3 — Rates, Ratios and Comparisons

Practice Exercise

1. 75 km/h **2.** 15 pages/h **3.** 45 L/h
4. 60 km **5.** \$0.70 per 100 g
6. 2 L: \$2.20/L; 1.5 L: \$2.30/L; 2 L is better value
7. 5 : 3 **8.** 2 : 3
9. (a) 3 : 5 (b) 2 : 3 (c) 3 : 4 : 5
10. 12 : 16 (scale factor 4)
11. 4 cups sugar
12. Cordial 0.5 L; Water 2 L
13. $\frac{7}{20}$; 35% **14.** 0.375; 37.5%

Application Exercise

1. 400 g: \$0.90/100 g; 700 g: \$0.85/100 g; 700 g better value
2. 30 km/h
3. (a) 17 h (b) 18 h total
4. Flour 6 cups, milk 3 cups, eggs 9
5. (a) 4 L blue (b) 14 L total
6. Jess \$360; Sam \$240
7. (a) 6 kg (b) 30 kg (c) 54 kg
8. Plan A: $\frac{1}{3}$ GB/\$; Plan B: 0.308 GB/\$; Plan A better
9. A: \$266.67; B: \$400; C: \$533.33
10. (a) 250 mL: \$0.72; 600 mL: \$0.70 (b) 600 mL (c) \$1.20 more
11. (a) 1.5 L (b) 2.5 L (c) $\frac{3}{8}$
12. (a) 5 : 8 (b) 0.38 (c) 61.5%

Chapter Review

1. 80 km/h **2.** \$0.38 per 100 mL

3. 1 kg: \$0.62/100 g; 1.5 kg: \$0.58/100 g; 1.5 kg better

4. 3 : 2 **5.** 20 **6.** 0.72; $\frac{18}{25}$

7. 0.833; 83.3%

8. First part = 20; second part = 50

9. 36 L

10. Rice 7.5 cups; beans 5 cups; spice 2.5 cups

11. Tom \$72; Lucy \$120

12. (a) \$240 (b) \$180 (c) \$300 (d) $\frac{5}{12}$

13. Plan X: 0.833 GB/\$; Plan Y: 0.842 GB/\$; Plan Y better

14. (a) 2.4 L (b) 4 L (c) \$14.32

Chapter 4 — Length, Perimeter and Area

Practice Exercise

1. (a) 3 500 m (b) 4.5 m (c) 8.2 cm (d) 240 cm (e) 600 m (f) 1.25 km

2. (a) 26 m (b) 26 cm (c) 27 cm (d) 33 m

3. (a) 36 m^2 (b) 49 cm^2 (c) 30 m^2 (d) 21 cm^2

4. Student counting (grid-dependent)

Application Exercise

1. (a) 26 m (b) \$312

2. (a) 17.1 m^2 (b) \$769.50

3. (a) 17.5 m^2 (b) \$140

4. (a) 356 m (b) 7 480 m^2

5. (a) 10 m^2 (b) 1.08 m^2 (c) 8.92 m^2

6. (a) 15 tiles/side (b) 225 tiles (c) \$787.50

7. (a) 33.8 m (b) \$928.40

8. (a) 8.64 m^2 (b) 0.09 m^2 (c) 96 tiles (d) \$460.80

9. 9.36 m^2

10. 285 m^2

11. (a) 7 m (b) 49 m^2

12. (a) 27 m × 12 m (b) 324 m^2 (c) 74 m^2

Chapter Review

1. Student conversion table

2. 23 m **3.** 38 cm **4.** 21 cm

5. 66 m^2 **6.** 72.25 cm^2

7. 63 m^2 **8.** 1 560 cm^2

9. (a) 20.8 m^2 (b) \$1 040 total

10. (a) 216 m^2 (b) \$1 512 (c) 60 m (d) \$1 320 (e) \$2 832

11. 576 cm^2

12. Bed A: 28 m^2; Bed B: 30 m^2; Bed B larger by 2 m^2

Chapter 5 — Shapes, Area and Maps

Practice Exercise

1. (a) Rectangle (b) Isosceles triangle (c) Rhombus (d) Circle

2. (a) 4 (b) 2 (c) 3 (d) Infinite

3. (a) 30 000 cm^2 (b) 8 m^2 (c) 5 000 000 m^2 (d) 2.5 cm^2 (e) 4 000 cm^2 (f) 2.5 km^2

4–6. [Springvale map] **7.** (a) 150 m (b) 320 m (c) 80 m

8. 4.5 cm on map

Application Exercise

1–7. [Springvale map — directions and distances depend on map measurements]

8. (a) 4.5 m × 3 m (b) 13.5 m^2

9. 0.0071 km^2

10. (a) 25 m (b) 6 250 000 cm^2

11. (a) 3.5 m × 2.5 m (b) 8.75 m^2

12. 6 m^2 = 60 000 cm^2

Chapter Review

1. 4 sides; 4 right angles; 2 lines of symmetry

2. Isosceles triangle (or similar valid answer)

3. 62 000 cm^2 **4.** 35 m^2 **5.** 1 500 000 m^2

6–7. [Springvale map]

8. 350 m **9.** 2 km

10. (a) 540 m^2 (b) 5 400 000 cm^2

11. (a) 6 m × 4 m (b) 24 m^2 (c) $1 248

12. [Springvale map — student route and measured distances]

Chapter 6 — Time and Angles

Practice Exercise

1. (a) 07:30 (b) 13:15 (c) 23:50 (d) 00:00 (e) 12:30

2. (a) 6:45 am (b) 2:30 pm (c) 9:05 pm (d) 12:20 am (e) 12:00 pm

3. (a) 2 h 30 min (b) 2 h 30 min (c) 2 h 35 min (d) 3 h 35 min

4–7. [Route 72 timetable]

8. 16 days **9.** 15 days **10.** June 17

11. Wednesday

12. (a) Acute (b) Straight (c) Reflex (d) Right (e) Obtuse (f) Full rotation

13. Student drawings

Application Exercise

1. 6 h 25 min

2. (a) 7 h 45 min (b) $189.88

3–4. [Route 72 timetable]

5. 16:20 (4:20 pm)

6. (a) June 23 (b) Tuesday

7. 17 days **8.** June 26

9. (a) 180°, straight (b) 90°, right

10. (a) Acute (b) 90°, right angle

11. (a) 180° (b) Yes, now facing opposite direction

12. [Route 72 timetable]

Chapter Review

1. 16:20 **2.** 00:15

3. 5:45 pm **4.** 3:30 am

5. 2 h 35 min

6. 9:50 pm (21:50)

7–8. [Route 72 timetable]

9. 5 Mondays (1, 8, 15, 22, 29 June)

10. 18 days

11. (a) Acute; e.g. knife blade angle (b) Right; corner of room (c) Reflex; clock hand past 6

12. [Route 72 timetable]

13. (a) 6 h 35 min (b) 5 h 50 min remaining

Chapter 7 — Data and Graphs

Practice Exercise

1–3. From data (3, 7, 2, 5, 8, 3, 7, 10, 2, 5, 7, 3, 8, 5, 7, 2, 10, 3, 5, 7):
Mode = 7; Pr(7 or more) = $\frac{7}{20}$

4–11. [Graph in workbook]

12. Student line graph construction (check SALT)

13. Accept reasonable trend description

14. Mean = $\frac{50}{5} = 10\,\text{km}$

15. (a) Mean = $\frac{92}{6} \approx 15.3$ (b) $19 > 15.3$, raises the mean

Application Exercise

1. Coffee: 10; Tea: 5; Juice: 5; Smoothie: 4 (total 24)

2. (a) $\frac{10}{24} = \frac{5}{12}$ (b) $\frac{5}{24} \approx 20.8\%$

3. Student bar graph

4–7. [Data tables in workbook — student calculations and graphs]

Chapter Review

1. (a) 8 (b) 11

2. Student frequency table

3. (a) 1 pet (mode) (b) $\frac{7}{20}$ (2 or more pets)

4. [Graph in workbook — add 15 to Tennis total]

5. (a) Mean = $\frac{210}{7} = 30°\text{C}$ (b) Week 2 (31.5°C) was hotter

6. A

7. (a) Line graph (change over time) (b) Bar graph (comparing categories)

8. (a) Student line graph (b)–(c) [Data in workbook] (d) Mean = $\frac{3810}{12} = 317.5\,\text{kWh}$ (e) January cost = January kWh × \$0.28 [data in workbook]

Chapter 8 — Chance and Probability

Practice Exercise

1. (a) Certain (b) Impossible (c) Even chance (d) Likely (e) Impossible (f) Even chance

2. Student probability scale

3. (a) {1,2,3,4} (b) {Mon,Tue,Wed,Thu,Fri} (c) {M,A,T,H,S} (d) {Summer,Autumn,Winter,Spring}

4. 8 outcomes

5. (a) 12 (b) $\frac{1}{3}$ (c) 0.5 (d) $16.\overline{6}\%$ (e) 0 (no orange counters) (f) $\frac{1}{2}$

6. (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) 1 (certain)

7. Student conversion table

8. Pr(A)=0.4; Pr(C)=0.48; Pr(B)=0.65; Pr(D)=0.75 (least to most likely)

9. (a) $\frac{32}{50} = \frac{16}{25}$ (b) $\frac{9}{25}$ (c) Yes, $\frac{16}{25} + \frac{9}{25} = 1$

10. (a) $\frac{8}{60} = \frac{2}{15}$ (b) $\frac{1}{6}$ (c) Close but slightly below theoretical

Application Exercise

1. [Data table in workbook]

2. (a) $\frac{5}{200} = \frac{1}{40}$; 2.5% (b) $\frac{1}{10}$; 10% (c) Sofia; by 7.5% (d) \$40

3. (a) Student spinner diagram (b) $\frac{1}{10}$; 0.1; 10% (c) $\frac{2}{5}$ (d) $\frac{7}{10}$ (e) 40 times

4. (a) $\frac{1}{4}$ (b) $\frac{1}{13}$ (c) $\frac{1}{2}$ (d) $\frac{1}{52}$ (e) Red card ($\frac{1}{2}$) more likely than Club ($\frac{1}{4}$)

5. (a) $\frac{6}{400} = \frac{3}{200}$; 1.5% (b) 48 faulty/shift (c) 1.5% < 2%; within target

6. (a) $\frac{14}{20} = \frac{7}{10}$ (b) $\frac{128}{200} = \frac{16}{25}$ (c) Tom's (more trials give better estimate) (d) 0.64 vs theoretical 0.5; biased toward Heads

Chapter Review

1. (a) Unlikely (b) Certain (c) Even chance (d) Impossible (e) Likely

2. (a) 10 (b) $\frac{2}{5}$ (c) 0.2 (d) 30% (e) $\frac{4}{5}$

3. (a) {1,2,3,4,5} (b) $\frac{1}{5}$ (c) $\frac{2}{5}$ (d) $\frac{3}{5}$

4. 20%, $\frac{1}{3}$, 45%, 0.70, $\frac{9}{10}$

5. (a) 35% (b) Unlikely (< 50%) (c) $1 - 0.35 = 0.65$

6. [Die-roll table in workbook]

7. (a) $\frac{11}{20}$ (b) $\frac{4}{5}$ (c) Frog $\frac{1}{4}$ more likely than bear $\frac{1}{5}$

Topic 2 — Workplace and Employment (Chapters 9–15)

Chapter 9 — Numbers at Work

Practice Exercise

1. (a) \$82.20 (b) \$134.00 (c) \$189.80 (d) \$276.00 (e) \$108.45

2. \$179.20 **3.** \$172.20

4. Mon: 4.5 h; Tue: 5 h; Wed: 5 h; Thu: 5.5 h; Fri: 5 h

5. Total 25 h; Gross = \$620.00

6. \$1 240.00

7–9. [Timesheet/payslip in workbook — student completion]

10. Total gross pay from Q9 data

Application Exercise

1. (a) Tue: \$141.00; Thu: \$141.00; Sat: \$211.50 (b) \$493.50 (c) \$441.50 (d) \$987.00

2. (a) Job A: \$496.00; Job B: \$422.40 (b) Job A by \$73.60 (c) Job B (\$26.40/h) (d) Job A pays more weekly despite lower hourly rate

3. [Timesheet in workbook]

4. (a) Riley \$1 398.40; Sam \$1 177.60; Taylor \$1 472.00 (b) \$4 048.00 (c) \$1 349.33

5. (a) \$371.40/week (b) 25.5 h in fortnight (c) \$674.80

6. [Payroll table in workbook]

Chapter Review

1. (a) \$152.40 (b) \$144.00 (c) \$192.80

2. Shift 1: 6 h; Shift 2: 6 h; Shift 3: 6 h

3. Tue: \$136.00; Thu: \$122.40; Sat: \$163.20; Sun: \$170.00; Total: \$591.60

4. \$535.60

5. \$1 183.20

6. 11.5% × \$591.60 = \$68.03

7. \$487.00

8. \$504 ÷ \$31.50 = 16 hours

9. Gross = total earnings before deductions; net = take-home pay after tax and other deductions

Chapter 10 — Workplace Numbers

Practice Exercise

1. (a) 6 parts (b) 2 L/part (c) 2 L fertiliser (d) 10 L water (e) Yes: 2+10=12

2. 6 cups: soy 2, honey 3, vinegar 1;
12 cups: soy 4, honey 6, vinegar 2;
18 cups: soy 6, honey 9, vinegar 3

3. White 15 L; Blue 5 L

4. Toolbox: GST \$6.50, total \$71.50;
Safety boots: GST \$14.80, total \$162.80;
Work gloves: GST \$2.25, total \$24.75;
First aid kit: GST \$8.40, total \$92.40;
Hi-vis vest: GST \$3.68, total \$40.48

5. (a) pre-GST \$40, GST \$4 (b) \$90, \$9 (c) \$120, \$12 (d) \$250, \$25

6. (a) \$72.00 (b) \$175.00 (c) \$51.75 (d) \$162.00 (e) \$160.00

7. (a) \$86.40 (b) \$95.04

8–9. [Invoice/price list in workbook]

10. \$187.60 remaining

Application Exercise

1. (a) Paint 12 L, water 3 L (b) Paint 18 L, water 2 L (c) 18 × \$28.50 = \$513.00

2. [Concentrate table in workbook]

3. [Invoice in workbook]

4. (a) Supplier A: 50×\$38×1.10 + \$25×1.10 = \$2 117.50 (b) Supplier B: 50×\$44 = \$2 200.00 (c) Supplier A cheaper by \$82.50

5. (a) [Table in workbook] (b) \$72×1.60×0.80×1.10 = \$101.38 (c) Yes, selling at \$92.16 vs wholesale \$72; profit \$20.16

6. [Catering price list in workbook]

Chapter Review

1. Concentrate 2 L; Water 6 L

2. Cement 5 bags; Sand 15 bags; Gravel 25 bags

3. (a) \$82.50 (b) \$264.00 (c) \$20.35

4. (a) Pre-GST \$50.00, GST \$5.00 (b) Pre-GST \$330.00, GST \$33.00

5. (a) \$432.00 (b) \$388.80 (c) \$427.68

6. [Invoice in workbook]

7. Incorrect. \$407 ÷ 1.10 = \$370 pre-GST; GST = \$37 (not \$40)

Chapter 11 — 3D Shapes and Volume

Practice Exercise

1. (a) Rectangular prism (b) Triangular prism (c) Cylinder (d) Cube

2. [Properties table — student completion]

3. Faces: 8; Edges: 18; Vertices: 12

4. (a) 60 cm^3 (b) 48 m^3 (c) 420 cm^3 (d) 0.75 m^3 (e) 15 000 cm^3

5. $H = 3\,000 \div (25 \times 12) = 10$ cm

6. (a) $A = 12$ cm^2; $V = 120$ cm^3 (b) $A = 22.5$ m^2; $V = 180$ m^3 (c) $A = 56$ cm^2; $V = 1\,120$ cm^3

7. (a) 5 000 000 cm^3 (b) 8 L (c) 2 500 mL (d) 3 000 L (e) 0.45 m^3

8. 160 000 cm^3 = 160 L

Application Exercise

1. (a) 5.04 m^3 (b) \$932.40 (c) \$1 025.64

2. (a) 33.12 m^3 (b) 2.16 m^3 (c) 15 pallets

3. (a) Student diagram (b) 0.54 m^3 (c) 3 bags

4. (a) Small 3 000 cm^3; Large 13 125 cm^3 (b) 4 small boxes (c) Room = 28 800 000 cm^3; 500 boxes = 1 500 000 cm^3; yes, all fit

5. (a) 9 m^3 (b) 9 000 L (c) 3 600 L currently in (d) 108 minutes to fill

6. (a) A: 45 m^3; B: 56 m^3 (b) B larger (c) A: \$1.89/m^3; B: \$1.61/m^3 (d) B offers better value

Chapter Review

1. Triangular prism; triangular base

2. 10 faces (2 octagons + 8 rectangles)

3. (a) 216 cm^3 (b) 1.875 m^3 (c) 450 cm^3 (d) 42 m^3

4. 2 400 000 cm^3; 15 L

5. (a) 162 000 cm^3 (b) 162 L (c) 129.6 L

6. Accept reasonable explanation with workplace example

7. (a) 0.12 m (b) 36 m^3 (c) 6 truck loads (d) \$7 020.00 (e) \$7 722.00 (f) 12 hours (g) \$1 824.00

8. (a) 384 m^3 (b) 0.12 m^3 (c) 0.24 m^3 (d) 3 200 Type X (e) 1 600 Type Y (f) 1 600 Type X; 800 Type Y (g) Total value \$532 000

Chapter 12 — Mass, Capacity and Temperature

Practice Exercise

1. [Mass conversion table — student completion]

2. (a) 4 700 g (b) 12.5 kg (c) 3 000 kg (d) 0.65 g (e) 2 100 000 g (f) 8 000 mg

3. [Capacity conversion table — student completion]

4. (a) 3 600 mL (b) 4.2 L (c) 5 000 L (d) 750 mL (e) 8 000 kL (f) 3.5 kL

5. (a) mg (b) L (c) t (d) ML (e) kg (f) mL (g) ML (h) mg

6–10. [Thermometer / food-safety table in workbook]

Application Exercise

1. (a) 1 000 mg (b) 10 000 mg (c) 10 g (d) 40 tablets

2. (a) 4 500 kg (b) 3 060 kg (c) Within limit; 1 440 kg spare (d) 22 more boxes

3. (a) Scale factor 15 (b) 9 000 g (c) 9 kg (d) 12 000 mL (e) 12 L (f) 6 cartons

4. (a) 51 000 L (b) 51 kL (c) ≈14.1 min (d) 45 000 000 L

5. Prawn cocktail 9°C — unsafe (danger zone), discard; Beef stew 63°C — safe; Chocolate mousse 3°C — safe; Fried rice 52°C — unsafe (danger zone), discard; Grilled fish 74°C — safe

6. [Temperature log in workbook]

Chapter Review

1. (a) 6 400 g (b) 0.9 g (c) 7.5 t (d) 350 kg (e) 2 500 000 mg

2. (a) 2 800 mL (b) 5.6 L (c) 4 000 L (d) 600 kL (e) 3.2 kL

3. (a) kL (b) mg (c) mL (d) t

4. (a) No; 14°C is in danger zone (5°C–60°C) (b) Below 5°C (c) Move to proper refrigeration below 5°C; discard if in danger zone > 2 hours

5. (a) 18 000 g (b) 18 kg (c) 4 bags needed

6. (a) 13 500 mL (b) 13.5 L (c) 9 bottles

7. (a) 24 000 mL (b) 24 L (c) 3 pots (d) 18 000 g (e) 18 kg (f) 10 trays (g) 68°C safe (above 60°C) (h) 45°C in danger zone; must be reheated to above 60°C before serving, or discarded

8. (a) A: 20 m^3; B: 16.2 m^3 (b) A: 20 000 L; B: 16 200 L (c) 36.2 kL (d) A: 12 000 L; B: 5 670 L (e) 18 530 L to pump; 2 h 34 min (f) 17 670 ÷ 4 500 ≈ 3.9 days (approximately 3 days)

Chapter 13 — Workplace Measurement

Practice Exercise

1. (a) 3 000 mm^3 (b) 12 cm^3 (c) 2 500 000 cm^3 (d) 3.4 m^3 (e) 8 000 cm^3 (f) 750 000 mm^3

2. (a) 3 000 mm^3 (b) 3 cm^3

3–5. [Warehouse floor plan in workbook]

6. Real: 20 m × 8 m; Area = 160 m^2

7. Real: 20 m × 16 m; Area = 320 m^2

8–12. [Delivery map in workbook]

Application Exercise

1. (a) 4.8 cm^3 (b) 15 000 cm^3 (c) 3 600 L (d) 0.75 m^3

2. (a) Cold Store: 16 m × 16 m = 256 m^2 (b) 256 m^2 = 2 560 000 cm^2; tiles needed = 2 560 000 ÷ 900 ≈ 2 845 tiles (c) 44 m ÷ 2.4 = 18 rack sections (d) Real perimeter ≈ 64 m × 2 + 48 m × 2 = 224 m [confirm with map]

3. [Delivery map in workbook — student route and distances]

4. (a) 16 cm (b) 10 cm (c) 40 m^2 (d) Real desk: 2 m × 1 m (e) $\frac{2}{40} \times 100 = 5\%$

5. (a) 36 pallets (b) 38 pallets (c) 1 368 pallets (d) 1 026 pallets (75%)

Chapter Review

1. (a) $70\,m^3$ (b) $108\,cm^3$

2. $192\,000\,cm^3$

3. (a) 3 500 g (b) 2.8 L (c) 4 000 L (d) 0.6 g (e) 1 200 kg

4. 28°C in danger zone (5–60°C); discard

5. (a) $4\,000\,mm^3$ (b) $9\,000\,000\,cm^3$ (c) $0.5\,m^3$

6. Real: $8\,m \times 24\,m = 192\,m^2$

7–9. [Floor plan / Delivery map in workbook]

10. (a) $256\,m^2$ (b) 6 units (each unit + aisle = 2.4 m; $16 \div 2.4 = 6.67 \to 6$) (c) $30\,000\,cm^3$ (d) 6 units × 8 shelves × 12 boxes = 576 boxes (e) 12:00 reading (6°C) in danger zone; investigate and restore to below 5°C

11. (a) [Map measurement] (b) B: 432 kg; C: 48 kg; D: 120 kg; E: 330 kg; F: 48 kg; Total: 978 kg < 1 t — load is within limit (c) [Map distance and driving time]

Chapter 14 — Data at Work

Practice Exercise

1. (a) 12 pm (busiest) (b) 8 am (quietest) (c) 61−24 = 37 more (d) Mean = 304 ÷ 7 ≈ 43.4

2–7. [Bakery table / bar graph in workbook]

8. Student bar graph (SALT)

9–10. [Spreadsheet in workbook]

Application Exercise

1. (a) Antihistamine (19 units) (b) Antihistamine (19 < 20) (c) 63 × \$6.80 = \$428.40

2. [Sandwich sales data in workbook — student graph and calculations]

3. (a) Supplier A: 10×\$3.20 + 8×\$2.80 + 5×\$4.50 + 4×\$8.40 + 12×\$1.90 = \$133.30; Supplier B: 10×\$3.60 + 8×\$2.50 + 5×\$4.20 + 4×\$7.80 + 12×\$2.10 = \$133.40 (b) Supplier A cheaper by \$0.10 (c) Accept: quality, reliability, delivery terms

4–5. [Staffing table / temperature log in workbook]

6. (a) Product X: consistently increasing (b) Product Y: consistently decreasing (c) Order more Product X; demand is growing

Chapter Review

1. (a) Student table (b) Thursday (15 requests) (c) Mean = 52 ÷ 5 = 10.4

2. Student bar graph

3–5. [Spreadsheet in workbook]

6. (a) Quote A: 5×\$6.40 + 3×\$8.80 + 10×\$2.20 + 4×\$4.60 = \$98.80;
Quote B: 5×\$5.90 + 3×\$9.60 + 10×\$2.50 + 4×\$4.10 = \$99.70 (b) Quote A cheaper by \$0.90

7. [Revenue tables in workbook]

8. (a) Wk1: \$4 700; Wk2: \$5 250; Wk3: \$4 450; Wk4: \$5 650 (b) Monthly revenue: \$20 050 (c) Manager: \$1 311/wk; Barista: \$834/wk; Wait staff: \$1 488/wk; Weekly total: \$3 633; Monthly: \$14 532 (d) Profit = \$20 050 − \$14 532 − \$3 200 = \$2 318 (e) Week 4 highest food revenue; accept reasonable explanation (e.g. end-of-month traffic) (f) Accept reasonable summary using figures

Chapter 15 — Probability and Trends at Work

Practice Exercise

1. (a) $\frac{10}{250} = \frac{1}{25}$; 0.04; 4% (b) Accept: 4 in every 100 tins are expected to be faulty

2. 2 000 × 0.04 = 80 faulty tins

3. (a) $\frac{5}{12}$; 0.42; 41.7% (b) $\frac{7}{12}$; 0.58; 58.3% (c) Yes: $\frac{5}{12} + \frac{7}{12} = 1$

4. $\frac{1}{5}$; 0.2; 20%

5. (a) $\frac{8}{20} = \frac{2}{5}$ (b) $\frac{5}{20} = \frac{1}{4}$; 25% (c) $\frac{15}{20} = \frac{3}{4}$

6–10. [Call volume line graph in workbook]

Application Exercise

1. (a) $\frac{12}{480} = \frac{1}{40}$; 0.025; 2.5% (b) $720 \times 0.025 = 18$ (c) $12 \times 5 \times \$3.50 = \$210.00$

2. (a) $\frac{5}{15} = \frac{1}{3}$; 0.33; 33.3% (b) $\frac{11}{15}$; 0.73; 73.3%

3. [Seasonal sales graph in workbook]

4. (a) Student line graph (b) Consistently decreasing trend (c) Accept 3–4 sentence conclusion noting decrease from 24 to 11 minutes

5. (a) $162 - 145 = 17$ units (b) ≈179 units (similar annual increase) (c) Accept recommendation to increase November stock order

6. (a) July: $\frac{36}{1200} = \frac{3}{100}$, 3%; Aug: $\frac{30}{1500} = \frac{1}{50}$, 2%; Sep: $\frac{18}{1800} = \frac{1}{100}$, 1% (b) Defect rate decreasing each month (c) ≈0.5%–1% (d) Accept reasonable recommendation to continue current improvements

Chapter Review

1. (a) 60 (b) Clothing (c) $\frac{6}{60} = \frac{1}{10}$

2. (a) $\frac{21}{6} = 3.5$ incidents (b) $\frac{21}{7} = 3.0$ incidents

3. (a) $\frac{18}{600} = \frac{3}{100}$; 0.03; 3% (b) $1\,000 \times 0.03 = 30$ expected errors

4. (a) $\frac{4}{10} = \frac{2}{5}$; 0.4; 40% (b) $\frac{7}{10}$

5. (a) Student line graph (b) Strongly increasing trend (c) $290 - 120 = 170$ units

6. No; 3% is the *expected* rate — actual number varies due to chance. "Exactly 9" is not guaranteed.

7. Percentages are easier to communicate (e.g. "3% defect rate" vs "$\frac{3}{100}$"); accept any valid example

Multi-Step 1:

(a) 2023: $48+45+52+58+60+62 = \$325\,000$; 2024: $55+52+60+66+70+75 = \$378\,000$ (b) \$53 000 increase (c) Student two-line graph (d) Both show increasing trend; 2024 consistently higher (e) Accept 3–4 sentence manager's recommendation

Multi-Step 2:

(a) $\frac{15}{500} = \frac{3}{100}$; 0.03; 3% (b) $3\,600 \times 0.03 = 108$ defective/day (c) $108 \times \$1.85 = \$199.80$ (d) New rate 1.5%; $3\,600 \times 0.015 = 54$ defective; reduction = 54 bottles; saving = $54 \times \$1.85 = \$99.90$/day (e) Accept conclusion noting defect rate halved (3% to 1.5%), 54 fewer defective bottles daily, saving \$99.90/day (≈\$36 464/year), recommend making new process permanent

Numeracy Short Course

Student Workbook

This workbook is the property of the student named on the cover.

Permitted in Assessment Task 2 (open-book examination).